Do Markets Dream of Artificial Intelligence?

Macroeconomic Effects and International Spillovers of AI News Shocks*

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This draft: November 2026

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Abstract

This paper identifies AI news shocks from financial markets and assesses their macroeconomic effects in the United States and the euro area. We combine a daily equity factor based on differences in firms' AI exposures with sign, magnitude, and narrative restrictions in a Bayesian VAR to isolate revisions in AI's expected economic value. Monthly local projections reveal an investment-led adjustment in the US: capital-goods orders rise early and persistently, while aggregate activity responds more gradually and with greater uncertainty. Producer prices increase before consumer prices, and the expansion spreads from sectors closely connected to the AI production network to the broader economy. In the euro area, equity prices rise and the euro depreciates; import and energy costs feed into inflation before real activity strengthens. Inflation expectations and nominal rates also increase. The results highlight a monetary policy dilemma: financial spillovers and imported price pressures arrive before the broader productive gains from AI adoption.

Keywords: AI news shocks, macroeconomic dynamics, international spillovers, production networks

JEL codes: C32, E32, E44, F41, O33

*The author would like to thank Kai Arvai, Niccolò Battistini, Claus Brand, Kai Christoffel, Paola Di Casola, Claudio Lissona, Giacomo Mangiante, Falk Mazelis, Davide Porcellacchia, Annukka Ristiniemi and seminar participants at the ECB and Newcastle University for useful comments and suggestions. The author also gratefully acknowledges excellent data support from Viola Maria Billerbeck, Sofia Caputo and Chiara Scenini. A previous version of this paper was circulated as DNB WP n. 871. The views expressed in this paper are those of the author only and should not be attributed to the European Central Bank, De Nederlandsche Bank or the ESCB.

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1 Introduction

"The future is already here – it's just not evenly distributed."

— William Gibson

AI is increasingly viewed as a potentially transformative general-purpose technology, with implications for productivity, investment, labour markets, inflation, and the conduct of macroeconomic policy. Yet its aggregate effects remain difficult to quantify. A growing microeconomic literature documents significant, though heterogeneous, effects of AI adoption on worker productivity, firm growth, and innovation (Brynjolfsson et al., 2025; Babina et al., 2024), while the magnitude and timing of the associated aggregate productivity gains remain debated (Acemoglu, 2025). This distinction is important for macroeconomic analysis: advances in the technological frontier do not necessarily translate immediately into measured productivity, as their effects may depend on adoption, complementary investment, organisational change, and the reallocation of resources across firms and sectors.

This paper approaches the measurement problem through financial markets. We use daily relative equity valuations to identify changes in expectations about AI's economic value and trace their subsequent domestic and international macroeconomic effects.

Recent theoretical work has begun to investigate these propagation mechanisms. Aldasoro et al. (2024) show in a multi-sector model that AI-driven productivity gains raise output, consumption, and investment, while their inflationary implications depend on the interaction between supply and demand forces. Cerutti et al. (2025) further emphasise that the aggregate consequences of AI can differ across countries, reflecting differences in exposure, adoption, and economic structure. A related strand highlights the capital accumulation required to realise the productive potential of AI. Fernández-Villaverde (2026) shows that the growth effects of automation depend critically on the response of saving and capital accumulation: even for a given path of technological progress, differences in capital supply can generate markedly different growth transitions. Similarly, Melina and Villa (2027) model ICT as a distinct form of capital and show that AI-related investment can affect output, inflation, and the natural rate of interest through intertemporal resource reallocation, with the effects becoming substantially larger when ICT deepening is accompanied by productivity gains and

greater substitutability between ICT capital and labour. From a monetary policy perspective, Hartmann and Maver (2025) distinguish between indirect effects operating through changes in macroeconomic conditions and direct effects on the transmission of monetary policy, highlighting the roles of productivity, investment, labour markets, inflation, and financial intermediation. Empirically, however, identifying an AI-specific macroeconomic shock remains challenging. Technological progress is gradual, its effects are difficult to separate from broader innovation trends, and the information relevant for economic decisions may become available well before it materialises in conventional measures of productivity or technology adoption.

Existing empirical work has addressed this measurement challenge primarily through patent data. At the firm level, David et al. (2026) link AI patenting to higher market valuation and subsequent increases in firms' capital, revenues, employment, and total factor productivity, while Yang (2022) identifies AI-related innovation from patent text and documents positive effects on productivity and employment. Patent information has also been used to measure the exposure of occupations and tasks to advances in AI (Septiandri et al., 2024; Webb, 2020), and more recent work develops increasingly refined classifications of AI patents to study their organisation, diffusion, and economic value (Fang et al., 2026). These approaches provide a natural measure of realised technological invention and are particularly well suited to studying where AI innovation originates and how it diffuses across firms and workers.

The closest empirical contribution to our paper is Gazzani and Natoli (2026), who exploit changes in the AI content of US patents to identify variation in the AI intensity of innovation and document a delayed expansion in economic activity and productivity. Our paper takes a complementary approach by identifying AI news directly from financial markets. Patent-based measures capture realised invention, while asset prices capture revisions in the *expected economic value* of AI. The daily frequency of market data also provides sharper timing, allowing us to combine valuation movements with precisely dated announcements and narrative information. This distinction is particularly relevant for the recent generative AI episode, in which market expectations about adoption, commercialisation, regulation, and profitability can change well before their aggregate effects become observable.

This approach connects the empirical literature on AI to the macroeconomic literature on technology news. In that literature, forward-looking variables reveal changes in expectations

about future productive opportunities before those changes appear in measured productivity or output (Barsky and Sims, 2011; Jaimovich and Rebelo, 2009; Beaudry and Portier, 2006). This distinction is particularly relevant for AI, whose productive effects may require substantial complementary capital accumulation before becoming visible in aggregate data. Fernández-Villaverde (2026), for example, shows that expectations of future automation can generate a strong capital-accumulation response and substantial movements in long real yields, while Melina and Villa (2027) show that AI-related ICT investment can generate time-varying effects on the natural rate as resources are reallocated towards digital capital. In complementary empirical work, Christensen and Rudebusch (2026) examine US Treasury yields and estimates of the natural interest rate around major generative AI model releases.

Our analysis focuses instead on relative equity valuations, exploiting cross-sectional differences in firms' AI exposure to identify news about capabilities, adoption, complementary investment, regulation, and commercial value. We construct a daily equity factor that captures the relative valuation performance of firms that stand to benefit more from advances in AI, building on the insight of Eisfeldt et al. (2023), who show that firms' valuations respond differentially to their exposure to generative AI. We combine these cross-sectional valuation movements with sign, magnitude, and narrative restrictions around precisely dated AI-related events to isolate news that is specific to AI rather than generic equity-market, technology-sector, or semiconductor shocks. The resulting disturbance is a high-frequency revision in the expected economic value of AI.

The paper's methodological contribution is to construct and identify a daily, financial-market-based measure of AI news, linking revisions in relative equity valuations to subsequent macroeconomic outcomes. Using this measure, we make three substantive contributions. First, we document its domestic macroeconomic transmission in the US. The clearest and earliest real-side response is a persistent increase in capital-goods orders, whereas aggregate output responds more modestly and with greater uncertainty. Capacity utilisation temporarily declines before recovering, employment expands at medium horizons, and average hours per worker fall. Price pressures build gradually: producer prices respond before headline and core consumer prices. This sequencing is consistent with AI news triggering a capital-deepening and implementation phase before its productive effects become clearly visible in aggregate

activity. It also distinguishes the shock from a conventional, immediately cost-reducing productivity innovation. The shock instead combines expected future productivity gains with near-term investment and resource demand.

Second, we show that these aggregate responses conceal substantial sectoral heterogeneity. Industries with greater upstream or downstream exposure to the AI production network initially record stronger relative production and more muted producer-price increases. Over time, this relative production advantage fades and the expansion rotates towards less-exposed industries, where price pressures are also larger. Employment rises particularly strongly among highly exposed upstream suppliers, consistent with demand for inputs and complementary capacity. The sectoral evidence therefore characterises the shock as a hybrid technology-adoption disturbance: favourable supply effects are most visible close to the AI ecosystem, while investment and resource-demand effects spread more broadly through the production network.

Third, we estimate sizeable spillovers from US-originated AI news shocks to the euro area. Import, energy, and headline consumer prices respond relatively early, while core inflation increases more gradually and real activity responds only with a delay. Euro area equity prices rise and the euro depreciates against the dollar; short-horizon inflation expectations and nominal rates also increase. The EA–US interest rate differential displays a phased adjustment, initially moving in favour of the US before reversing as euro area inflation becomes more persistent. These results show that AI-related news is transmitted internationally through financial prices and external costs well before its effects on euro area real activity are fully apparent. They point to a two-speed mechanism in which the global financial cycle carries the news rapidly across borders, while productive gains spread more slowly as the technology is adopted and integrated by domestic firms.

Viewed together, the three sets of results describe a common transition from expectations to adoption. Within the US, market news is followed by capital spending and demand for complementary inputs before its effects spread to aggregate production, employment, and consumer prices. Across industries, the initial supply-side gains close to the AI ecosystem coexist with broader resource pressures transmitted through production networks. Across borders, financial prices and external costs adjust more rapidly than productive capacity.

The central empirical finding is therefore the gap between the speed at which the economic value of AI is repriced and the time required for that value to be realised throughout the economy.

The remainder of the paper is structured as follows. Section 2 discusses the construction and validation of the AI news factor; Section 3 presents the identification of the structural AI news shock; Section 4 estimates its domestic, sectoral, and international macroeconomic effects; and Section 5 concludes.

2 The AI news factor

In this Section, we construct an equity-based AI news factor that captures changes in investors' expectations about the future economic relevance of AI. Market prices respond immediately to information about technological capabilities, adoption, investment, competitive advantages, and regulatory constraints. This forward-looking dimension is especially useful in the recent AI episode, where expectations can adjust well before the broader economic effects become visible. We identify a universe of companies relevant to AI and categorise them into a *long* portfolio, consisting of firms at the forefront of AI semiconductors, big-tech platforms, and software, and a *short* portfolio comprising legacy technology companies and non-AI peers.[1]

The AI news factor is the difference between the returns on the long and short portfolios. The *long portfolio* consists of companies directly involved in AI development or positioned to benefit from its adoption. These include firms producing AI semiconductors and compute hardware (e.g., NVIDIA, AMD, and ASML), big-tech platforms investing heavily in AI capabilities (e.g., Microsoft, Alphabet, Amazon, and Meta), and software or enterprise companies providing AI-related solutions (e.g., Palantir, Salesforce, and Adobe).

The *short portfolio* comprises legacy technology companies and non-AI peers. It includes semiconductor firms with more limited AI exposure (e.g., Intel and Texas Instruments), traditional IT service providers (e.g., Accenture and Hewlett Packard Enterprise), and established enterprise technology firms (e.g., IBM and Cisco). This portfolio provides a benchmark against which AI-specific relative returns can be measured.

[1] Section A of the Appendix provides additional details on the construction of the index, including the list of constituents.

The long–short construction captures the relative performance of companies driving or benefiting from AI development and adoption. The portfolios use *time-varying market-capitalisation weights*, allowing changes in firms' market values and capital structures to enter the factor composition.[2] Financial data for both portfolios, including adjusted closing prices, cover January 2012 to October 2026.

To construct the time-varying weights, we obtain historical shares outstanding from SEC filings, adjust them for stock splits, and interpolate them to the daily frequency. This procedure incorporates stock splits, buybacks, and share issuances into daily market capitalisations. We rebalance the long and short portfolio weights quarterly.

Figure 1 examines the factor's ability to capture AI-related repricing rather than broad equity-market movements. Panel a shows the daily series alongside major AI-related events, while Panel b uses one-minute data around OpenAI's GPT-4o presentation on 13 May 2024 to examine the timing of relative valuation changes.

OpenAI had scheduled its Spring Update for 10:00 PT (13:00 ET), during US trading hours, without fully disclosing its content. The combination of a known release time and potentially unanticipated information about model capabilities and commercial availability provides a useful event window, paralleling the timing logic of high-frequency monetary-policy studies (Gürkaynak et al., 2005; Kuttner, 2001).

The AI portfolio initially underperforms during the product, access and API announcements, with NVIDIA and Microsoft accounting for the largest negative contributions. The differential begins to recover during the voice demonstrations, with positive contributions becoming more widespread during the vision and mathematics demonstrations, including Alphabet, Microsoft, Meta, Amazon and NVIDIA.[3] By approximately 30 minutes after the event begins, the differential reaches around 20 basis points, while the S&P 500 remains below its pre-announcement level. The alignment of relative repricing with the presenta-

[2] Section A.1.1 in the Appendix provides further details and descriptive statistics. Section A.1.2 validates the constituent classification using an independent, real-time measure of firm-level AI exposure constructed from SEC annual filings.

[3] In Section A.1.3 of the Appendix, we discuss the construction and coverage of the high-frequency GPT-4o exercise. Figure A.2 decomposes the intraday differential into stock-level contributions over four non-overlapping presentation phases: announcement, access and API economics ($[0, 9)$ minutes), voice demonstrations ($[9, 14)$), vision and mathematics demonstrations ($[14, 18)$), and coding, translation and closing ($[18, 26)$). The phase markers are also shown in Panel b.

tion's progression supports the interpretation that the factor captures market responses to AI-related information, complementing the daily-event evidence.[4]

To assess whether the factor also moves in the anticipated direction across a broader set of announcements, Table 1 compares expected signs with realised event-day returns for 33 AI-related events. We distinguish three categories: i) technological breakthroughs, i.e. moments when new AI capabilities become publicly known, such as the launch of large language models or the announcement of new chip architectures; ii) adoption and commercialisation signals (hereafter adoption events), including major corporate investment announcements and earnings releases from key AI infrastructure companies; iii) regulatory and industrial policy interventions, ranging from national bans on AI applications to executive orders and export restriction decisions. For each event, we assign an ex ante directional expectation: positive when the event is expected to raise the relative value of AI-intensive firms, negative when it signals adverse competitive or regulatory effects, and ambiguous when neither direction can be assigned with confidence. The realised factor return matches the expected direction for 25 of the 26 signed events; the remaining seven have ambiguous expected effects and are excluded from this comparison. The single signed-event miss is the GPT-4o release on 13 May 2024. This descriptive evidence motivates the event study below, which examines the factor's response jointly with other financial variables while controlling for concurrent macro-financial conditions.

2.1 Event study

Following the literature[5], we provide additional evidence that the AI factor captures AI-related surprises through an event study of the dates reported in Table 1. Exploiting the ex ante sign classification of these events, we study the joint behaviour of the AI factor and a set of daily financial variables: broad equity returns (S&P 500), implied volatility (VIX), the US 2-year Treasury yield, the slope of the yield curve (the US 10-year minus 2-year yield spread), 2-year and 10-year inflation-linked swap (ILS) rates, and the broad US dollar nominal effective exchange rate (USDNEER). We estimate separate regressions for innovation, adoption and

[4] In Sections A.1.4 and A.1.5, we compare the proposed factor with an independent, market-priced measure based on the first principal component of five AI-themed ETFs. In Section A.1.6, we also examine alternative weighting, semiconductor-cycle exposure, stock selection. Conclusions are unchanged.

[5] See Gürkaynak et al. (2005); Kuttner (2001); MacKinlay (1997)

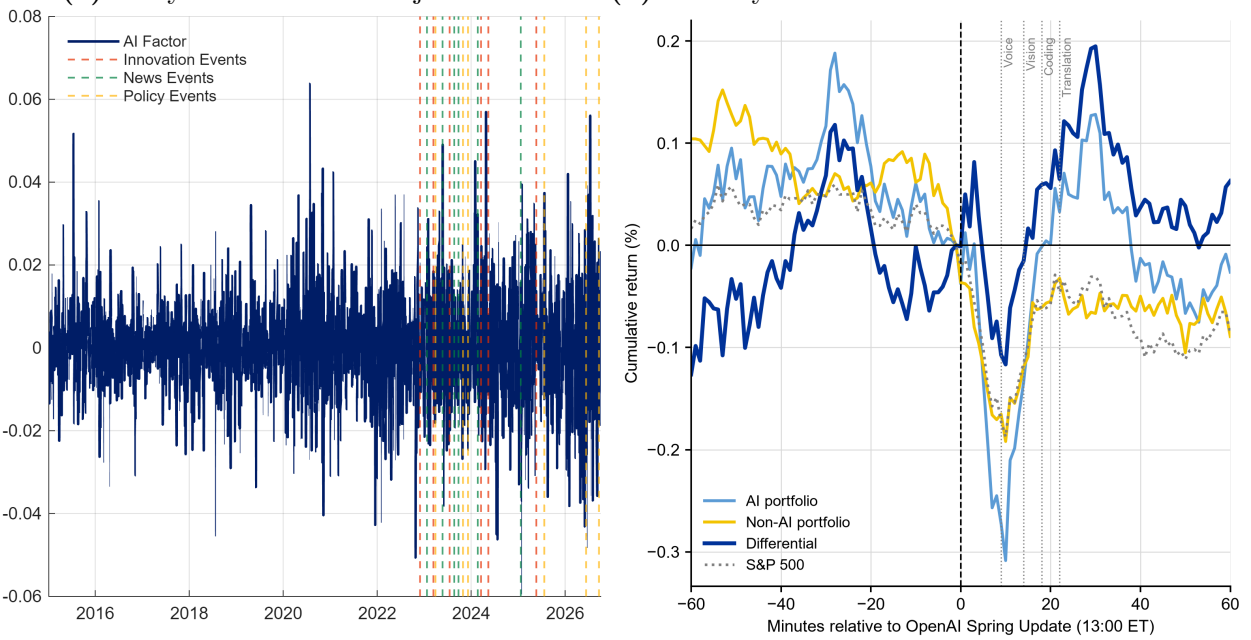
Table 1: AI-related events, expected signs, and sign validation

Date	Event	Expected Sign	Validation	AI factor return
Innovation Events (Technological Breakthroughs)				
30 Nov 2022	OpenAI ChatGPT (GPT-3.5)	+	✓	+2.03%
14 Mar 2023	OpenAI GPT-4 release	+	✓	+1.84%
18 Jul 2023	Meta Llama 2	+	✓	+0.48%
06 Dec 2023	Google Gemini	−	✓	-0.47%
18 Mar 2024	NVIDIA Blackwell architecture	+	✓	+1.48%
18 Apr 2024	Meta Llama 3	−	✓	-0.29%
13 May 2024	OpenAI GPT-4o release	+	×	-1.13%
27 Jan 2025	DeepSeek R1 release/mkt reass.	−	✓	-5.91%
24 Feb 2025	Anthropic Claude 3.7 Sonnet	?	–	-1.23%
04 Apr 2025	Meta Llama 4	?	–	+1.61%
16 Apr 2025	OpenAI o3 and o4-mini	−	✓	-2.09%
22 May 2025	Anthropic Claude 4 (Opus and Sonnet)	+	✓	+1.36%
10 Jun 2025	OpenAI o3-pro release	−	✓	-0.51%
Adoption and Commercialisation Events				
23 Jan 2023	Microsoft $10bn OpenAI investment	+	✓	+0.64%
24 May 2023	NVIDIA Q1 FY24 earnings	+	✓	+0.91%
23 Aug 2023	NVIDIA Q2 FY24 earnings	+	✓	+0.17%
25 Sep 2023	Amazon $4bn Anthropic investment	+	✓	+0.64%
22 Feb 2024	NVIDIA FY24 Q4 earnings	+	✓	+3.22%
23 May 2024	NVIDIA Q1 FY25 earnings	+	✓	+2.74%
10 Jun 2024	Apple Intelligence announced	+	✓	+0.32%
29 Aug 2024	NVIDIA Q2 FY25 earnings	?	–	-2.11%
21 Jan 2025	Stargate Project announced	+	✓	+0.34%
27 Feb 2025	NVIDIA FY25 Q4 earnings	?	–	-3.17%
29 May 2025	NVIDIA Q1 FY26 earnings	+	✓	+0.79%
28 Aug 2025	NVIDIA Q2 FY26 earnings	+	✓	+0.12%
Policy Events (Regulatory and Industrial Policy)				
30 Mar 2023	Italy bans ChatGPT	−	✓	-0.76%
30 Oct 2023	Biden Executive Order on AI	?	–	+1.70%
08 Dec 2023	EU AI Act political agreement	?	–	+0.05%
23 Jul 2025	US AI Action Plan	+	✓	+2.87%
23 Jul 2025	Trump AI Executive Orders	+	✓	+2.87%
30 Jul 2025	US restrictions on advanced AI chip exports	?	–	+1.16%
12 Jun 2026	US export restrictions on Anthropic frontier models	−	✓	-2.25%
21 Sep 2026	Texas freezes new data-center permits	−	✓	-1.82%
# of events				33
# of ambiguous events				7
# of signed events				26
# of matched signed events				25 (96%)

Notes: The table lists the AI-related events used in the analysis, grouped into innovation, adoption and commercialisation, and policy events. "Expected sign" denotes the ex ante anticipated direction of the market-cap-weighted AI factor (+ positive, − negative, ? ambiguous). For signed events, ✓ indicates that the realised event-day AI factor return matches the expected sign, while × denotes a mismatch. The "AI factor return" column reports the realised market-cap-weighted AI factor return on the event day. For events with an ambiguous prior, the validation column is marked with – and only the empirical AI factor return is reported. Event returns are measured on the event date when it is a trading day and otherwise on the nearest trading day. Ambiguous events are excluded from the hit-rate calculation.

Figure 1

(a) Daily AI factor and major events **(b)** Intraday returns around GPT-4o announcement



Sources: Yahoo Finance, Dukascopy, author's computations.
Notes: Panel a reports the daily market-capitalisation-weighted AI factor; innovation, adoption and commercialisation, and policy events (vertical dashed lines) are listed in Table 1. Panel b reports cumulative one-minute returns over the 60 minutes before and after the start of OpenAI's Spring Update on 13 May 2024 at 13:00 ET. Intraday returns are normalised to zero immediately before the event. The black vertical dashed line marks the event start; grey vertical dotted lines mark the approximate starts of voice, vision, coding and translation demonstrations at minutes 9, 14, 18 and 22, respectively. "Differential" denotes the return on the AI portfolio minus that on the synthetic non-AI portfolio. The S&P 500 is shown only as a market benchmark and does not enter the construction of the differential. Details on the intraday data, portfolio coverage, and construction are reported in Section A.1.3.

policy events, controlling for concurrent macro-financial conditions through the US economic growth surprise index (EGS).[6] We also control for the US monetary policy surprise measure of Jarociński and Karadi (2020).

Let $y_{j,t}$ denote variable j on day t, where the event-study vector includes the AI equity factor, broad equity returns, changes in the VIX, changes in the 2-year yield, changes in the slope of the yield curve, changes in the 2-year and 10-year ILS rates, and changes in the USDNEER. All dependent variables are standardised before estimation. For each event

[6]US surprise indices are computed by Haver Analytics and are normalised, equally weighted standard deviations of data surprises based on Action Economics median forecasts versus their respective first-release indicator value or growth rate. A positive reading of the surprise index suggests that economic releases are beating the consensus. As a robustness check, we augment the baseline event-study specification with Haver's inflation surprise measure. The estimated responses are virtually unchanged, indicating that the results are not driven by omitted contemporaneous macroeconomic news. In additional checks, we substitute the EGS with Citigroup Economic Surprise Indices. The results are unchanged.

category $c \in \{\text{innovation}, \text{adoption}, \text{policy}\}$, we define a signed event dummy

$$D_t^c = \sum_{e \in \mathcal{E}_c} s_e \, \mathbf{1}\{t = \tau_e\},$$

where $\mathcal{E}_c$ is the set of events in category c, τ_e is the corresponding trading day, and $s_e \in \{+1, -1, 0\}$ is the ex ante expected sign of the event. Events with ambiguous sign, $s_e = 0$, are retained in the event classification but do not contribute to the signed dummy. Positive events therefore enter with $+1$, negative events with -1, and ambiguous events with zero weight. For each variable j and event category c, we estimate

$$\tilde{y}_{j,t} = \alpha_j^c + \beta_{j,\text{EGS}}^c \, \widetilde{egs}_t + \beta_{j,\text{MP}}^c \, mp_t + \gamma_j^c D_t^c + u_{j,t}^c, \tag{1}$$

where $\tilde{y}_{j,t}$ denotes the standardised dependent variable, $\widetilde{egs}_t$ is the standardised change in the US economic growth surprise index, and mp_t denotes monetary policy surprises, demeaned over non-zero observations. The coefficient γ_j^c is the signed average abnormal response (SAAR). A positive value of γ_j^c indicates that variable j moves, on average, in the same direction as the event sign, while a negative coefficient indicates an average response in the opposite direction. Table 2 reports the estimates for the three event categories.[7]

The AI equity factor responds positively and significantly across all three types of events. The strongest response is observed on policy event days, where the SAAR reaches 1.54 standard deviations, followed by innovation events at 1.25 and adoption events at 0.83. The uniform positive and highly significant response across categories provides direct validation of the factor as a high-frequency measure of AI-related news and strongly supports a positive sign restriction on the AI factor in the structural BVAR.

Broad equity returns also respond positively to innovation and adoption events. The response is large and statistically significant for innovation events (0.73 standard deviations) and positive and marginally significant for adoption events (0.43), whereas the response to policy events is negative but statistically insignificant (-0.38). Importantly, the AI factor responds more strongly than the aggregate equity market for innovation and adoption events.

[7] Table B.7 in the Appendix reports the full set of results for each event in the sample. Policy events that occur on the same trading day are represented by a single event dummy in the regression.

Table 2: Event study results

Variable	Innovation	Adoption and commercialisation	Policy	Sign Restriction
AI factor	$+1.2451^{***}$	$+0.8340^{***}$	$+1.5422^{***}$	+
EQ returns	$+0.7290^{***}$	$+0.4298^{*}$	-0.3805	+
ΔVIX	-0.4241^{**}	-0.0537	$+0.0778$	−
Δ2Y	$+0.1399$	$+0.2583$	-0.0177	·
ΔSlope	$+0.1228$	-0.3547	$+0.6958^{**}$	·
ΔILS2Y	$+0.3149^{*}$	-0.2785^{**}	$+0.2299$	·
ΔILS10Y	$+0.4933$	-0.4678^{**}	-0.6494^{***}	−
ΔNEER	-0.0459	-0.2977	$+0.6189^{*}$	+

Notes: Coefficients are in standardised units. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$, based on Newey–West HAC standard errors, with bandwidth $L = [4(T/100)^{2/9}]$ (Andrews, 1991). This allows for heteroskedasticity and serial correlation in the daily regression residuals. All regressions control for the Haver US economic growth surprise index and the monetary policy surprise measure of Jarociński and Karadi (2020). The last column reports the sign restrictions suggested by the event-study evidence for the daily BVAR. The 2-year ILS response is left unrestricted because statistically significant innovation and adoption responses have opposite signs. By contrast, the 10-year ILS response is negative and statistically significant for both adoption and policy events, motivating a negative restriction on long-term inflation compensation. The positive ΔNEER restriction is supported by the policy event study and is used as an additional identifying dimension.

The ratio of the AI-factor response to the broad-equity response is approximately 1.7 for innovation events and 1.9 for adoption events. This relative-response pattern is useful for identification: AI-related announcements generate a disproportionately large repricing of AI-exposed firms relative to the broad market, helping distinguish AI-specific news from generic risk-on or aggregate growth shocks. This interpretation is consistent with the literature on technology and news shocks, in which information about future productivity is capitalised immediately in asset prices (Barsky and Sims, 2011; Alexopoulos, 2011; Jaimovich and Rebelo, 2009; Beaudry and Portier, 2006).

The response of implied volatility provides an additional dimension of identification. For innovation events, the VIX falls by 0.42 standard deviations and the response is statistically significant at the 5% level. This is consistent with major capability breakthroughs being associated with higher equity valuations and lower aggregate market volatility. The VIX response is instead close to zero for both adoption and policy events. Cross-category Wald tests indicate some heterogeneity: the difference between adoption and policy responses is statistically significant at the 5% level ($p = 0.049$), while the joint null of equal responses across the three categories is rejected only at the 10% level ($p = 0.065$). We therefore retain a negative VIX restriction as an innovation-driven identifying restriction, while not interpreting it as a uniform feature of all AI-related announcements. This pattern is consistent with the

broader literature linking uncertainty shocks, risk premia and financial volatility (Bekaert et al., 2013; Bloom, 2009).

The responses of Treasury yields and the yield-curve slope remain heterogeneous. The 2-year yield rises modestly after innovation and adoption events but is essentially unchanged after policy events; none of the signed coefficients is individually statistically significant. Pairwise Wald tests nevertheless show that the innovation and policy responses differ ($p = 0.019$), while the difference between adoption and policy responses is marginally significant ($p = 0.074$); the joint equality test rejects homogeneous responses across categories at the 5% level ($p = 0.041$). The slope of the yield curve also changes sign across event types. The policy response is positive and statistically significant, whereas the innovation and adoption responses are not. These results provide no stable basis for imposing a contemporaneous restriction on either Treasury-rate variable.

Related evidence is provided by Christensen and Rudebusch (2026), who document declines in five- and ten-year US Treasury yields around major AI model releases. While our innovation events do not generate a significant decline in the two-year yield, they likewise provide no evidence of a significant increase. Despite differences in maturities and event windows, both studies caution against assuming that favourable AI news necessarily raises US yields.

Inflation compensation provides a more informative additional dimension. The 2-year ILS response is positive after innovation events (0.31 standard deviations, or about 2.0 basis points) and negative after adoption events (-0.28 standard deviations, or about -1.8 basis points), with both responses statistically significant at conventional levels. The policy response is positive but imprecisely estimated. The conflicting signs of the significant innovation and adoption responses therefore provide no robust basis for imposing a restriction on short-term inflation compensation. We accordingly leave the 2-year ILS rate unrestricted and exclude it from the baseline structural BVAR.

The 10-year ILS rate displays a clearer pattern among the event categories that are informative about its response. The innovation coefficient is positive but statistically insignificant (0.49 standard deviations; $p = 0.137$). By contrast, the response is negative and statistically significant for adoption events (-0.47 standard deviations, about -1.4 basis points) and

strongly negative for policy events (-0.65 standard deviations, about -2.0 basis points). The adoption and policy responses differ significantly from one another in magnitude ($p = 0.024$), but share the same sign. Thus, across the event categories that yield statistically informative responses for long-term inflation compensation, AI news is associated with a decline in the 10-year ILS rate. In augmented specifications that include long-term inflation compensation, we use this evidence to impose a negative contemporaneous restriction on ΔILS10Y. The baseline six-variable BVAR excludes the ILS rates for parsimony. The restriction in the augmented system is based on the concordant and statistically significant evidence from the adoption and policy studies; the insignificant innovation response is treated as uninformative for the sign decision.

The USDNEER response remains heterogeneous across categories. It is close to zero for innovation events (-0.05 standard deviations), negative and statistically insignificant for adoption events (-0.30), and positive and marginally significant for policy events ($+0.62$). The innovation–policy difference is itself marginally significant ($p = 0.094$), while the joint equality test does not reject common responses across the three categories. We therefore use the positive NEER response as an additional identifying dimension in the baseline BVAR, while recognising that its direct event-study support is concentrated in policy events and is weaker than for the AI factor itself. This mixed exchange-rate response is consistent with the broader evidence that exchange rates jointly reflect interest-rate expectations, risk premia and global financial conditions (Miranda-Agrippino and Rey, 2020; Engel, 2016).

As an additional validation, we assess whether the AI factor responds systematically to salient technology news unrelated to AI. Section B.1 in the Appendix reports a placebo event study based on 24 non-AI technology events, including generic big-tech product announcements, digital innovations, technology disruptions, semiconductor-cycle news and broad technology-policy announcements. The signed average abnormal response of the AI factor on the 12 signed placebo dates is close to zero and statistically insignificant (-0.05 standard deviations, $p = 0.874$), whereas the corresponding pooled response on AI-event dates is about 1.18 standard deviations. A Newey–West HAC Wald test strongly rejects equality of the two responses ($p = 0.002$). The same conclusion holds after orthogonalising the factor with respect to semiconductor returns: the AI–placebo difference remains significant for both

the SMH- and SOXX-orthogonalised factors. These robustness checks further indicate that the factor captures AI-specific information rather than generic mega-cap or semiconductor-market news.

Overall, the event-study evidence supports a sparse identification scheme. The most robust contemporaneous responses are a positive AI-factor response, a positive broad-equity response for innovation and adoption events, and a negative VIX response concentrated in innovation events. The 2-year Treasury yield and the slope of the yield curve are left unrestricted, while the positive NEER response on policy event days provides an additional identifying dimension. The baseline structural BVAR does not include the ILS rates because their signed responses are not uniform across event categories.[8] Together with the narrative classification of event dates and the relative magnitude of the AI-factor response, these results provide the empirical basis for the identification strategy used below, which combines sign, magnitude and narrative restrictions to isolate AI news shocks within the structural BVAR (Arias et al., 2018a; Antolín-Díaz and Rubio-Ramírez, 2018; Uhlig, 2005).

3 Identifying AI news shocks from financial markets

Our identification strategy exploits both the forward-looking content and the daily frequency of financial-market data. Relative equity valuations provide information about changes in the expected economic value of AI, while their joint movements with broader financial variables help distinguish AI-specific news from risk-on and monetary policy shocks. Narrative restrictions connect this variation to dated AI-related announcements.

We estimate a *daily* Bayesian vector autoregression (BVAR) with the following endogenous variables:

$$y_t = [AI_t \; EQ_t \; \Delta VIX_t \; \Delta 2Y_t \; \Delta Slope_t \; \Delta NEER_t] \, .$$

Here, AI_t denotes the AI equity long–short factor, EQ_t broad equity returns, ΔVIX_t the daily change in implied volatility, $\Delta 2Y_t$ the daily change in the US 2-year yield, $\Delta Slope_t$ the change in the 10-year minus 2-year yield spread, and $\Delta NEER_t$ the daily change in the US

[8] In unreported analyses, we also estimate seven- and eight-variable BVARs that include the 10-year ILS rate and both the 2- and 10-year ILS rates, respectively. The results are qualitatively similar to the baseline but are estimated with greater uncertainty. We therefore retain the six-variable specification.

dollar nominal effective exchange rate. All variables are standardised before estimation. The sample runs from January 2015 to October 2026. The reduced-form BVAR is

$$y_t = c + \sum_{l=1}^{p} A_l y_{t-l} + u_t, \quad u_t \sim \mathcal{N}(0, \Sigma_u), \tag{2}$$

where the Bayesian information criterion selects $p = 1$.[9] We use a conjugate normal-inverse-Wishart prior for the autoregressive coefficients and the variance–covariance matrix. The structural representation is

$$A_0 Y_t = c^* + \sum_{l=1}^{p} A_l^* Y_{t-l} + \varepsilon_t, \tag{3}$$

where A_0 contains the contemporaneous structural relationships. The structural shocks satisfy $u_t = A_0^{-1}\varepsilon_t$, with $\varepsilon_t \sim \mathcal{N}(0, I_N)$, and are recovered as

$$\varepsilon_t = A_0 u_t, \quad (A_0^{-1})(A_0^{-1})' = \Sigma_u,$$

Admissible impact matrices A_0 are selected using sign, magnitude, and narrative restrictions, following Arias et al. (2018b); Antolín-Díaz and Rubio-Ramírez (2018). We identify three shocks: i) an AI news shock; ii) a risk-on shock, capturing general risk-on movements in financial markets; iii) a monetary policy shock. Table 3 reports the main identification scheme, consisting of fourteen sign restrictions and one magnitude restriction.

Table 3: Identification Restrictions

Shock	AI **Factor**	**EQ Returns**	**ΔVIX**	**Δ2Y**	**ΔSlope**	**ΔNEER**
AI news shock	++	+	−	·	·	+
Risk-on shock	+	+	−	+	·	−
Monetary policy shock	+	+	−	−	·	−

Notes: + denotes a positive impact, − a negative impact, and · an unrestricted impact. ++ denotes a positive response whose magnitude exceeds that of broad equity returns.

The AI news shock raises both the AI factor and broad equity returns, with a larger response of the former, lowers market volatility, and appreciates the dollar. A risk-on shock raises both equity measures, lowers volatility, raises the 2-year yield, and depreciates the dollar. A monetary easing shock raises equity returns, lowers volatility and the 2-year yield,

[9] Re-estimating the BVAR with two lags produces results very similar to the baseline.

and depreciates the dollar.[10]

We conduct the analysis using $R = 10{,}000$ draws from the posterior distribution. To distinguish the AI news shock further from generic technology, growth, or financial-market disturbances, we complement the sign and magnitude restrictions with a set of narrative restrictions based on the event-study evidence discussed in Section 2.1. The identification proceeds in three steps. First, for each posterior draw of the reduced-form BVAR, we construct candidate structural decompositions and retain those satisfying the contemporaneous sign and magnitude restrictions. Second, we impose a set of narrative restrictions on the historical AI news shock realisations. Specifically, we focus on those events in Table 4 where the AI factor experienced particularly large movements relative to its historical distribution (exceeding 1.5 standard deviations), and require the identified AI news shocks to have the direction implied by the corresponding event. This second-stage restriction exploits information that is external to the BVAR dynamics and helps sharpen the economic interpretation of the identified disturbance. Third, in the spirit of Antolín-Díaz and Rubio-Ramírez (2018), we use the event-study information to assign differential posterior weight to the structural decompositions that survive the sign, magnitude, and narrative restrictions. In particular, on the selected AI-event dates we evaluate the extent to which the unexpected movement in the AI factor is attributable to the identified AI news shock rather than to the other structural disturbances in the BVAR. Structural decompositions in which the AI news shock accounts for a larger share of the AI-factor innovation, and dominates competing shocks more clearly on these dates, receive a higher posterior weight. Importantly, this step does not impose an additional hard dominance restriction: decompositions in which other shocks also contribute materially to the observed market movement remain part of the posterior, but receive relatively less weight. The resulting event-informed posterior therefore exploits the information content of the event chronology while retaining uncertainty about the structural decomposition of individual

[10]The VIX restriction in the first row of Table 3 is used as an auxiliary identifying restriction rather than as a substantive prediction of the AI news shock. Its purpose is to rule out rotations in which favourable AI news is confounded with a contemporaneous deterioration in aggregate risk sentiment. In set-identified VARs, additional economically motivated restrictions help reduce identification uncertainty and narrow the set of admissible structural shocks (e.g. Baumeister and Benati, 2013; Arias et al., 2018a). The restriction therefore sharpens the identification of AI-specific news shocks without playing a direct role in the subsequent macroeconomic transmission.

announcement days.[11] We implement this re-weighting through importance resampling and use the resulting posterior distribution of AI news shocks throughout the subsequent analysis.

Table 4: Narrative Restrictions

Date	Event	Sign
30 Nov 2022	ChatGPT launch	+
23 Jan 2023	Microsoft $10bn OpenAI investment	+
14 Mar 2023	GPT-4 release	+
30 Mar 2023	Italy bans ChatGPT	−
24 May 2023	NVIDIA Q1 FY24 earnings	+
18 Jul 2023	Meta Llama 2	+
22 Feb 2024	NVIDIA FY24 Q4 earnings	+
18 Mar 2024	NVIDIA Blackwell architecture	+
18 Apr 2024	Meta Llama 3	−
23 May 2024	NVIDIA Q1 FY25 earnings	+
10 Jun 2024	Apple Intelligence announced	+
27 Jan 2025	DeepSeek R1 release	−
23 Jul 2025	US AI Action Plan	+
12 Jun 2026	US export restrictions on Anthropic frontier models	−
21 Sep 2026	Texas freezes new data-center permits	−

Notes: '+': shock is positive; '-': shock is negative.

3.1 Results

Figure 2 reports the posterior impact responses of the endogenous variables to AI news, risk-on, and monetary policy shocks. Results are economically coherent and distinguish the AI news shock from the two competing financial disturbances.

An AI news shock that raises the AI factor by 10% produces i) an increase in equity returns of about 2.9%; ii) a decrease in market volatility of about 2.8%; iii) an appreciation of the USD broad exchange rate of about 1.3%. The posterior intervals exclude zero for all three responses. These results are consistent with the view that favourable AI-specific news has effects beyond the firms directly represented in the long–short portfolio. The response of the market-wide equity return is in line with the idea of AI as a general-purpose technology whose productivity implications extend beyond the small set of firms directly involved in its development (Brynjolfsson et al., 2019; Acemoglu and Restrepo, 2019a; Aghion et al., 2017). The accompanying decline in volatility suggests that positive AI news resolves, rather than amplifies, aggregate uncertainty, complementing the uncertainty-shock literature typically focused on the reverse case of negative shocks raising volatility (Bloom, 2009). Finally, the

[11] See Section C.1 in the Appendix for details.

appreciation of the dollar following a positive AI news shock is consistent with AI-related optimism being disproportionately concentrated in US firms and attracting capital towards US assets, in line with the broader evidence on US asset returns, capital flows and the dollar (Miranda-Agrippino and Rey, 2020; Gourinchas and Rey, 2007).

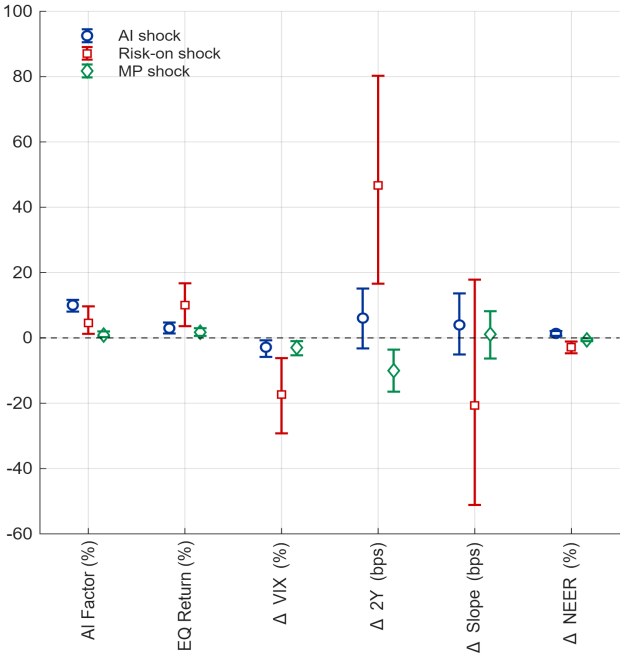
A risk-on shock that raises broad equity returns by 10% produces i) an increase in the AI factor of about 4.6%; ii) a decrease in market volatility of about 17.4%; iii) an increase in the US 2-year yield of about 46.7 basis points; iv) a depreciation of the USD broad exchange rate of about 2.9%. The risk-on shock therefore generates a substantially smaller response of the AI factor relative to the broad market than does the AI news shock. This contrast supports the identification of an AI-specific impulse despite the two shocks' likely co-movement in the raw data. The joint rise in equities and short-term yields, together with the decline in volatility, is characteristic of the risk-on and risk-off dynamics documented in the global financial-cycle and exchange-rate literature (Miranda-Agrippino and Rey, 2020; Ready et al., 2017). The dollar *depreciates* following the risk-on shock, in contrast to its appreciation after AI news. This pattern is consistent with the safe-haven dollar literature (Jiang et al., 2021; Engel, 2016): generic risk-on episodes shift capital towards riskier global assets, whereas the AI news shock contains a more specifically US component.

Finally, a monetary easing shock that lowers the 2-year yield by 10 basis points produces i) an increase in the AI factor of about 0.9%; ii) an increase in equity returns of about 1.7%; iii) a decrease in market volatility of about 3.0%; iv) a depreciation of the USD broad exchange rate of about 0.6%. The posterior intervals exclude zero for these responses. The decline in volatility and the depreciation of the dollar are standard features of monetary policy transmission and are consistent with the extensive literature on high-frequency identified monetary policy shocks (Jarociński and Karadi, 2020; Gürkaynak et al., 2005; Kuttner, 2001). The response of equity returns is also consistent with the duration channel of monetary policy, as growth assets are typically more sensitive to changes in the risk-free rate (Gormsen and Lazarus, 2023).

Overall, the impact responses distinguish AI news most clearly through the joint behaviour of relative equity valuations and the dollar. AI-related assets also behave as long-duration, growth-sensitive assets following monetary easing, but their response is small relative to

the 10% normalization imposed on the AI shock. The dollar appreciates after positive AI news but depreciates following risk-on and monetary easing shocks. This contrast points to a US-specific component of AI-related optimism and complements evidence on the global financial cycle (Miranda-Agrippino and Rey, 2020) and the safe-haven role of the dollar (Jiang et al., 2021).

Figure 2: Impact of shocks at horizon 0



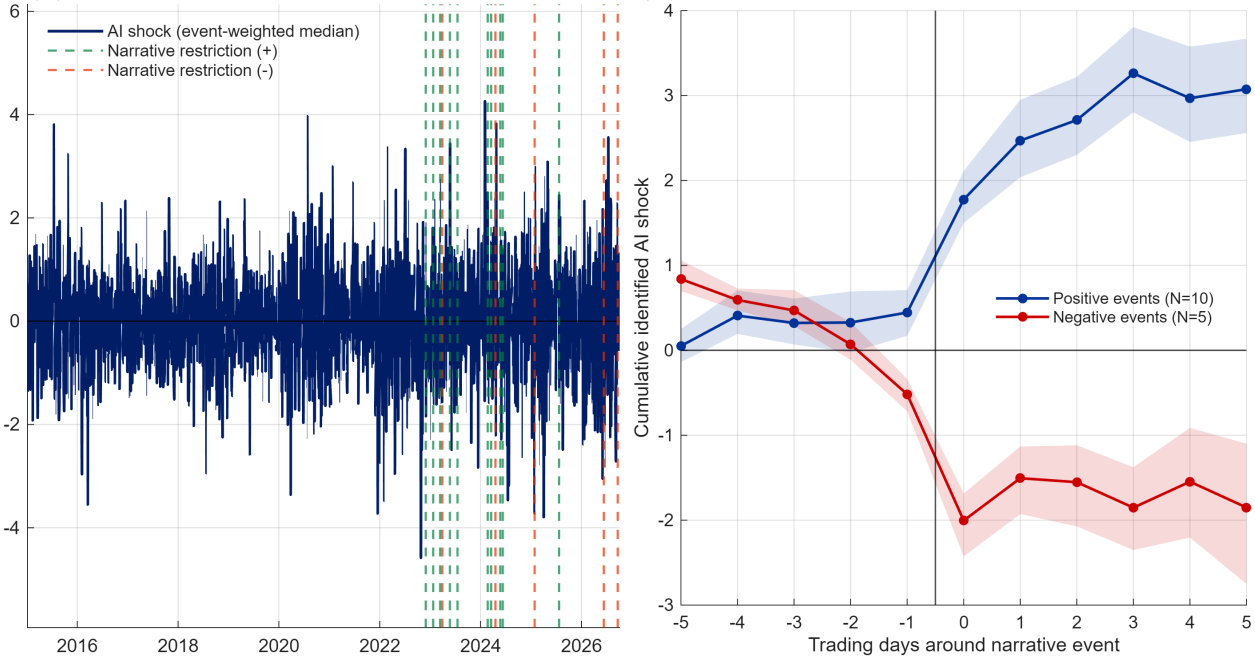
Notes: Impact responses of the endogenous variables to scaled structural shocks. The AI news shock is scaled to raise the AI factor by 10%; the risk-on shock to raise broad equity returns by 10%; and the monetary easing shock to lower the US 2-year yield by 10 basis points, all on impact. Point estimates are event-weighted posterior medians; whiskers indicate 68% highest-density posterior sets.

Having distinguished the AI news shock from risk-on and monetary policy shocks, we next examine its daily history. For each day t, we obtain the posterior distribution of ε_t^{AI} across accepted rotations and use its cross-rotation median as the point estimate. Panel a plots this series together with the events subject to narrative sign restrictions. These include positive developments, such as major model releases, chip-architecture announcements, and large investment deals, as well as adverse developments, such as regulatory bans and export restrictions. The largest shock realisations align closely with the narrative dates, providing an additional check on the economic plausibility of the identification. As an additional diagnostic, we examine the cumulative identified AI shocks in a $[-5, +5]$ trading-day window around

the events used for the narrative restrictions (Table 4). For each admissible structural draw, shocks are cumulated within each event window and then averaged separately across positive and negative events; the resulting distributions are summarized using the event-weighted posterior. Results are reported in Figure 3. The two groups display a marked divergence around the event date, with cumulative shocks rising sharply for positive events and falling for negative events, while remaining clearly separated in the subsequent days. The shaded 68% highest-density posterior sets indicate that this pattern is robust across admissible structural rotations.[12]

Figure 3

(a) Daily AI news shocks with major events **(b)** Cumulative shocks around narrative events



Notes: In Panel a, shocks are standardised and vertical lines mark the event days reported in Table 4. Panel b reports cumulative identified AI shocks in a $[-5, +5]$ trading-day window around the events used for the narrative restrictions. For each admissible structural rotation, daily AI shocks are cumulated from $t-5$ through each relative trading day and then averaged separately across positive and negative narrative events. Solid lines show the event-weighted posterior medians across admissible rotations, while shaded areas denote 68% highest-posterior-density sets. The vertical line marks the narrative-event date $t = 0$. Since the underlying data are measured at the daily close, the movement between $t-1$ and t captures the contemporaneous market reaction to the event. Positive and negative events are classified according to the sign imposed in the narrative restrictions.

[12]Additional checks appear in the Appendix: variance and historical decompositions (Figures C.3 and C.4), predictability tests (Section C.4), and validation against the release dates of Christensen and Rudebusch (2026), conditional on the raw-factor salience screen (Section C.2).

4 The macroeconomic impact of AI news shocks

We assess the macroeconomic implications of US-originated AI news shocks by first aggregating the daily structural shocks to the monthly frequency. The aggregation is performed separately for each structural draw m,

$$\varepsilon_{q,m}^{AI} = \sum_{t \in q} \varepsilon_{t,m}^{AI}, \tag{4}$$

where q indexes calendar months and the sum runs over all trading days t in month q. Before aggregation, the daily shocks are expressed in the original units of the AI factor using its daily standard deviation. The posterior weights attached to the structural draws are preserved, and posterior summaries are computed only after aggregation. This procedure retains the full distribution of monthly shock histories instead of replacing the structural posterior with a single point estimate.

For each outcome variable y and horizon $h = 0, 1, \ldots, H$, we estimate a cumulative local projection following Jordà (2005):

$$y_{t+h} - y_{t-1} \;=\; \alpha_h \;+\; \beta_h\, \varepsilon_t^{AI} \;+\; \sum_{j=1}^{p} \gamma_{h,j}\, y_{t-j} \;+\; \sum_{j=1}^{p} \delta_{h,j}\, \varepsilon_{t-j}^{\mathrm{AI}} \;+\; \theta_h'\, \mathbf{X}_t \;+\; u_{t+h}, \tag{5}$$

where β_h is the response of y at horizon h to an AI news shock in month t, $\mathbf{X}_t$ is a vector of controls,[13] p is the lag length, and u_{t+h} is the projection residual. The sequence $\{\beta_h\}_{h=0}^{H}$ traces the *cumulative* response of y_{t+h}, relative to the pre-shock month, to a monthly AI news shock at time t.

To propagate the event-informed BVAR posterior through the local projections stage while retaining the existing simulation-based inference procedure, we construct an equally weighted Monte Carlo representation of Equation (C.29) through systematic importance resampling. BVAR shock histories are sampled with replacement according to their probabilities ω_m, and each resampled monthly history is treated as an equal-weight draw in the local projection estimation.[14] We use systematic rather than independent multinomial resampling to reduce

[13] To account for the macroeconomic disruption associated with the COVID-19 pandemic, the baseline includes an indicator equal to one whenever the cumulative response window $[t-1, t+h]$ overlaps March–June 2020 (Lenza and Primiceri, 2022).

[14] Repeating the local projections with equal weights on the original admissible BVAR shock histories,

Monte Carlo noise. The normalisation to a one-standard-deviation AI news shock is computed directly from the weighted source distribution rather than from the resampled approximation.

Rather than estimating each horizon independently, we regularise the sequence of impulse-response coefficients across horizons using a hierarchical second-order random-walk prior. In particular, letting $\boldsymbol{\beta} = (\beta_0, \ldots, \beta_H)'$, the prior penalises changes in the slope of the impulse response,

$$\Delta^2 \beta_h = \beta_h - 2\beta_{h-1} + \beta_{h-2}, \tag{6}$$

while leaving the overall shape and sign of the response unrestricted. The strength of the smoothing is not fixed ex ante, but is governed by a variance parameter that is itself estimated hierarchically from the data. Hence, the procedure borrows information across neighbouring horizons while allowing for non-monotonic and hump-shaped responses.[15]

An additional complication is that $\varepsilon_t^{\mathrm{AI}}$ is a *generated regressor*: it is inferred from the daily BVAR rather than directly observed. Conditioning the local projections on a single estimated shock series would therefore discard the first-stage uncertainty associated with the identification of the structural shock and would generally understate uncertainty in the impulse responses (Pagan, 1984). We address this problem by integrating the local projection posterior over the posterior distribution of the identified AI news shock.

More specifically, let $\{\varepsilon_t^{\mathrm{AI},(r)}\}_{r=1}^{R}$ denote draws from the event-informed posterior distribution of the structural AI news shock. The daily draws are first aggregated to the monthly frequency draw by draw, preserving their posterior weights. We then use systematic importance resampling to obtain R shock histories distributed according to the event-weighted posterior. Conditional on each history r, we estimate the Bayesian smooth local projection and draw B realisations from the conditional posterior of the impulse response,

$$\beta_h^{(r,b)} \sim p\Big(\beta_h \mid \varepsilon_{1:T}^{\mathrm{AI},(r)}, \mathbf{Y}\Big), \qquad r = 1, \ldots, R, \quad b = 1, \ldots, B. \tag{7}$$

rather than event-informed weights, yields broadly similar median responses but wider posterior intervals that generally include zero at the reported credibility levels.

[15] Details of the Bayesian smooth local projection estimator and the hierarchical smoothing prior are provided in Section D.1.

The reported posterior distribution is therefore the mixture

$$p(\beta_h \mid \mathbf{Y}) \simeq \frac{1}{R}\sum_{r=1}^{R} p\Big(\beta_h \mid \varepsilon_{1:T}^{\mathrm{AI},(r)}, \mathbf{Y}\Big), \tag{8}$$

Following importance resampling, the histories entering the local projections carry equal weights, while their frequency reflects the original event-informed probabilities. Posterior medians and credible intervals are computed from the resulting pooled distribution

$$\Big\{\beta_h^{(r,b)} : r = 1, \ldots, R,\ b = 1, \ldots, B\Big\}. \tag{9}$$

This procedure propagates both layers of uncertainty: variation across r captures uncertainty about the structural shock generated in the first-stage BVAR, while variation across b conditional on a given r captures estimation uncertainty in the local projection. In this way, the generated AI news shock is not treated as an observed regressor, and inference incorporates rather than conditions away first-stage identification uncertainty.

In the baseline specification we set $H = 48$ and $p_y = p_\varepsilon = 12$. We use $R = 10{,}000$ event-weighted shock histories and draw $B = 100$ posterior realisations of the local projection coefficients for each history. The sample coverage goes from January 2015 to September 2026 for the US, and from January 2015 to August 2026 for the euro area. The hierarchical smoothing parameter is estimated using the procedure described in Section D.1 of the Appendix.

4.1 US economy

Aggregates. We begin with the effects of AI news shocks on the US economy. The outcomes include headline and core CPI, core PCE prices, the headline PPI, real GDP, industrial production, nonfarm employment, average weekly hours, hourly earnings, core capital-goods orders (CAPEX), and total capacity utilisation. Figure 4 reports their monthly responses to a one-standard-deviation shock – around +5% in the AI factor – over a four-year horizon.[16]

The responses point to an adjustment process in which investment and resource demand

[16] Section D.2.1 shortens the lag length to six, while Section D.2.2 removes smoothing across horizons. The results for both the US and the euro area are similar to the baseline.

precede the full expansion of productive capacity. Core capital-goods orders display the largest real-side response, rising strongly and reaching a cumulative peak of about 1.53% around month 30. Real GDP responds earlier but more moderately, peaking at approximately 0.30% around month 17, while industrial production reaches about 0.36% around month 29. This sequencing is consistent with firms responding to favourable news about future AI-related opportunities by expanding investment and reorganising production before the resulting increase in productive capacity is fully reflected in aggregate output.

Capacity utilisation provides further evidence of an implementation phase. It declines over the medium run and reaches a trough of about -0.34% around month 23, despite the strong increase in capital expenditure. The combination of rising investment and temporarily lower utilisation is consistent with adjustment costs, reallocation frictions, and the installation of new capacity during periods of technological change (Acemoglu, 2025; Caballero, 2019; Fujita, 2014). Firms devote resources to capital deepening, organisational change, and complementary inputs before the new capacity is used intensively.

The labour-market response develops more gradually. Nonfarm employment rises over the medium and long horizon and reaches approximately 0.49% around month 45, while hourly earnings increase by about 0.33% at a similar horizon. Average weekly hours instead decline, reaching roughly -0.14% around month 44. The combination of rising employment and falling hours per worker suggests that labour adjustment operates predominantly through the extensive margin, consistent with firms expanding headcount while reorganising the allocation of tasks and production processes as AI adoption proceeds.

Price dynamics complement this interpretation. Producer prices respond before consumer prices, with the cumulative PPI response reaching approximately 0.96% around month 35. Headline CPI, core CPI, and core PCE prices increase more gradually, reaching peaks of roughly 0.74%, 0.57%, and 0.52%, respectively, around month 45. The ordering from producer to consumer prices is consistent with a gradual pass-through of upstream cost pressures (Blanchard, 2010; Galí, 2008).

The positive price response is informative about the nature of the disturbance. A conventional positive productivity shock would tend to reduce marginal costs and prices relatively quickly. By contrast, AI news raises investment and resource demand before its

full productivity effects materialise. Expectations of future productivity gains can therefore coexist with near-term pressure on capital goods, intermediate inputs, energy, labour, and production capacity. This distinction also helps reconcile our findings with the disinflationary effects reported by Gazzani and Natoli (2026) for patent-based AI technology shocks. Patent measures are more directly associated with realised technological innovation, whereas the market-based measure used here captures revisions in the expected economic value, adoption, and commercialisation of AI. Financial markets can therefore respond to anticipated future productivity well before the corresponding productive capacity is in place.

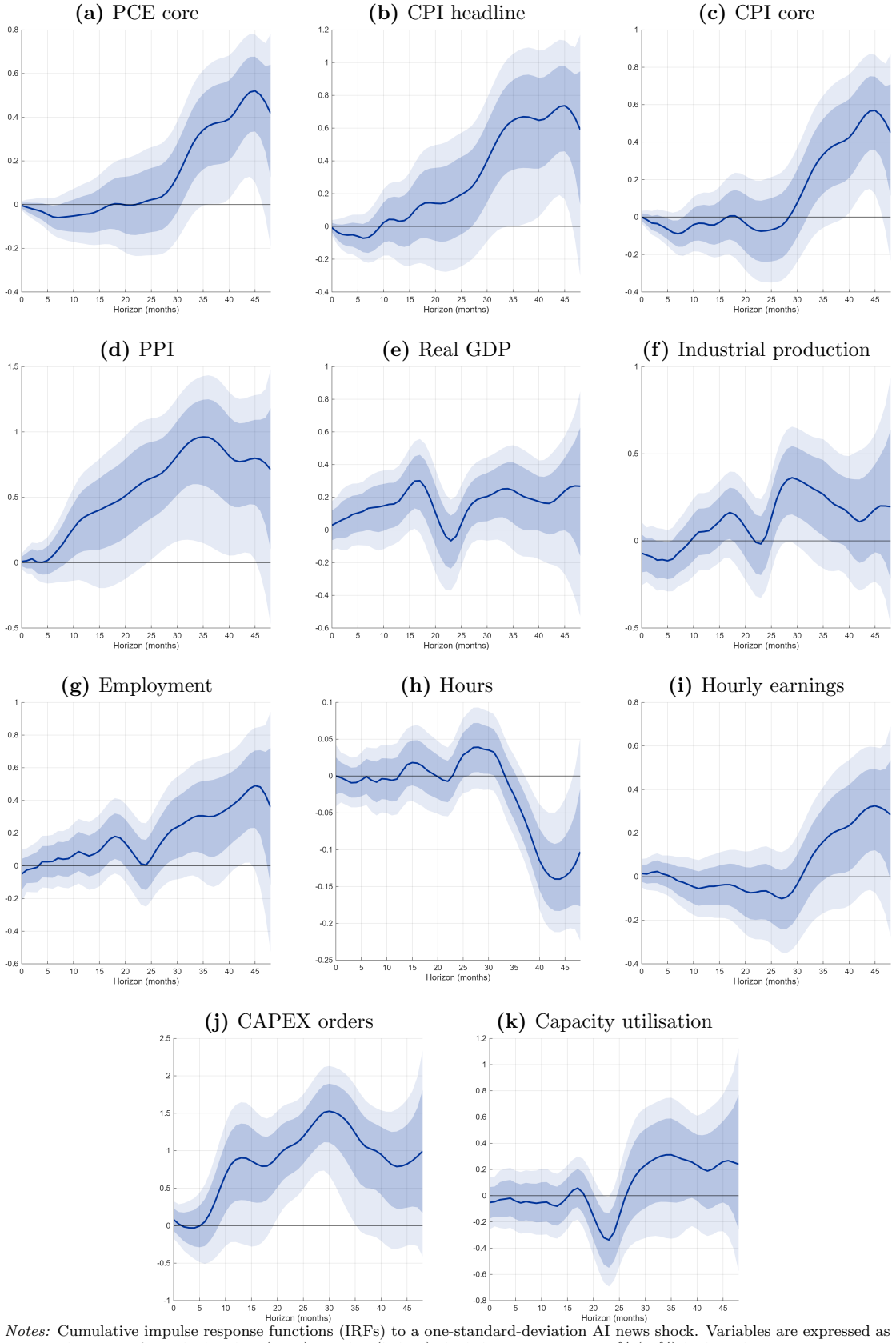
Taken together, the aggregate responses characterise the AI news shock as an investment- and implementation-intensive technology-news disturbance. The initial response is dominated by capital deepening and resource reallocation, rather than by an immediate economy-wide increase in productive efficiency. Output expands more gradually, capacity utilisation initially falls, labour demand strengthens with a delay, and producer-price pressures eventually pass through to consumer prices. The macroeconomic effects therefore emerge through a transition in which expected future productivity gains first induce investment and implementation expenditure and only subsequently translate into broader productive capacity (Acemoglu, 2025; Carvalho and Gomes, 2017; Gordon, 2010; Mokyr, 2015; Jaimovich and Rebelo, 2012).

Sectoral effects. We next examine sectoral prices, production, and employment to assess whether the aggregate responses conceal heterogeneous adjustment across industries with different degrees of AI exposure.

To construct the sectoral aggregates, we map individual statistical series into a common set of BEA industries and assign each industry upstream and downstream AI-network exposure measures.[17] Figure 5 illustrates the production linkages underlying these measures. Upstream exposure is highest in industries such as software publishing, broadcasting and telecommunications, motion pictures and sound recording, and computer and electronic products. Downstream exposure is also high in software publishing and computer and electronic products, as well as in broadcasting and telecommunications, data processing, and internet services. AI-related activity is therefore embedded in a broad production network

[17] Section A.2.3 provides details.

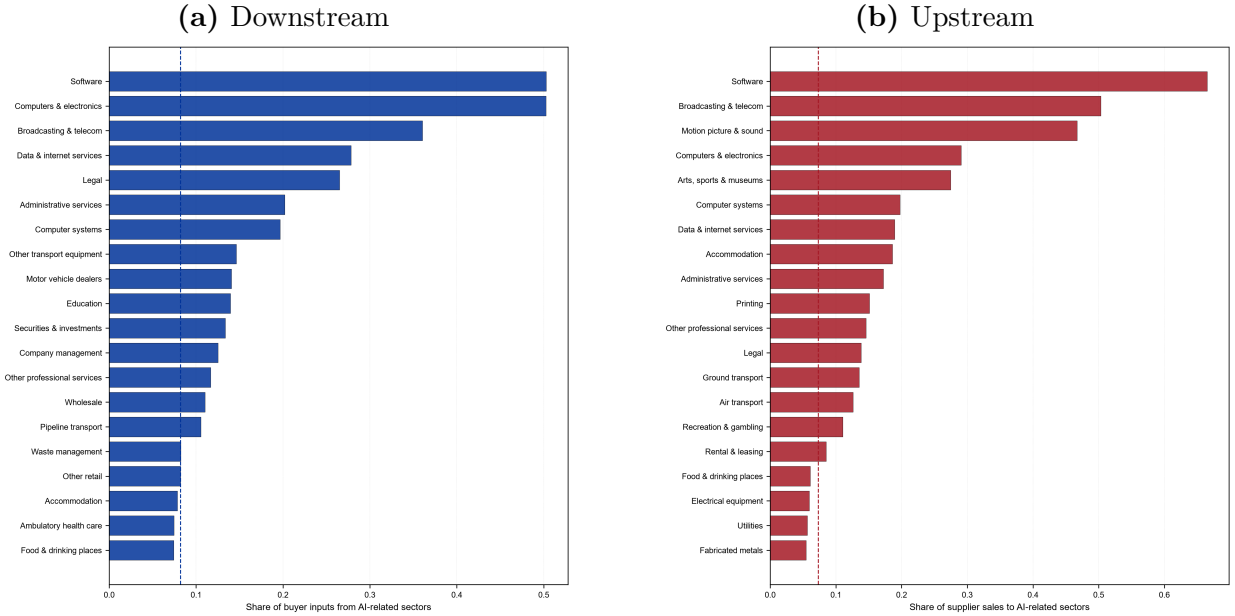
Figure 4: Impact of AI news shocks on US aggregates



Notes: Cumulative impulse response functions (IRFs) to a one-standard-deviation AI news shock. Variables are expressed as percent deviations from the baseline ($t-1$). Darker (lighter) shaded areas represent 68% (90%) highest-density posterior sets.

through which news shocks can propagate.

Figure 5: Sectoral exposures to AI



Sources: BEA and author's computations. *Notes:* Vertical dashed lines indicate the 75th percentile.

For each economic domain, we then split sectors into high- and low-exposure groups using the corresponding exposure distribution and construct fixed-composition aggregate indices for each group. The same sectoral classification is kept throughout the sample, so changes in the aggregates reflect within-sector dynamics rather than time variation in group membership. We separately construct upstream and downstream aggregates, and for each dimension consider both the responses of the high- and low-exposure groups. This yields a direct comparison of AI-related shocks' effects between sectors that are more and less exposed to AI-related activity through the production network.

Figures 6 and 7 show that the aggregate adjustment has a clear sectoral counterpart. Production initially performs relatively better in industries with greater AI-network exposure, while producer-price pressures are systematically weaker in those industries. Over time, however, the production advantage of highly exposed sectors dissipates and eventually shifts towards less-exposed industries. The sectoral responses therefore trace a transition from an initially concentrated expansion around the AI ecosystem to a broader production response as the shock diffuses through the economy.

The relative price response is particularly informative. Producer prices rise in both high-

and low-exposure groups, but the increase is consistently larger among sectors with lower AI-network exposure. Upstream, the cumulative PPI response peaks at about 1.32% for the high-exposure group and 1.83% for the low-exposure group, both around month 33. The high-minus-low differential is negative over almost the entire horizon and reaches roughly -0.38 percentage points around month 32. Downstream, the contrast is even stronger: producer prices peak at approximately 0.94% among highly exposed industries and 1.87% among low-exposure industries, with the differential widening to about -0.92 percentage points by the end of the reported horizon.

This configuration is consistent with favourable supply effects being strongest close to AI-related activity. Industries more directly connected to the AI production network experience smaller price increases at the same time as their relative production initially improves. By contrast, price pressures become larger as the shock spreads towards less-exposed industries. The positive aggregate PPI response therefore masks an important composition effect: the sectors most closely linked to AI display a more supply-like quantity–price configuration, while the broader economy is more strongly affected by the resource demands associated with investment, implementation, and diffusion. This mechanism is consistent with production-network models in which sector-specific disturbances propagate through intermediate-input demand, complementarities, and general-equilibrium reallocation (Baqaee and Farhi, 2018; Carvalho, 2014; Carvalho and Voigtländer, 2014).

Production responses make the diffusion mechanism especially visible. Upstream, the high-minus-low response becomes positive at relatively short horizons and reaches approximately 0.50 percentage points around month 24. The advantage subsequently dissipates and ultimately reverses, with the differential falling to around -0.64 percentage points at long horizons. Highly exposed upstream industries therefore benefit relatively more during the initial implementation phase, while production eventually expands more strongly outside the sectors closest to AI activity.

The downstream response follows the same sequence even more clearly. The high-minus-low production differential becomes positive during the first year, reaches about 0.61 percentage points around month 25, and then declines sharply. At longer horizons the differential turns negative as production among less-exposed sectors catches up and ultimately exceeds that of

the more-exposed group. The shock therefore does not generate a permanently concentrated expansion in AI-intensive industries. Instead, the initial increase in activity near the AI ecosystem is progressively transmitted towards a wider set of industries.

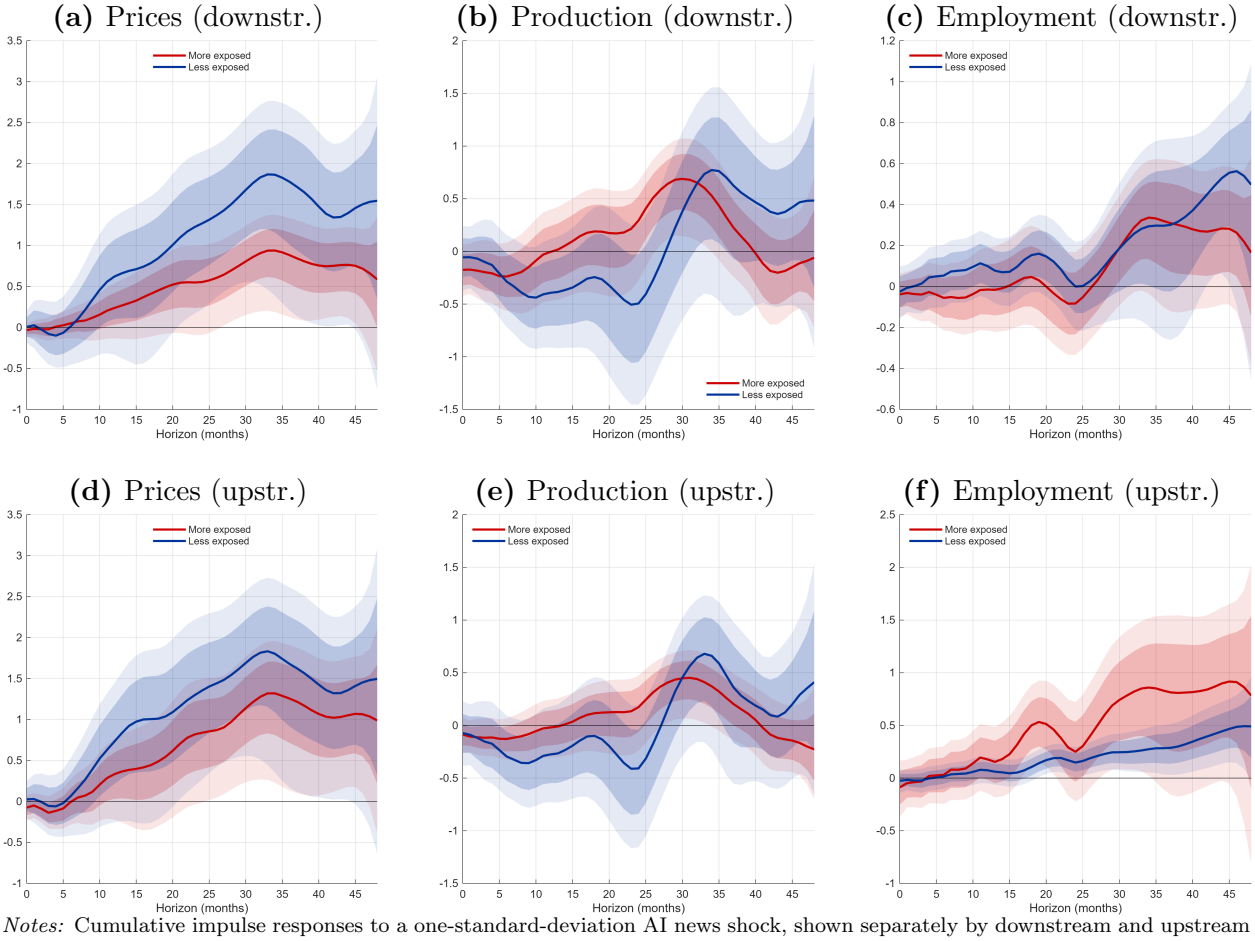
Employment provides complementary evidence on the direction of this propagation. Among upstream suppliers, employment increases considerably more in highly exposed industries. Their cumulative response reaches approximately 0.92% at long horizons, compared with roughly 0.49% among low-exposure suppliers. The high-minus-low employment differential becomes positive over the medium run and reaches around 0.63 percentage points. This suggests that the investment and implementation phase generates substantial demand for labour among industries supplying inputs and complementary capacity to the AI ecosystem. The upstream propagation of the shock is therefore not simply a process of capital substituting for labour; it is accompanied by an expansion of employment in the sectors most closely involved in supplying the new technology.

Downstream employment follows a different pattern. Highly exposed industries initially experience the stronger response, but the differential is modest and eventually reverses. Employment among less-exposed downstream sectors continues to strengthen at longer horizons, while the high-minus-low differential turns negative. The downstream labour-market response thus mirrors the broader diffusion of production: as AI adoption spreads beyond the sectors most directly connected to the technology, employment growth also becomes more widely distributed across the economy.

The sectoral evidence therefore provides a mechanism for the aggregate responses. The AI news shock initially raises activity in sectors closest to the AI production network, where stronger relative production is accompanied by comparatively muted price pressures. At the same time, upstream employment increases strongly, consistent with demand for complementary inputs and implementation capacity. As the adjustment proceeds, activity spreads towards less-exposed sectors, while producer-price pressures become increasingly concentrated outside the industries benefiting most directly from AI-related supply effects. The production network thus acts as a diffusion mechanism, transforming an initially concentrated technology-related impulse into a broader macroeconomic expansion.[18]

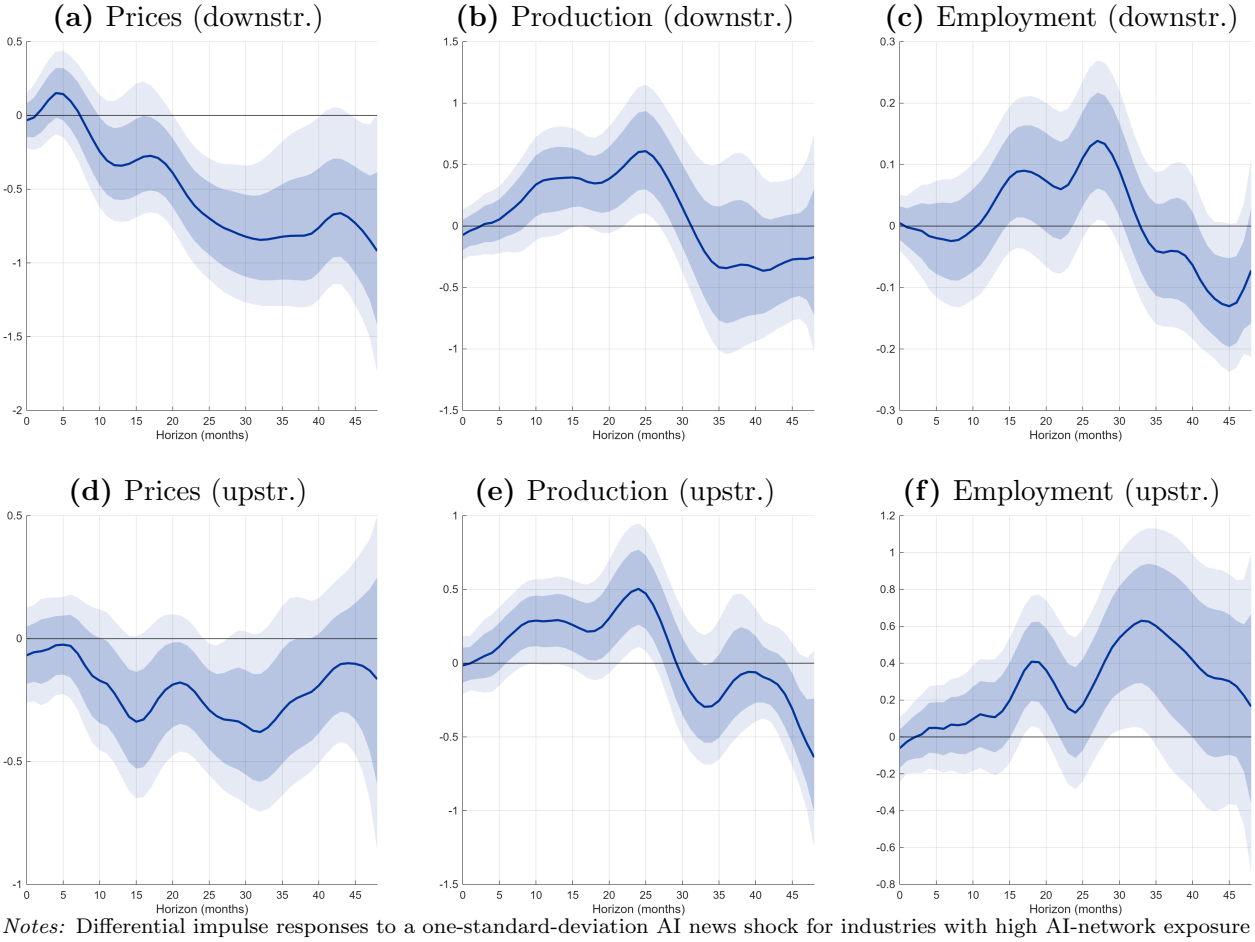
[18]As a robustness exercise, we reconstruct the aggregates using the $10^{th}/90^{th}$ percentiles of the empirical

Figure 6: Impact of AI news shock on sectors



Notes: Cumulative impulse responses to a one-standard-deviation AI news shock, shown separately by downstream and upstream input–output exposure to AI-related sectors. Darker (lighter) shaded areas represent 68% (90%) highest-density posterior sets. Variables are expressed as percentage deviations from the baseline $(t-1)$.

Figure 7: *Differential* impact of AI news shock on sectors



Notes: Differential impulse responses to a one-standard-deviation AI news shock for industries with high AI-network exposure relative to industries with low exposure. Darker (lighter) shaded areas represent 68% (90%) highest-density posterior sets. Responses are expressed in percentage points.

AI news shocks versus conventional productivity shocks. The aggregate and sectoral evidence together help distinguish the identified AI news shock from a conventional aggregate productivity innovation. A favourable TFP shock is typically associated with an immediate increase in productive efficiency and a decline in marginal costs. The responses documented here instead involve a substantial implementation phase: capital expenditure rises strongly, capacity utilisation initially falls, producer prices increase, and the expansion in aggregate production and employment develops gradually.

At the sectoral level, however, the industries closest to AI activity display a more conventional supply-side signature. Highly exposed sectors initially experience stronger relative production together with weaker relative price pressures. This configuration is consistent with favourable technology or productivity effects being concentrated initially in the parts of the economy most directly connected to AI. These local supply effects coexist with broader demand for capital, intermediate inputs, energy, and labour generated by the process of adoption.

The direction of propagation through the production network reinforces this interpretation. Standard network models imply that supply-side productivity disturbances propagate predominantly downstream, whereas demand disturbances transmit upstream towards input suppliers (Acemoglu et al., 2016). In our estimates, the AI news shock generates economically meaningful responses along both dimensions. Relatively strong production and muted price pressures in highly exposed industries point to a local supply-side component, whereas the pronounced employment response among upstream suppliers indicates substantial demand for complementary inputs and implementation capacity. At longer horizons, the relative production advantage moves away from the sectors closest to AI as the technology is adopted more broadly.

The resulting configuration is therefore consistent with a hybrid technology-adoption shock. News about future AI productivity raises the value of adopting the technology and induces investment before those productivity gains are fully realised. During the transition, the economy must accumulate complementary capital, reorganise production,

exposure distribution, rather than the $25^{th}/75^{th}$ percentiles used in the baseline, and re-estimate the local projections. The results, reported in Section D.2.4, are qualitatively similar.

expand relevant inputs, and reallocate labour. These implementation costs generate demand and price pressures even as productivity begins to improve in the sectors closest to the new technology. Only subsequently do the productive benefits diffuse through the broader production network. In this respect, AI news differs from a generic TFP innovation precisely because the technology must first be embodied in new capital, organisational structures, and complementary tasks before its full productivity effects become visible, generating potentially important displacement, capital-deepening, and reinstatement effects along the transition (Acemoglu and Restrepo, 2019b,a).

4.2 Spillovers to the Euro Area

We next examine how US-originated AI news shocks propagate to the euro area. Figure 8 reports the responses of real and price variables, while Figure 9 presents the corresponding financial responses, including two- and ten-year EA–US interest rate differentials.[19] The estimates point to a pronounced difference in timing between financial and external-price adjustment on the one hand and domestic real activity and inflation on the other. This sequencing is consistent with evidence that US technology and uncertainty shocks travel internationally through financial, trade, and expectations channels (e.g., De Gasperi and Ricco, 2023; Miranda-Agrippino and Rey, 2020; Jordà et al., 2019).

Real and price responses. The response of euro area prices is strongly hump-shaped and heterogeneous across components. Headline HICP eventually rises by about 0.86%, reaching its peak around month 42. Core HICP displays a more complex pattern: it initially falls, reaching a trough of roughly -0.20% around month 19, before reversing and increasing to about 0.45% by month 43. Producer prices respond much more strongly, peaking at approximately 2.83% around month 32. This sequencing suggests that the initial transmission operates through external and upstream costs, while broader underlying inflation emerges only with a substantial lag.

External price pressures are sizeable. Import prices rise by around 2.35% at their peak near month 32, while the broader energy-price index increases by about 8.09% around month 30. Gas prices display the largest price response, reaching approximately 14.29% around

[19] See Section A.2.2 for the variables and data sources.

month 29. These movements are consistent with a channel in which AI-related investment and computing demand increase the demand for energy and other complementary inputs, with the resulting external cost pressure subsequently feeding into domestic prices (IEA, 2025; De Gasperi and Ricco, 2023; Känzig, 2021). The temporary decline in core HICP before its later increase is also consistent with an initially mixed supply-and-demand response. Imported cost pressures coexist with relatively weak domestic labour-market conditions in the first stage of the transmission, so that external price pressures need not immediately translate into underlying inflation. This sequencing is consistent with recent theoretical work on AI and technology news. Aldasoro et al. (2024) show that an AI-driven productivity expansion can initially be disinflationary when future productivity gains are not fully anticipated, while stronger consumption and investment eventually raise demand, wages and inflation; when productivity gains are anticipated, these demand effects are brought forward. Fornaro and Wolf (2026) similarly emphasise that AI-induced automation can raise productive capacity while weakening labour income and aggregate demand in the short run, generating a non-trivial inflation-employment trade-off. A related mechanism arises in the recent work by Nispi Landi (2026), where anticipated AI-related productivity gains stimulate investment but transition and adjustment costs absorb resources before the productivity gains are fully realised, weakening consumption and inflation initially. More broadly, the result echoes the technology-news literature, in which favourable news about future productivity can be disinflationary at first even when forward-looking demand components strengthen (Jinnai, 2013; Barsky and Sims, 2011).

Real activity responds only gradually. Industrial production reaches a cumulative peak of about 0.65% around month 29. Employment is initially weak but subsequently rises, peaking at approximately 0.26% around month 37, while hours worked increase more strongly and peak at about 0.79% around month 41. The delayed real response relative to financial prices and external costs is consistent with the euro area acting primarily as an adopter rather than an originator of the technology shock: the productive effects materialise only as AI is gradually integrated into domestic production processes.

Financial spillovers. Euro area equity prices increase significantly after the initial months and reach a cumulative peak of about 2.71% around month 33. The euro depreciates against the dollar, with the USD/EUR response reaching approximately −1.11% around month 39. Thus, favourable US AI news is associated with a global rise in equity valuations but also with a persistent strengthening of the dollar relative to the euro. This combination is consistent with a global financial-cycle channel in which technology-related news raises valuations across markets while simultaneously increasing the relative attractiveness of US assets (Miranda-Agrippino and Rey, 2020; Rey, 2015).

Inflation expectations and nominal rates. Inflation compensation rises materially, particularly at shorter maturities. The two-year inflation-linked swap rate reaches a peak response of about 23 basis points around month 30, while the ten-year inflation-linked swap rate peaks at roughly 13 basis points around month 46. The larger response at the two-year maturity is consistent with markets pricing a stronger near- to medium-term inflationary impulse than a permanent shift in long-run inflation expectations.

Nominal rates display an even larger response. The two-year OIS rate initially falls, but subsequently rises sharply, reaching a peak of about 37 basis points around month 36. The ten-year OIS rate peaks at roughly 27 basis points around month 34. Since the increase in nominal rates exceeds that in inflation-linked swap rates, the adjustment cannot be attributed solely to inflation compensation; it also reflects a sizeable increase in real rates and possibly term premia. The delayed reversal from initially lower to substantially higher short-term OIS rates is consistent with the gradual transition from an early external financial shock to a more persistent domestic inflationary and real activity response.

Interest rate differentials and the policy dilemma. The EA–US interest rate differentials point to a clearly phased adjustment in relative monetary and financial conditions. At the two-year maturity, the differential falls almost immediately and becomes significantly negative from around month 3, reaching a trough of about 5.6 basis points around months 7–8. It remains significantly negative through roughly month 14 before gradually closing and moving back towards zero. The ten-year differential also declines at short horizons, indicating

that US rates initially respond more strongly than euro area rates across the term structure. Thus, the early adjustment is not one in which euro area rates rise relative to US rates; rather, favourable US AI news initially shifts relative financial conditions in favour of the United States.

This short-run movement in the differentials contrasts with the subsequent behaviour of euro area nominal rates. In particular, the euro area two-year OIS rate initially declines before reversing and rising strongly at longer horizons. The same broad pattern is visible, though less sharply, at the ten-year maturity. The differentials therefore capture an asynchronous international adjustment: in the first stage, the US interest-rate response dominates, while euro area rates remain comparatively subdued; only later, as domestic inflationary pressures and real activity strengthen, do euro area rates rise more substantially.

The interpretation should not be that these differentials identify a mechanical policy rule. Rather, they illustrate the policy dilemma created by global financial spillovers (Rey, 2015). The euro area initially faces a deterioration in relative financial conditions, alongside currency depreciation and higher imported and energy costs, before the full domestic real and inflation effects of AI adoption materialise. As producer-price pressures feed through more broadly, core inflation eventually turns positive and domestic activity strengthens, euro area nominal rates rise markedly. Monetary conditions therefore adjust only gradually to a shock whose financial and external-price effects arrive much earlier.

These results provide an empirical counterpart to the conceptual assessment of Hartmann and Maver (2025), who emphasise that AI can affect the monetary policy stance indirectly through its joint effects on productivity, investment, aggregate demand, labour markets, energy costs, and inflation. The estimates sharpen the distinction between stages of transmission. In the first stage, equity prices, the exchange rate, external input prices, and relative interest rates adjust rapidly, while domestic real activity remains weak and core inflation initially declines. In the second stage, industrial production, employment and hours strengthen, headline and core inflation rise, and euro area nominal rates increase substantially.

Taken together, the results point to a two-speed open-economy transmission mechanism. At short horizons, US AI news propagates through the global financial cycle and external input prices. Euro area equity valuations rise, the euro depreciates, imported and energy

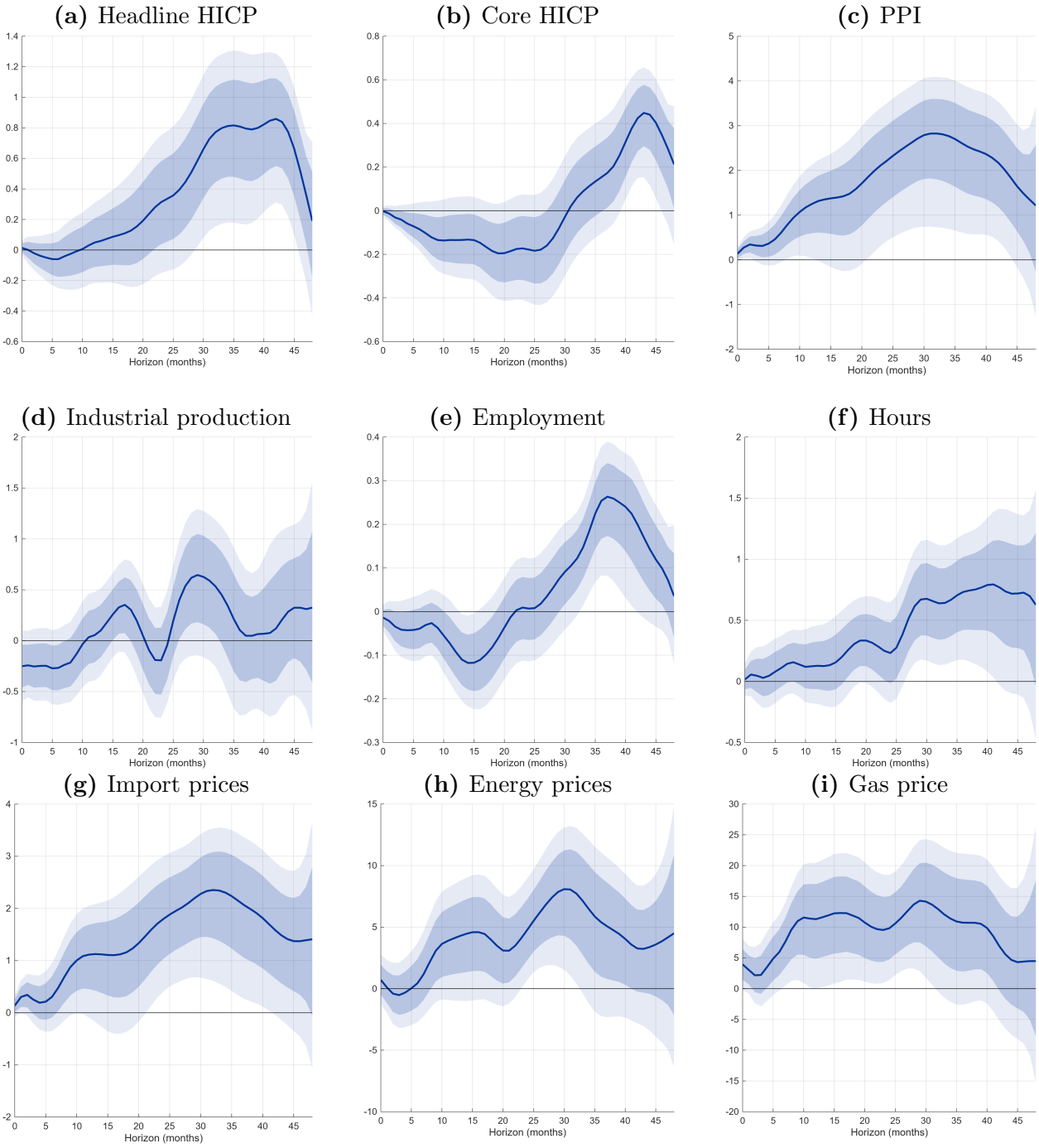
costs increase, and EA–US rate differentials move in favour of the United States, while domestic activity remains subdued. At longer horizons, the expansion becomes more broadly based: industrial production, employment and hours increase, inflation pressures broaden, and euro area nominal rates rise. The initial widening and subsequent closing of relative rate differentials is therefore part of the same transition from an externally driven financial shock to a more domestic inflationary and real response.

The distinction between AI innovator and AI adopter helps explain this timing. The US is more directly exposed to the development and commercialisation of frontier AI and therefore experiences an earlier response in valuations, capital spending, and demand for complementary inputs. The productive benefits for the euro area depend more heavily on subsequent adoption and integration of the new technology by domestic firms. Related firm-level evidence is provided by Aldasoro et al. (2026), who estimate that AI adoption raises labour productivity in EU firms by around 4%, with no significant employment effect. The gains are larger among firms undertaking complementary investments in software, data and workforce training. These findings support the relevance of adoption and complementary investment for realising productivity gains in Europe, although they do not establish the timing of the aggregate response to AI news documented here.

This pattern echoes the history of earlier general-purpose technologies: the gains from steam, electricity, and ICT emerged gradually as firms accumulated complementary capital and reorganised production, while international diffusion lagged invention and varied with countries' capacity to absorb and deploy new technologies (Crafts, 2021; Brynjolfsson et al., 2019; Comin and Hobijn, 2010; Keller, 2004; Eaton and Kortum, 1999). The global-financial-cycle and innovator–adopter interpretations are therefore complementary rather than competing. The former explains the rapid transmission of valuations, relative prices, and financial conditions; the latter accounts for the slower diffusion of real gains. Monetary and financial conditions adjust as the spillover moves from its initial external-price and financial phase towards a broader domestic real and inflation response.[20]

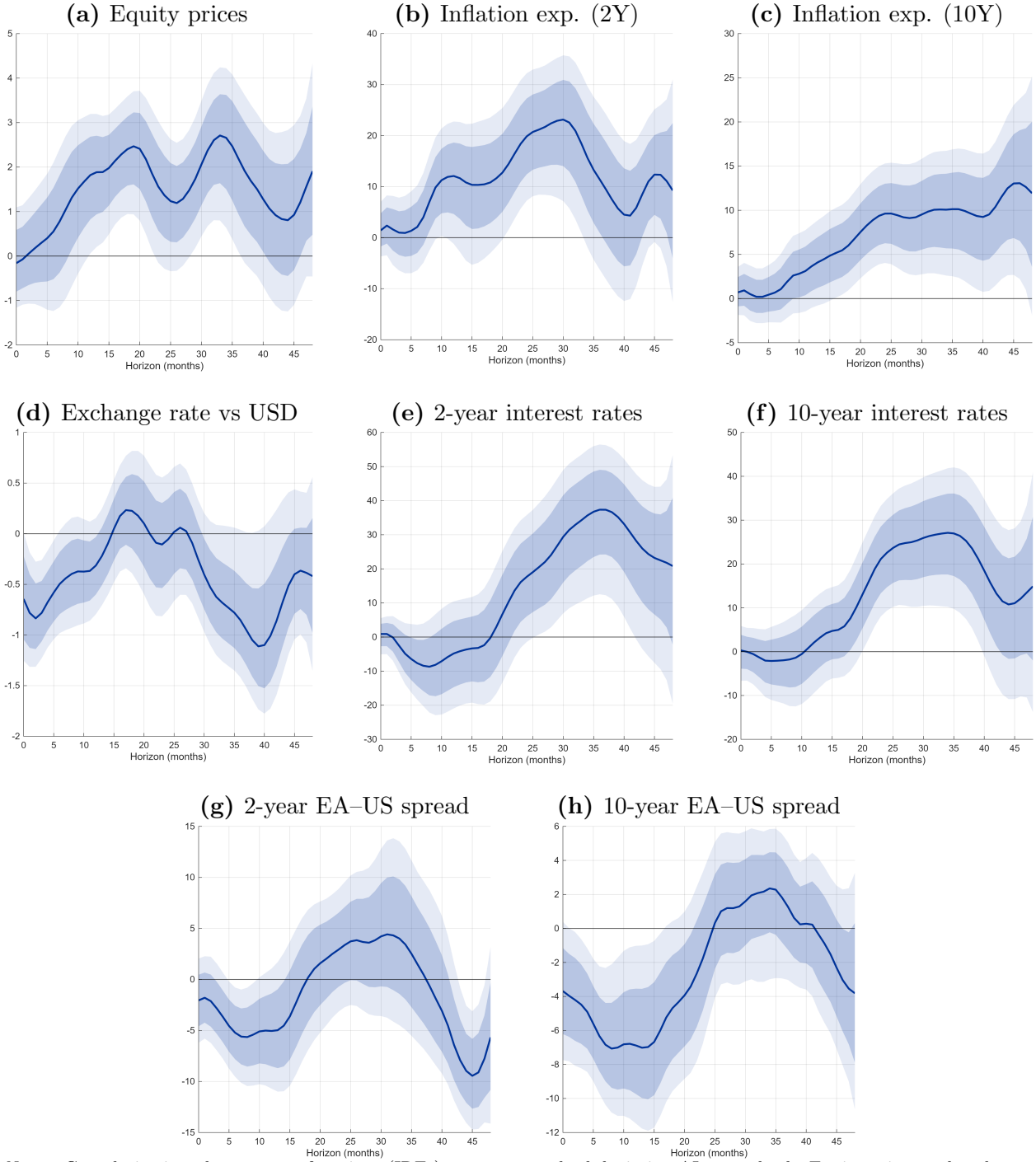
[20]The results excluding the 2020–2022 period are reported in Section D.2.3.

Figure 8: Impact of AI news shocks on euro area *real and price* aggregates



Notes: Cumulative impulse response functions (IRFs) to a one-standard-deviation AI news shock. Variables are expressed as percent deviation from the baseline $(t-1)$. Darker (lighter) shaded areas represent 68% (90%) highest-density posterior sets.

Figure 9: Impact of AI news shocks on euro area *financial* aggregates



Notes: Cumulative impulse response functions (IRFs) to a one-standard-deviation AI news shock. Equity prices and exchange rate are expressed as percent deviation from the baseline $(t-1)$, rates, spreads and inflation expectations in basis points. Darker (lighter) shaded areas represent 68% (90%) highest-density posterior sets.

5 Conclusions

This paper develops a financial-market-based approach to measuring and identifying AI news shocks. By linking daily relative equity valuations to monthly macroeconomic outcomes, it characterises the adjustment following changes in the expected economic value of AI, rather than changes in realised AI adoption or measured productivity. The results reveal a gradual and uneven adjustment rather than an immediate productivity-led expansion. News about AI first stimulates investment and demand for the resources needed to adopt the technology; production, employment, and consumer prices respond only as this process unfolds.

In the US, core capital-goods orders provide the clearest and most persistent real-side response. Producer prices also rise relatively early, whereas GDP and industrial production expand more modestly and with greater uncertainty. Capacity utilisation temporarily declines before recovering, consistent with firms building capacity and reorganising production while existing resources are being reallocated. The labour-market response emerges later: employment and, less precisely, hourly earnings increase, but average hours per worker fall. Price pressures follow a similarly ordered path, passing from producer prices to headline and then core consumer prices. The LPs therefore describe an investment-led transition in which adoption costs and resource demand precede the broader productive effects of AI.

The production-network results show why this aggregate adjustment contains both supply and demand forces. Industries with greater upstream or downstream exposure to AI initially record stronger relative production and smaller producer-price increases, revealing a favourable supply-side effect close to the AI ecosystem. At the same time, the pronounced employment response among highly exposed upstream suppliers points to demand for complementary inputs and implementation capacity. As the shock diffuses, the initial production advantage of highly exposed industries fades and the effects spread towards less-exposed sectors. AI news thus combines localised technological gains with an investment and resource-demand impulse that propagates through the wider economy.

International transmission follows the same logic but operates at two speeds. In the euro area, equity prices, the exchange rate, import costs, energy prices, inflation expectations, and nominal rates respond before industrial production and hours worked. Headline inflation

rises ahead of core inflation, indicating that external price pressures gradually pass through to underlying prices. The EA–US interest rate differential initially moves in favour of the US and later reverses as domestic inflation broadens and financial conditions tighten. The rapid financial and nominal adjustment is consistent with transmission through the global financial cycle, while the delayed real response reflects the time required for an adopter economy to integrate a technology developed and commercialised primarily abroad.

Taken together, the findings reconcile rapid market repricing with the slower macroeconomic effects of AI. Expected future gains can generate substantial capital spending, input demand, employment adjustment, and inflation before their full effect on productive capacity is visible. They can also create cross-border price and financial pressures before productive benefits diffuse internationally. For monetary policy, the central issue is therefore not only the eventual supply potential of AI, but also the timing and composition of the demand, cost, and financial effects generated along the transition.

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A Data

A.1 Construction of the AI equity factor and robustness

This section describes the construction of the AI equity factor used in the main analysis and presents a battery of robustness checks. We build a primary long–short factor from a hand-selected universe of AI-exposed equities and then compare it with i) an equal-weighted variant; ii) orthogonalisation against semiconductor returns; iii) a fully independent, market-priced factor derived from AI-themed exchange-traded funds (ETFs). Throughout, the sample runs at the daily frequency and all returns are continuously compounded log returns of adjusted closing prices obtained from Yahoo Finance.

A.1.1 Benchmark factor: market-cap-weighted long–short portfolio

The primary factor, $\mathrm{AI}_t^{\mathrm{MC}}$, is the return of a long–short portfolio that is long firms whose business models are exposed to AI and short firms in adjacent technology segments with more limited AI exposure. The long leg spans three tiers: AI compute and semiconductors (`NVDA`, `AMD`, `AVGO`, `SMCI`, `TSM`, `MU`, `ASML`, `QCOM`), large-cap AI platforms (`MSFT`, `GOOGL`, `AMZN`, `META`, `ORCL`), and AI-oriented enterprise software (`PLTR`, `AI`, `CRM`, `NOW`, `ADBE`, `ANET`). The short leg comprises non-AI semiconductor peers (`INTC`, `TXN`, `STX`, `WDC`), legacy enterprise hardware (`IBM`, `HPE`, `CSCO`), and traditional IT services (`ACN`, `GLW`). The short leg therefore serves as a technology-sector hedge, reducing broad technology beta and isolating relative AI-related valuation news.

The daily factor is available from 4 January 2012 to 5 October 2026, for a total of 3,709 return observations. The constituent universe is allowed to evolve with listings: firms that are not yet publicly traded receive zero portfolio weight, and weights within each leg are normalised across securities available at each rebalancing date. This is particularly relevant for later entrants such as `ANET`, `HPE`, `PLTR`, and `AI`.

Within each leg, portfolio weights are market-capitalisation weighted and rebalanced on the first trading day of each calendar quarter. Let $\mathcal{L}_t$ and $\mathcal{S}_t$ denote the sets of available long- and short-leg securities at date t, and let $w_{i,t}^{\mathcal{L}}$ and $w_{j,t}^{\mathcal{S}}$ be the corresponding weights. Market capitalisations are constructed from daily prices and time-varying shares outstanding; annual

shares outstanding are linearly interpolated between reporting dates. The factor return is

$$\mathrm{AI}_t^{\mathrm{MC}} \;=\; \underbrace{\sum_{i\in\mathcal{L}_t} w_{i,t}^{\mathcal{L}}\, r_{i,t}}_{R_t^{\mathcal{L}}} \;-\; \underbrace{\sum_{j\in\mathcal{S}_t} w_{j,t}^{\mathcal{S}}\, r_{j,t}}_{R_t^{\mathcal{S}}}, \tag{A.1}$$

where $r_{i,t}$ is the log return of security i on day t. As a robustness variant we also construct an equal-weighted factor $\mathrm{AI}_t^{\mathrm{EW}}$. In the extended sample, equal weights are computed across the constituents actually trading on each date, so that pre-IPO observations are not treated as zero returns.

Table A.1: Benchmark AI factor constituents: composition and sample-end market data

Ticker	Tier	Weight (%)	Shares (bn)	Mkt cap ($bn)
		Long leg		
NVDA	Semis	20.81	24.147	5,768.7
GOOGL	Big tech	15.44	12.230	4,237.3
MSFT	Big tech	14.21	7.426	3,899.7
AMZN	Big tech	9.99	10.786	2,711.7
TSM	Semis	8.89	5.186	2,519.6
META	Big tech	6.90	2.548	1,890.0
AVGO	Semis	6.12	4.774	1,730.5
MU	Semis	4.62	1.129	1,201.4
AMD	Semis	3.75	1.632	1,031.3
ASML	Semis	2.59	0.384	713.4
PLTR	Software	1.70	2.403	455.1
ORCL	Big tech	1.56	3.032	430.3
ANET	Software	0.96	1.261	260.9
QCOM	Semis	0.73	1.068	193.1
CRM	Software	0.73	0.823	189.1
NOW	Software	0.53	1.034	140.7
ADBE	Software	0.36	0.389	92.9
SMCI	Semis	0.10	0.647	27.9
AI	Software	0.01	0.161	1.8
		Short leg		
INTC	Hedge	28.00	5.286	614.2
CSCO	Hedge	18.93	3.943	444.8
TXN	Hedge	11.28	0.913	268.0
IBM	Hedge	9.38	0.942	208.8
STX	Hedge	9.49	0.227	201.7
WDC	Hedge	7.36	0.361	159.2
GLW	Hedge	6.09	0.860	137.0
ACN	Hedge	5.69	0.612	118.4
HPE	Hedge	3.78	1.327	90.7

Notes: The table reports the constituents of the AI long–short factor. "Weight" is the market-capitalisation weight at the end of the sample, 5 October 2026. "Shares" (billions) and "Mkt cap" ($bn) are measured on the same date. Within each leg, constituents are ordered by weight. The long leg is concentrated in the largest AI-related firms: NVIDIA and Alphabet together account for approximately 36.2 % of the long portfolio at the end of the sample.

Table A.2: Benchmark AI factor summary statistics by specification and sub-period

Specification	Period	N	Ann. mean (%)	Ann. std (%)	Sharpe	Corr(AI,EQ)	Skew.	Kurt.
Market-cap (TV)	Full	3,709	9.22	17.13	0.54	0.143	0.066	3.088
	Pre-Nov 2022	2,745	7.43	14.82	0.50	0.108	0.161	4.040
	Post-Nov 2022	964	14.31	22.45	0.64	0.229	-0.043	1.154
Equal-weight	Full	3,709	5.31	16.76	0.32	0.117	-0.127	2.630
	Pre-Nov 2022	2,745	6.93	14.77	0.47	0.057	-0.066	2.876
	Post-Nov 2022	964	0.70	21.43	0.03	0.262	-0.151	1.330
Long-only (TV)	Full	3,709	20.93	24.11	0.87	0.859	-0.284	6.398
	Pre-Nov 2022	2,745	15.85	23.34	0.68	0.865	-0.496	7.415
	Post-Nov 2022	964	35.43	26.19	1.35	0.858	0.124	4.256

Notes: The table reports summary statistics for three specifications of the AI factor: market-cap time-varying (TV) long–short, equal-weight long–short, and long-only (market-cap TV). The full sample runs from 4 January 2012 to 5 October 2026. Sub-periods split the sample at the launch of ChatGPT on 30 November 2022. Annualised means and standard deviations use 252 trading days. "Corr(AI,EQ)" is the contemporaneous correlation with the S&P 500. Kurtosis is reported in excess of the normal. Over the extended sample, the benchmark market-cap long–short factor has an annualised mean of 9.22 %, volatility of 17.13 %, and a correlation of only 0.143 with the S&P 500. The corresponding long-only portfolio has a correlation of 0.859 with the market, highlighting the role of the short leg in removing broad equity exposure.

Table A.3: Evolution of long-leg market-cap weights (selected constituents)

Date	NVDA	MSFT	GOOGL	AMZN	AMD
2012	0.3	26.2	32.3	0.6	0.6
2013	0.2	21.0	29.6	0.7	0.2
2014	0.2	20.6	32.8	0.7	0.2
2015	0.2	22.7	25.5	0.5	0.1
2016	0.3	23.1	31.4	0.9	0.1
2017	0.7	20.9	26.2	8.7	0.5
2018	1.2	23.1	14.0	11.1	0.4
2019	0.8	27.1	13.4	14.5	0.7
2020	0.9	29.5	11.8	12.3	1.5
2021	1.5	27.4	10.1	14.3	2.0
2022	2.0	25.1	19.2	18.0	2.5
2023	5.8	28.9	18.5	15.0	1.7
2024	11.1	25.5	15.9	14.9	2.1
2025	20.3	18.6	13.9	14.1	1.2

Notes: The table reports the market-capitalisation weight (in %) of selected long-leg constituents on the first trading day of January of each year. Weights are rebalanced on the first trading day of January, April, July, and October. The extended series makes explicit that the composition of the long leg changes substantially over time, rather than imposing the end-of-sample AI portfolio retrospectively with fixed weights.

A.1.2 Filing-based validation of constituent classification

A separate concern is that the constituent classification of the benchmark factor necessarily involves an element of researcher judgement and may therefore be influenced, at least partly, by knowledge of how firms' exposure to the AI ecosystem subsequently evolved. We address this concern by complementing the market-based robustness exercises in Sections A.1.4 to A.1.6 with an independent measure of firm-level AI exposure constructed from corporate filings. The purpose of this exercise is not to claim that the benchmark portfolio could have been mechanically selected from textual information in real time, but rather to assess whether its economically motivated long–short classification is corroborated by an independent

measure of firms' realised AI exposure. Importantly, the filing-based measure is constructed without using stock returns or the narrative-event chronology employed elsewhere in the paper.

For each firm and quarterly vintage, we use the most recent annual SEC filing available strictly before the vintage date. The source documents are 10-K, 20-F and 40-F filings. We construct a weighted dictionary-based measure of AI exposure designed to capture both direct references to artificial intelligence and exposure through the technologies that enable its development and deployment. The dictionary is organised into five groups. First, *core AI and machine-learning terms*, including expressions such as "artificial" intelligence", "machine learning, "deep learning", "neural networks", "natural language processing", "computer vision", and "reinforcement learning", receive unit weight. Second, references to *generative and frontier AI*, including "generative AI", "large language models, "foundation models", "transformer models", "multimodal models", "AI agents", "retrieval-augmented" generation" and related concepts, receive a weight of 1.25, reflecting their greater specificity to the recent wave of AI technologies. Third, terms capturing *AI adoption and commercialisation* – for example, "AI-powered", "AI-enabled", "AI platform", "AI services", AI products", "AI workflows", "enterprise AI", and "AI copilots" – receive unit weight. References to *model development and deployment*, including model training, inference, fine-tuning, model serving and deployment, also receive unit weight when they appear in an explicitly AI-related form.

A separate part of the dictionary captures *AI-specific computing infrastructure*. Expressions such as AI "accelerator", "AI chip", "AI processor", "AI server", "AI cluster", "AI compute", "AI networking", "GPU computing", "accelerated computing", "inference accelerator", and "neural processing unit" receive a weight of 0.75. Finally, we include a broader set of *enabling-hardware terms*, such as "GPUs", "high-bandwidth memory", "advanced packaging", "CoWoS", "EUV lithography", "foundry technology", "high-performance computing", "InfiniBand, "optical interconnects", "silicon photonics", "liquid cooling" and "rack-scale computing". Because these expressions are not intrinsically AI-specific, they receive a lower weight of 0.50 and are counted only when they occur within a narrow textual window around an explicit AI-related term. This contextual restriction is intended to capture firms whose exposure operates through the AI supply chain while limiting false positives from

generic semiconductor, networking or data-centre discussion.

For each filing, the weighted number of dictionary matches is normalised by document length and expressed per 10,000 words. We deliberately exclude highly generic standalone expressions—including bare AI", algorithm", automation", cloud", data centre" and semiconductor"—unless they form part of a more specific AI-related expression or satisfy the contextual criterion above. The resulting score therefore places the greatest weight on direct and highly specific AI disclosure, while still recognising economically important exposure through computing and semiconductor infrastructure.

Panel a in Figure A.1 provides a descriptive decomposition of the filing-based measure by comparing the frequency of selected AI-related terms across the benchmark long and short legs. For each term category, positive values identify expressions that are relatively more prevalent in filings of long-leg firms and negative values those that are relatively more prevalent in filings of short-leg firms. The largest positive differentials are associated with terms such as "AI applications", "machine learning", "AI platforms", "generative AI and "AI-related GPU" references, providing a transparent indication of the textual features underlying the higher aggregate exposure scores of the long leg.

Panel b compares the cross-sectional medians of this filing-based score for the firms assigned to the benchmark long and short legs. The scores of the two groups are relatively close during the earlier part of the sample but diverge substantially as AI adoption broadens.

Table A.4 formalises the visual comparison. Over the period 2015-2021, the firm-level median score is 0.82 for the long leg and 0.43 for the short leg, implying a difference of 0.39; a Wilcoxon rank-sum test fails to reject the null of equality between the two cross-sectional distributions. Over the period 2022-2026, the corresponding medians rise to 5.72 and 1.86, respectively, and the long-short gap widens to 3.85. The post-2022 cross-sectional difference is statistically significant at approximately the 5% level.

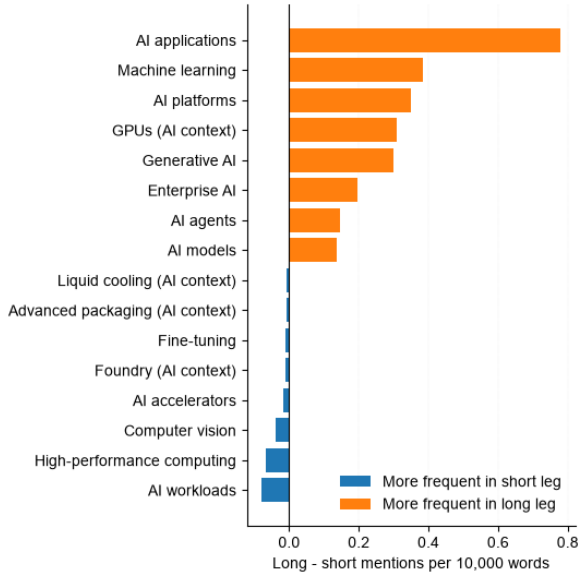
As an additional test, we exploit the full quarterly panel and estimate

$$S_{i,t} = \alpha_t + \beta \, \text{Long}_i + \delta \left(\text{Long}_i \times \text{Post2022}_t \right) + \varepsilon_{i,t}, \tag{A.2}$$

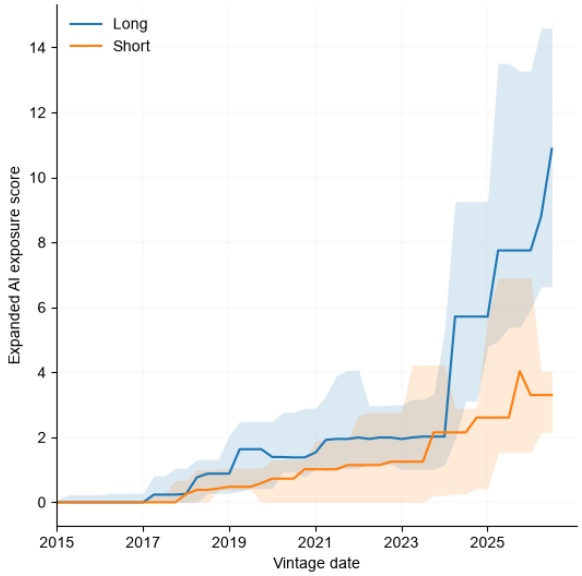
where $S_{i,t}$ is the filing-based AI-exposure score, α_t denotes quarter fixed effects, Long_i identifies

Figure A.1

(a) Differential AI-related terminology



(b) AI exposure scores



Notes: Panel a reports differences in the frequency of selected AI-related terms between the benchmark long and short legs, measured as mentions per 10,000 filing words. Positive values indicate terms that occur more frequently in filings of long-leg firms, while negative values indicate terms that occur more frequently in filings of short-leg firms. Each annual filing is counted once, and closely related singular, plural and synonymous expressions are grouped into common categories. Panel b reports the cross-sectional median of the filing-based AI-exposure score for each portfolio leg; shaded regions denote the corresponding 25th–75th percentile range across firms. Scores are based on the most recent annual filing available strictly before each quarterly vintage. The long and short classifications are those used to construct the benchmark factor. Sample coverage: 2015Q1–2026Q2.

Table A.4: Filing-based scores differences

Period	Long leg	Short leg	Difference	p-value
2015–2021	0.82	0.43	0.39	0.395
2022–2026	5.72	1.86	3.85	0.049

Notes. The table compares firm-level median filing-based AI-exposure scores across the benchmark long and short portfolios. For each firm and period, quarterly scores are first summarised by their within-period median. The final column reports the two-sided Wilcoxon rank-sum test of equality between the long- and short-leg firm-level distributions.

constituents of the benchmark long portfolio, and Post2022_t equals one from 2022Q1 onwards.

Table A.5: Filing-based AI exposure: long–short differences

	Coefficient	Standard error	p-value
Long: pre-2022 difference	0.91	1.05	0.390
Long × Post-2022	4.68	2.40	0.051
Implied long-short difference, 2022-2026	5.58	2.80	0.046
Quarter fixed effects	Yes		
Firm-clustered standard errors	Yes		
Observations	1,260		
Firms	28		
Quarters	47		

Notes: Standard errors are clustered at the firm level using the CR1 finite-sample correction.

Results, reported in Table A.5, show that firms assigned to the benchmark high-AI-exposure portfolio subsequently exhibit a substantially stronger increase in independently measured textual AI exposure than firms assigned to the comparison portfolio.

This exercise should not be interpreted as an alternative portfolio-selection rule. Textual disclosure intensity does not map one-for-one into the market sensitivity of a firm's equity value to AI news, particularly for firms whose exposure operates through semiconductor fabrication, lithography, memory, servers or other enabling infrastructure. Rather, the filing-based evidence provides an external validation of the economic constituent classification: the firms assigned to the high-AI-exposure leg display a markedly stronger subsequent increase in AI-related corporate disclosure than the firms used as the technology-sector hedge.

A.1.3 High-frequency validation around the GPT-4o announcement

We complement the daily event validation with a high-frequency event study around the release of OpenAI's GPT-4o on 13 May 2024. The OpenAI Spring Update started at 13:00 ET, while US equity markets were open, providing a setting in which the relative valuation response of AI-exposed firms can be examined over a narrow intraday window. The exercise is intended as a validation of the information content of the daily AI factor rather than as an alternative high-frequency identification strategy.

Intraday data. One-minute bid and ask quotes are obtained from the Dukascopy historical datafeed for 13 May 2024. For each stock i, we construct the one-minute midquote as

$$P_{i,\tau}^{\text{mid}} = \frac{P_{i,\tau}^{\text{bid}} + P_{i,\tau}^{\text{ask}}}{2}, \tag{A.3}$$

and compute intraday log returns as

$$r_{i,\tau} = \Delta \log P_{i,\tau}^{\text{mid}}. \tag{A.4}$$

The analysis uses the regular US trading session and focuses on the interval from 12:00 to 14:00 ET, corresponding to a $[-60, +60]$ minute window around the beginning of the OpenAI event. When no new quote is observed within a given minute, the most recent midquote is carried forward, implying a zero return until the next price update. Dukascopy

prices correspond to stock-CFD quotes rather than consolidated US transaction prices; the exercise should therefore be interpreted as a high-frequency robustness check rather than as a microstructure analysis based on the consolidated tape.

Portfolio construction and coverage. The intraday portfolios reproduce as closely as possible the construction of the daily AI factor. We use the same long and short stock universes and the same market-capitalisation weights that are in force in the second quarter of 2024. The daily benchmark contains 19 stocks in the AI-exposed long portfolio and 9 stocks in the non-AI short portfolio. Dukascopy provides usable one-minute data for 16 of the 19 long-leg stocks and 8 of the 9 short-leg stocks. The unavailable long-leg constituents are SMCI, ASML and C3.ai (ticker `AI`), while STX is unavailable in the short leg.

Despite the missing securities, coverage in market-capitalisation terms is very high. The available stocks account for 97.11% of the original long-leg weight and 98.10% of the original short-leg weight. We therefore retain the original daily-factor weights and renormalise them within each leg over the subset of securities for which intraday data are available. Denoting by $\mathcal{L}_{HF}$ and $\mathcal{S}_{HF}$ the available long and short constituents, respectively, the intraday portfolio returns are

$$r_\tau^L = \sum_{i \in \mathcal{L}_{HF}} \widetilde{w}_i^L r_{i,\tau}, \tag{A.5}$$

$$r_\tau^S = \sum_{j \in \mathcal{S}_{HF}} \widetilde{w}_j^S r_{j,\tau}, \tag{A.6}$$

where $\widetilde{w}_i^L$ and $\widetilde{w}_j^S$ are the corresponding daily-factor weights rescaled to sum to one within each available portfolio. The high-frequency differential is then

$$r_\tau^{AI,HF} = r_\tau^L - r_\tau^S. \tag{A.7}$$

Thus, the intraday measure preserves the economic definition of the benchmark daily factor and differs only because a small fraction of the original portfolio is unavailable in the high-frequency data.

Event-study measure. For presentation, one-minute returns are cumulated and normalised to zero in the minute immediately preceding the event:

$$CAR_{\tau}^{k} = 100 \sum_{s=-59}^{\tau} r_{s}^{k} - 100 \sum_{s=-59}^{-1} r_{s}^{k}, \qquad k \in \{L, S, AI, HF\}. \tag{A.8}$$

The resulting series therefore measures the cumulative percentage log return relative to 12:59 ET. We additionally report the cumulative return on the S&P 500 over the same window as a broad-market benchmark. The S&P 500 does not enter either portfolio and is used only to distinguish relative AI repricing from aggregate equity-market movements.

Results. As shown in Panel b of Figure 1, the high-frequency exercise reveals a sizeable relative repricing of AI-exposed equities following the announcement, with a non-monotonic intraday response. The initial decline coincides with a fall in the broader market: the S&P 500 and both portfolios decline, but the AI portfolio falls more strongly than the non-AI portfolio. NVIDIA and Microsoft account for much of the negative contribution from the long leg to the differential. This relative underperformance subsequently reverses, initially through an Alphabet-led recovery during the voice-demo window and subsequently through more widespread gains among AI-exposed firms during the vision and mathematics demonstrations. Positive contributions moderate during the final phase. The differential reaches approximately 20 basis points around 30 minutes after the beginning of the event. The S&P 500 remains below its pre-announcement level over the same window. An equal-weighted version of the intraday differential produces a qualitatively similar initial response but a smaller subsequent increase, indicating that the subsequent positive repricing is concentrated among the larger firms in the AI portfolio. Since the benchmark daily factor is market-capitalisation weighted, the corresponding intraday specification is the closest high-frequency analogue of the measure employed in the main analysis.

Stock-level contributions. Figure A.2 decomposes the differential into the contributions of individual constituents over four non-overlapping windows corresponding approximately to successive presentation phases: announcement, access and API economics ($[0, 9)$ minutes);

voice demonstrations ([9, 14)); vision and mathematics demonstrations ([14, 18)); and coding, translation and closing ([18, 26)). Time is measured from 13:00 ET, and the windows last 9, 5, 4 and 8 minutes, respectively.[21] The timing markers in Panel b show the approximate starts of the demonstrations; translation at minute 22 falls within the final contribution window. Each contribution is the stock's cumulative log return over the indicated window multiplied by its intraday portfolio weight, with the sign reversed for short-leg stocks.[22] Blue bars identify AI-exposed long-leg stocks and yellow bars identify non-AI short-leg stocks. A positive yellow bar therefore indicates that a decline in a short-leg stock supports the differential, rather than that the stock itself appreciates.

Panel a shows negative contributions from most AI-exposed constituents during the initial announcement phase. NVIDIA and Microsoft together contribute approximately −14.5 basis points, with Meta, Broadcom and Amazon adding roughly another −6 basis points. Alphabet is the main positive contributor within the AI portfolio. Falling non-AI stocks partly offset these negative long-leg contributions, consistent with the broader market decline visible in Panel b. During the voice-demo window, Panel b shows an initially concentrated reversal: Alphabet contributes around 7 basis points, while NVIDIA remains negative and Microsoft's contribution is close to zero. Negative short-leg contributions indicate that non-AI stocks also recover, dampening the increase in the differential.

During the vision and mathematics window, Panel c shows more widespread positive contributions among AI-exposed firms. Alphabet, Microsoft, Meta, Amazon and NVIDIA contribute approximately 5.2, 2.7, 2.3, 1.7 and 1.3 basis points, respectively. A simultaneous recovery in non-AI stocks again offsets part of these gains in the differential. Finally, Panel d shows smaller and more concentrated gains during coding, translation and closing: Alphabet and NVIDIA remain the leading positive contributors, while Broadcom contributes negatively. The decomposition thus documents an initial decline, an Alphabet-led reversal, a broader

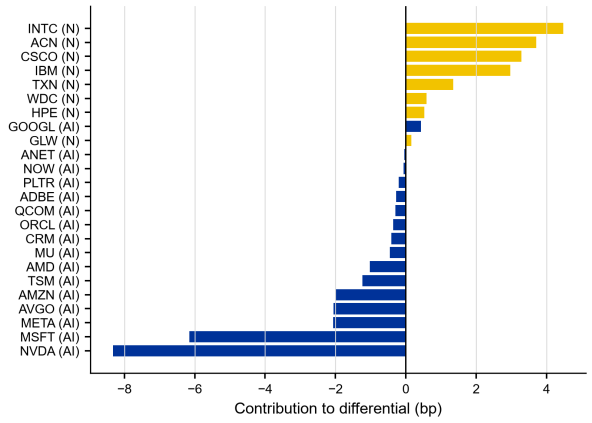
[21] The phase boundaries are approximate, at one-minute precision. These windows describe returns during successive presentation phases rather than separately identifying the causal effects of individual demonstrations. Contributions are measured over each window, not per minute; window lengths therefore matter when comparing their magnitudes.

[22] The weights are the renormalised second-quarter 2024 weights described above and are held fixed across the windows. Contributions are expressed in basis points and sum to the long–short differential over the corresponding window. They reflect both portfolio weights and stock returns; a larger contribution does not necessarily imply a larger individual-stock return.

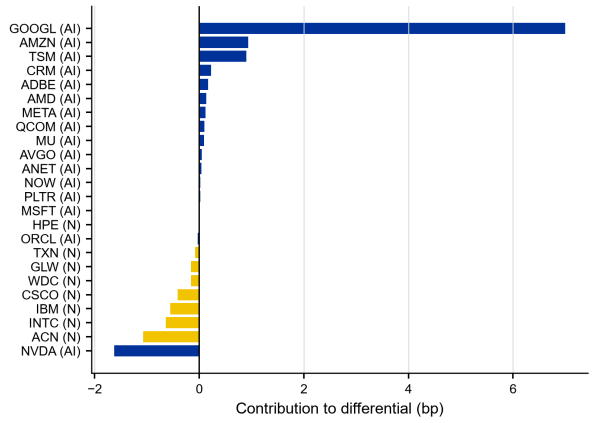
AI-portfolio recovery and then more modest, uneven gains.

Figure A.2: Stock-level contributions to the AI differential by presentation phase

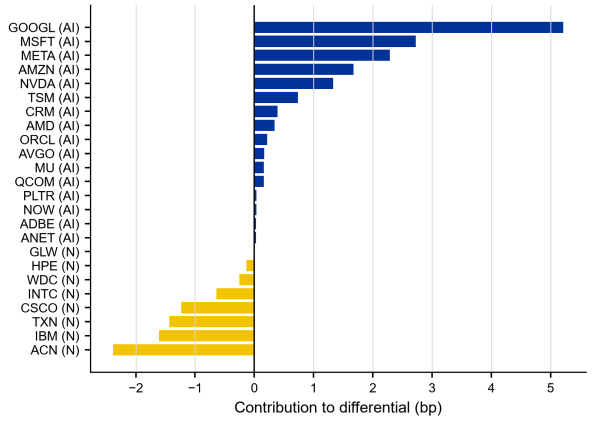
(a) Announcement, access and API: $[0, 9)$



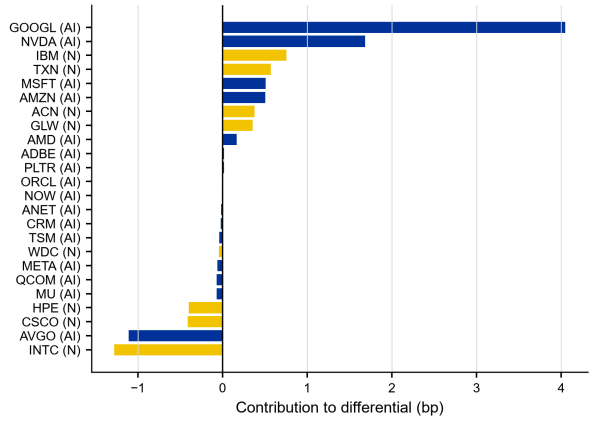
(b) Voice demonstrations: $[9, 14)$



(c) Vision and mathematics: $[14, 18)$



(d) Coding, translation and closing: $[18, 26)$



Sources: Dukascopy, author's computations.
Notes: Stock-level contributions to the AI-portfolio minus non-AI-portfolio cumulative log return, in basis points. Time is measured in minutes from 13:00 ET on 13 May 2024. Blue (AI) and yellow (N) bars denote long- and short-leg constituents, respectively; short-leg contributions enter with a negative portfolio sign. Each panel reports contributions over its own non-overlapping phase window, not cumulative contributions from the event start. The windows last 9, 5, 4 and 8 minutes, respectively, and their boundaries approximate the presentation's chronology. Positive short-leg contributions indicate falling non-AI stock prices; negative short-leg contributions indicate rising prices. Stocks are ranked separately in each panel; horizontal-axis scales differ across panels.

A.1.4 An independent, market-priced factor from AI ETFs

A natural concern is that the primary factor reflects discretionary stock selection rather than a priced AI-related return component. To address this we construct a fully independent measure from a basket of five AI- and robotics-themed ETFs that share no securities-selection methodology with our stock-level factor: Global X Artificial Intelligence & Technology (`AIQ`), Global X Robotics & Artificial Intelligence (`BOTZ`), ROBO Global Robotics & Automation (`ROBO`), First Trust Nasdaq AI & Robotics (`ROBT`), and Defiance Quantum (`QTUM`). The

common sample begins on 6 September 2018, when all five funds are available, and contains 2,030 daily observations through 5 October 2026.

Because the individual ETFs are highly collinear, we extract their common component by principal component analysis (PCA). Let $z_{k,t}$ denote the standardised (zero-mean, unit-variance) daily return of ETF k, and let $\mathbf{z}_t = (z_{1,t}, \ldots, z_{5,t})'$. The first principal component,

$$\mathrm{PC1}_t \;=\; \mathbf{v}_1' \mathbf{z}_t, \tag{A.9}$$

where $\mathbf{v}_1$ is the leading eigenvector of the return correlation matrix, explains 92.01 % of the basket's variance and loads almost identically on all five funds (loadings between 0.441 and 0.454). We rescale $\mathrm{PC1}_t$ into daily return units by projecting it onto the equal-weighted basket return, and denote the result $\widetilde{\mathrm{PC1}}_t$.

The preferred ETF-based AI factor is defined as the excess of this common AI-ETF component over the broad market,

$$\mathrm{AI}_t^{\mathrm{ETF}} \;=\; \widetilde{\mathrm{PC1}}_t - r_t^{\mathrm{SPY}}, \tag{A.10}$$

where r_t^{SPY} is the return on the S&P 500 ETF. The choice of market benchmark is consequential. A regression of $\widetilde{\mathrm{PC1}}_t$ on the Nasdaq-100 ETF (Invesco QQQ ETF – `QQQ`) return yields a slope of 0.972 and an R^2 of 0.843, indicating that most daily variation in the diversified AI-ETF basket is shared with the broader technology sector. Consequently, subtracting `QQQ` or residualising the ETF factor with respect to `QQQ` removes much of the component that responds on AI event days. We therefore use the broader S&P 500 as the benchmark and treat the Nasdaq-based variants as diagnostics.

A.1.5 Narrative validation of ETF factors

We validate the sign of each factor using a hand-collected chronology of AI-related events, classified as technological innovations, adoption and commercialisation events, or policy actions, and assigned an ex ante expected sign (+, −, or ambiguous). For each dated event we take the factor return on the nearest trading day and record whether its sign matches the prior. The updated chronology contains 33 events, seven of which are classified as ambiguous

Table A.6: Event-day hit rates by factor specification

Factor	Hits	Events	Hit rate
Market-cap long–short (primary)	25	26	96.2 %
Equal-weighted long–short	23	26	88.5 %
ETF: PC1 − SPY (independent)	18	26	69.2 %
MC factor orthogonal to SMH	24	26	92.3 %
MC factor orthogonal to SOXX	25	26	96.2 %
ETF: PC1 − QQQ (diagnostic)	13	26	50.0 %
ETF: residual of PC1 on QQQ	14	26	53.8 %
ETF: residual of PC1 on QQQ and SPY	13	26	50.0 %

Notes. Hit rate is the fraction of the 26 signed AI events on which the factor return has the ex ante expected sign; seven ambiguous events are excluded. The semiconductor-orthogonalised rows remove, respectively, the daily return on SMH or SOXX from the benchmark market-cap factor. The common AI-ETF component has a slope of 0.972 and $R^2 = 0.843$ on QQQ, so Nasdaq-based hedging removes a large share of the common ETF variation.

and excluded from the scoring, leaving 26 signed events. The event chronology is concentrated in the generative-AI era, while the daily market factor itself is available from 2012.

Table A.6 reports the resulting event-day hit rates. The primary market-cap factor recovers the expected sign on 25 of 26 signed events (96.2 %); the only miss is the GPT-4o announcement. The equal-weighted variant recovers 23 of 26 signs (88.5 %). The independent ETF factor in Equation (A.10) recovers the expected sign on 18 of 26 events (69.2 %). Under a fair-coin null of no directional content, 18/26 correct corresponds to a one-sided exact binomial p-value of approximately 0.038. By contrast, the Nasdaq-benchmarked ETF variants recover 13–14 of the 26 signs. The semiconductor-orthogonalised versions of the primary stock-level factor remain much closer to the benchmark, with 24/26 correct signs (92.3 %) after removing the VanEck Semiconductor ETF (SMH) and 25/26 (96.2 %) after removing the iShares Semiconductor ETF (SOXX).

A.1.6 Robustness checks of ETF factors

Composition and weighting. The event-day performance of the equal- and market-cap-weighted factors (88.5 % versus 96.2 %) indicates that the directional event signal is not specific to the market-cap weighting scheme. The extended 2012–2026 factor also uses a dynamically available constituent set, so later-listed firms enter only once their returns become observable.

Semiconductor cycle. Because semiconductors are a key transmission channel for AI, a natural concern is that the factor merely proxies the chip cycle. We regress the primary factor on semiconductor-sector ETFs over the full 2012–2026 sample. The exposure is small: the regression on `SMH` gives $\beta = 0.075$ and $R^2 = 1.69\,\%$, while the regression on `SOXX` gives $\beta = 0.044$ and $R^2 = 0.63\,\%$. Orthogonalising the benchmark factor with respect to each semiconductor ETF and repeating the narrative validation leaves the event-sign recovery at 24/26 (92 %) and 25/26 (96 %), respectively. Thus, the identifying event signal is not accounted for by generic semiconductor-sector variation.

Independent corroboration and its limitations. The independent ETF factor recovers the expected sign on 18 of 26 signed events (69.2 %) despite using a completely different portfolio construction. The ETF and stock-level measures agree on several broad AI repricing episodes, including the ChatGPT launch, the DeepSeek R1 release, the Stargate announcement, the US AI Action Plan, and the restrictions on Anthropic frontier models. At the same time, the ETF measure misses a number of model-release and issuer-specific events, as well as the Italy ChatGPT ban and the Texas data-centre permit freeze. The relatively low full-sample correlation between the preferred ETF and benchmark stock-level factors (0.129) therefore supports viewing the ETF measure as independent corroboration rather than as an interchangeable proxy.

Stability across the ChatGPT structural break. The benchmark market-cap factor remains only weakly correlated with the S&P 500 both before and after the launch of ChatGPT, with correlations of 0.109 and 0.229, respectively. Its annualised Sharpe ratio rises from 0.50 before the launch to 0.64 afterwards. The correlation between the independent ETF factor and the benchmark factor is 0.282 before 30 November 2022 but falls to 0.003 afterwards. This weaker post-ChatGPT comovement is consistent with the fact that diversified thematic ETFs retain substantial broad-technology exposure even as AI exposure diffuses across their holdings, and motivates treating the ETF measure as a robustness check rather than the baseline factor.

Taken together, these checks address four principal concerns about the factor: i) sensitivity

to the portfolio weighting scheme, ii) changes in the investable constituent set over the extended sample, iii) latent semiconductor-cycle exposure, and iv) discretionary stock selection, supporting its use as a market-based measure of AI-related news distinct from broad equity-market and semiconductor sector movements.

A.2 Macroeconomic data

A.2.1 US aggregates

Aggregate data for the US are taken from the Federal Reserve Economic Data (FRED) database. All series are available at monthly frequency and are seasonally adjusted at source. The real GDP series is interpolated at the monthly frequency using the Chow-Lin procedure with monthly industrial production as control. The sample used in the local projection exercise starts in January 2015 and ends in September 2026.

A.2.2 Euro area aggregates

Data for the euro area are taken from the following sources:

- **Employment, hours worked, industrial production, industrial producer price index (PPI), import prices**: Eurostat.
- **HICP core and headline, EUR/USD exchange rate**: ECB Statistical Data Warehouse.
- **Energy price index**: OECD.
- **Euro area equity prices (EUROSTOXX 50) and ILS 2-year/10-year (midpoint of bid/ask prices)**: Refinitiv.
- **Natural gas price (NL), two- and ten-year euro area OIS rates and differentials with the US**: Bloomberg.

All series are seasonally adjusted at source, except PPI and import prices, which we adjust using an STL decomposition. Quarterly employment and hours-worked series are converted to monthly frequency by linear interpolation. The sample starts in January 2015 and ends in August 2026.

A.2.3 US sectoral aggregates

This section describes the sectoral aggregates used to study the heterogeneous transmission of AI news shocks across US industries. We compare sectors with strong and weak production-

network connections to AI-related industries by constructing separate high- and low-exposure aggregates for producer prices, industrial production, and employment, distinguishing between upstream and downstream exposure.

Sectoral data. Sectoral price, production, and employment series are collected at monthly frequency from US statistical sources. Producer price indices are obtained from the Bureau of Labor Statistics (BLS). Industrial production indices are drawn from the Federal Reserve's G.17 *Industrial Production and Capacity Utilization* release, mostly through FRED. Employment series are obtained from the BLS Current Employment Statistics. The sample runs from January 2015 to June 2026.

Detailed BLS PPI series are not always available on a seasonally adjusted basis. We therefore seasonally adjust the relevant PPI series using an STL decomposition. For the construction of the sectoral aggregates, all series are expressed in log levels and rebased around January 2015. Since the rebasing amounts to subtracting a series-specific constant in logs, it does not affect the estimated local projection responses.

Input-output exposure measures. We construct production-network exposure measures using the Bureau of Economic Analysis (BEA) Input-Output Accounts. Specifically, we use BEA Table 259, "The Use of Commodities by Industries – Summary," which reports the value of commodities supplied by each row sector and used by each column industry. We use the 2023 table.

Let Z_{ij} denote the value of inputs supplied by sector i and used by sector j. We define a set of AI-related BEA industries, $\mathcal{A}^{BEA}$, consisting of computer and electronic products, computer systems design and related services, publishing industries except internet (including software), data processing and internet-related services, and broadcasting and telecommunications.

We consider two distinct measures of production-network exposure. First, upstream exposure captures the extent to which a sector supplies AI-related industries:

$$\text{Exp}_i^{up} = \frac{\sum_{j \in \mathcal{A}^{BEA}} Z_{ij}}{\sum_j Z_{ij}}. \tag{A.11}$$

Hence, Exp_i^{up} measures the share of sector i's sales that is directed towards AI-related

industries. A high value therefore identifies sectors that are important suppliers to the AI production ecosystem and may be particularly exposed to AI-induced demand pressures, capacity constraints, and input bottlenecks.

Second, downstream exposure captures the extent to which a sector relies on inputs supplied by AI-related industries:

$$\mathrm{Exp}_j^{down} = \frac{\sum_{i \in \mathcal{A}^{BEA}} Z_{ij}}{\sum_i Z_{ij}}. \tag{A.12}$$

A high value of Exp_j^{down} therefore identifies industries that intensively use AI-related inputs and is more closely related to the diffusion and adoption of AI-related technologies through the production network.

For ease of comparison, both measures are normalised by their respective maximum across BEA sectors,

$$\widetilde{\mathrm{Exp}}_i^{up} = \frac{\mathrm{Exp}_i^{up}}{\max_k \mathrm{Exp}_k^{up}}, \qquad \widetilde{\mathrm{Exp}}_i^{down} = \frac{\mathrm{Exp}_i^{down}}{\max_k \mathrm{Exp}_k^{down}}, \tag{A.13}$$

so that each exposure measure ranges between zero and one. This normalisation has no bearing on the sectoral ranking.

Mapping statistical series into BEA industries. The BEA input-output classification is coarser than the classifications underlying the BLS PPI, BLS CES, and Federal Reserve industrial-production series. We therefore map each statistical series to the closest BEA summary industry. For example, semiconductor and computer-related series are mapped to "Computer and electronic products," software publishers to "Publishing industries, except internet (includes software)," data processing to "Data processing, internet publishing, and other information services," computer systems design to "Computer systems design and related services," and electric power to "Utilities."

When several statistical series map into the same BEA summary industry, we first collapse them into a single BEA-sector series by taking their equal-weighted average in rebased log levels. This prevents an industry from receiving a larger weight merely because more detailed statistical series happen to be available for that sector. The production-network classification is therefore performed over unique BEA industries rather than over individual statistical

series.

Construction of high- and low-exposure aggregates. We construct the exposure groups separately for each outcome domain—producer prices, industrial production, and employment—and for upstream and downstream exposure. We first retain BEA industries for which at least 90% of the monthly observations are available over the local projection sample. Within each resulting cross-section, industries are ranked according to either $\widetilde{\text{Exp}}_i^{up}$ or $\widetilde{\text{Exp}}_i^{down}$.

Our baseline specification defines the *high AI-network-exposure* group as the set of industries whose network exposure lies at or above the 75th percentile of the relevant cross-sectional exposure distribution. Conversely, the *low AI-network-exposure* group contains industries whose exposure lies at or below the 25th percentile.[23] Industries belonging to the middle 50% of the distribution are omitted in order to sharpen the contrast between sectors with particularly strong and particularly weak production-network linkages to AI-related industries. The classification is performed separately for upstream and downstream exposure and for each outcome domain. Group membership is determined once from the time-invariant BEA input-output exposure measures and is held fixed throughout the sample.

For each domain d and exposure direction $g \in \{up, down\}$, we construct the high- and low-exposure indices as equally weighted geometric means of the constituent sectoral series. Let $\mathcal{H}_{d,g}$ and $\mathcal{L}_{d,g}$ denote the corresponding sets of BEA industries, and let $y_{i,t}$ denote the rebased log level of sector i. The high-exposure index is

$$Y^H_{d,g,t} = 100 \times \exp\left(\frac{1}{|\mathcal{H}_{d,g}|}\sum_{i\in\mathcal{H}_{d,g}} y_{i,t}\right), \tag{A.14}$$

with the low-exposure index $Y^L_{d,g,t}$ defined analogously. The equal weighting ensures that the results are not mechanically dominated by the largest industries. To maintain a stable composition over time, the sets of constituent industries are fixed; an aggregate is treated as missing in a given month if fewer than 80% of its constituent sectors are observed.

[23] In the robustness exercise reported in Section D.2.4 below, we set these thresholds to 90th and 10th percentiles respectively.

Finally, we also construct the relative high-to-low exposure index

$$Y^{H/L}_{d,g,t} = 100 \times \frac{Y^{H}_{d,g,t}}{Y^{L}_{d,g,t}}. \tag{A.15}$$

Because the local projections are estimated in logs, the response of this index is exactly the difference between the log responses of the high- and low-exposure aggregates,

$$\Delta_h \log Y^{H/L}_{d,g,t} = \Delta_h \log Y^{H}_{d,g,t} - \Delta_h \log Y^{L}_{d,g,t}. \tag{A.16}$$

It therefore provides a direct measure of the differential response of industries with high AI-network exposure relative to industries with low exposure (see Figure 7).

B Event study

Table B.7: Individual event study results

Type	Event date	Trading date	Event	Exp.	AI factor	EQ returns	ΔVIX	Δ2Y	ΔSlope	ΔILS 2Y	ΔILS 10Y	ΔNEER
Innovation	30-Nov-22	30-Nov-22	ChatGPT launch	+	+1.70***	+2.78***	−0.71***	−1.89***	+0.86***	+0.24***	+0.97***	−0.35***
Adoption	23-Jan-23	23-Jan-23	Microsoft 10bn OpenAI investment	+	+0.53***	+1.03***	−0.02	+1.32***	−0.83***	−0.32***	+0.52***	−0.55***
Innovation	14-Mar-23	14-Mar-23	GPT-4 release	+	+1.57***	+1.43***	−1.44***	+3.25***	−2.22***	+1.61***	+3.10***	+0.30***
Policy	30-Mar-23	30-Mar-23	Italy bans ChatGPT	−	−0.12***	+0.49***	−0.06***	+0.37***	−1.10***	−0.32***	+0.78***	−0.84***
Adoption	24-May-23	24-May-23	NVIDIA Q1 FY24 earnings	+	+0.77***	−0.70***	+0.78***	+0.94***	−0.55***	−0.32***	−0.70***	+0.38***
Innovation	18-Jul-23	18-Jul-23	Meta Llama 2	+	+0.31***	+0.67***	−0.16***	+0.10**	−0.23***	−0.01	−0.91***	−0.23***
Adoption	23-Aug-23	23-Aug-23	NVIDIA Q2 FY24 earnings	+	+0.12***	+0.96***	−0.51***	−1.38***	−2.21***	+0.01	−0.13***	−0.86***
Adoption	25-Sep-23	25-Sep-23	Amazon 4bn Anthropic investment	+	+0.51***	+0.34***	−0.16***	−0.21***	+3.35***	−0.86***	−0.25***	+1.48***
Policy	30-Oct-23	30-Oct-23	Biden Executive Order on AI	?	+1.44***	+1.03***	−0.79***	+0.75***	+0.01	−0.40***	−0.13***	−0.55***
Innovation	06-Dec-23	06-Dec-23	Google Gemini	−	−0.45***	−0.41***	+0.06***	+0.56***	−2.49***	−1.04***	−1.07***	−0.68***
Policy	08-Dec-23	08-Dec-23	EU AI Act political agreement	?	+0.11***	+0.27***	−0.27***	+2.31***	−1.17***	+0.64***	+1.16***	+0.32***
Adoption	22-Feb-24	22-Feb-24	NVIDIA FY24 Q4 earnings	+	+2.78***	+1.80***	−0.41***	+0.93***	−1.10***	+0.18***	−0.87***	+0.15***
Innovation	18-Mar-24	18-Mar-24	NVIDIA Blackwell architecture	+	+1.26***	+0.48***	−0.03**	+0.15***	+0.56***	+0.04***	−0.01	+0.43***
Innovation	18-Apr-24	18-Apr-24	Meta Llama 3	−	−0.24***	−0.27***	−0.06***	+0.85***	−0.02	−0.05**	+0.19***	−0.10***
Innovation	14-May-24	14-May-24	GPT-4o release	+	+0.10***	+0.36***	−0.09***	−0.81***	+0.28***	−0.69***	−0.30***	−0.28***
Adoption	23-May-24	23-May-24	NVIDIA Q1 FY25 earnings	+	+2.34***	−0.70***	+0.23***	+0.97***	−0.26***	−0.00	−1.38***	+0.51***
Adoption	10-Jun-24	10-Jun-24	Apple Intelligence announced	+	+0.26***	+0.23***	+0.28***	−0.04**	+1.11***	+0.24***	+0.10***	+0.89***
Adoption	29-Aug-24	29-Aug-24	NVIDIA Q2 FY25 earnings	?	−1.88***	−0.03*	−0.77***	+0.76***	−0.26***	+0.25***	+0.25***	+0.85***
Adoption	21-Jan-25	21-Jan-25	Stargate Project announced	+	+0.25***	+0.78***	−0.48***	+0.37***	−1.65***	−1.12***	−0.82***	−2.60***
Innovation	27-Jan-25	27-Jan-25	DeepSeek R1 release	−	−5.19***	−1.34***	+1.59***	−1.97***	+0.00	−0.35***	−0.71***	+1.51***
Innovation	24-Feb-25	24-Feb-25	Claude 3.7 Sonnet	?	−1.11***	−0.46***	+0.40***	−1.18***	+1.12***	+0.06***	+0.40***	+0.11***
Adoption	27-Feb-25	27-Feb-25	NVIDIA FY25 Q4 earnings	?	−2.80***	−1.50***	+1.05***	+0.37***	+0.57***	+0.79***	+0.08***	+1.66***
Innovation	05-Apr-25	04-Apr-25	Meta Llama 4	?	+1.36***	−5.49***	+7.92***	−0.59***	−0.54***	−2.72***	−2.90***	+2.67***
Innovation	22-May-25	22-May-25	Claude 4 Opus and Sonnet	+	+1.14***	−0.00	−0.31***	−0.01	−1.10***	+0.02	+0.09***	+0.74***
Adoption	29-May-25	29-May-25	NVIDIA Q1 FY26 earnings	+	+0.68***	+0.29***	−0.04***	−0.84***	−0.01	−0.59***	−0.03*	−1.13***
Innovation	10-Jun-25	10-Jun-25	OpenAI o3 and o4-mini	−	−0.48***	+0.46***	−0.11***	−0.03	−0.55***	−0.50***	−0.41***	+0.34***

Continued on next page

Type	Event date	Trading date	Event	Exp.	Sign	AI factor	EQ returns	ΔVIX	Δ2Y	ΔSlope	ΔILS 2Y	ΔILS 10Y	ΔNEER
Policy	23-Jul-25	23-Jul-25	US AI Action Plan	+		$+2.46^{***}$	$+0.73^{***}$	-0.60^{***}	$+0.96^{***}$	$+0.02$	-0.69^{***}	-0.73^{***}	-0.30^{***}
Policy	30-Jul-25	30-Jul-25	US restrictions AI chip exports	?		$+1.01^{***}$	-0.18^{***}	-0.23^{***}	$+1.47^{***}$	-1.12^{***}	$+0.17^{***}$	-0.07^{***}	$+1.08^{***}$
Adoption	28-Aug-25	28-Aug-25	NVIDIA Q2 FY26 earnings	+		$+0.09^{***}$	$+0.26^{***}$	-0.20^{***}	$+0.52^{***}$	-1.39^{***}	$+0.00$	-1.13^{***}	-1.24^{***}
Policy	13-Jun-26	12-Jun-26	US export restrictions Anthropic	−		-1.97^{***}	$+0.42^{***}$	-0.89^{***}	$+0.70^{***}$	-0.29^{***}	-0.70^{***}	$+0.71^{***}$	-1.62^{***}
Policy	21-Sep-26	21-Sep-26	Texas freezes new data-center permits	−		-1.61^{***}	$+1.34^{***}$	$+0.04^{**}$	-0.03^{**}	-1.38^{***}	-0.58^{***}	$+0.39^{***}$	-0.31^{***}

Notes: The table reports individual event-dummy coefficients from the event-study regressions, in standardised units. Stars denote statistical significance based on Newey–West HAC standard errors: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$. The column "Exp." reports the expected sign of the AI factor response. For events falling on non-trading days, the trading date is the nearest available trading day used in the regression.

B.1 Robustness checks

The event-study evidence in Section 2.1 shows that the AI factor responds systematically, and in the expected direction, around a hand-collected set of AI-related announcements. A potential concern, however, is that the factor may respond more generally to salient technology or US mega-cap news, rather than specifically to information related to AI. This concern is particularly relevant given that several large technology firms enter the long leg of the portfolio.

We address this issue through a placebo-event exercise. We collect 24 major technology-related events that are economically important but whose primary content is not related to AI. The events, reported in Table B.8, span generic big-tech product announcements, non-AI digital and corporate innovations, broad technology policy measures, semiconductor-cycle news, and major technology disruptions. The chronology deliberately includes announcements affecting firms and sectors represented in the AI factor, while excluding events whose main information content concerns AI capabilities, adoption, investment, or regulation.

To remain directly comparable with the baseline event study, we focus on the *signed average abnormal response* (SAAR). For placebo events for which a directional prior can be formulated ex ante, we define

$$D_t^P = \sum_{e \in \mathcal{E}^P} s_e \mathbf{1}\{t = \tau_e\}, \qquad s_e \in \{-1, +1\}, \tag{B.17}$$

where $\mathcal{E}^P$ denotes the set of signed placebo events, τ_e is the corresponding trading date, and s_e is the expected direction of the technology shock. We then re-estimate the specification in Equation (1), replacing the AI-event dummy with D_t^P. All variables are standardised, and the control variables and Newey–West HAC inference are identical to those used in the baseline event study.

For several generic product and corporate announcements, there is no unambiguous ex ante prediction for the direction of a *relative* AI long–short portfolio. These observations are retained in the placebo chronology but assigned no sign and therefore do not enter the SAAR. Importantly, we do not infer their sign from the realised movement of the factor, which would introduce ex post selection bias. The signed placebo sample consequently contains 12 events:

five semiconductor-cycle events, four broad technology-policy events, and three technology disruptions.

As an additional check, we construct a "market-moving" placebo subsample. An event is classified as market-moving whenever

$$\max\left\{\left|\widetilde{r}_t^{EQ}\right|, \left|\widetilde{\Delta VIX}_t\right|\right\} \geq 0.75,$$

where the two variables are expressed in standard-deviation units. This selection criterion is deliberately based only on broad market returns and the VIX, and therefore does not use the AI factor itself. 10 of the 24 placebo dates satisfy this criterion, of which 7 have a directional prior and therefore enter the market-moving SAAR.

Placebo responses in the baseline financial system. Table B.9 reports the signed placebo response for the same financial variables considered in the baseline event study. None of the SAAR coefficients is statistically different from zero. In particular, the AI factor displays a placebo SAAR of only -0.05 standard deviations ($p = 0.874$). Broad equity returns, the VIX, short-term yields, the slope of the yield curve, inflation compensation, and the dollar exchange rate likewise display no systematic signed response on these non-AI technology dates. The placebo SAARs of the 2- and 10-year ILS rates are $+0.43$ and $+0.12$ standard deviations, respectively, and are statistically insignificant ($p = 0.371$ and $p = 0.635$); in original units these correspond to approximately $+2.61$ and $+0.35$ basis points.

The complementary absolute-response specification leads to the same conclusion. For the AI factor, the placebo absolute-response AAR is $+0.07$ standard deviations ($p = 0.556$), compared with 0.61 on AI-event dates. A Newey–West HAC Wald test rejects equality of the two absolute responses at the 5% level ($p = 0.019$). Thus, the full set of non-AI technology announcements does not generate systematically larger AI-factor movements.

Removing semiconductor-market exposure. A related concern is that the factor may capture semiconductor-market developments rather than AI-specific news. **??** shows that the unconditional exposure of the benchmark factor to semiconductor-sector ETFs is small, with R^2 values of approximately 1.69% against `SMH` and 0.63% against `SOXX`. We complement

Table B.8: Non-AI technology placebo events

Date	Category	Event	Exp.	Market-moving
07-Sep-2016	Big-tech product	Apple iPhone 7 launch	?	No
04-Oct-2016	Big-tech product	Google Pixel phone launch	?	No
16-Jun-2017	Digital/corporate	Amazon Whole Foods acquisition announced	?	No
12-Sep-2017	Big-tech product	Apple iPhone X launch	?	No
14-Dec-2017	Broad tech policy	FCC net-neutrality repeal vote	?	No
15-Nov-2018	Chip cycle	NVIDIA gaming/crypto inventory shock	−	Yes
18-Dec-2018	Chip cycle	Micron memory-cycle/inventory warning	−	No
19-Mar-2019	Digital/corporate	Google Stadia cloud-gaming platform announced	?	No
18-Jun-2019	Digital/corporate	Facebook Libra/Calibra digital wallet announced	?	Yes
13-Oct-2020	Big-tech product	Apple iPhone 12 launch	?	No
20-Oct-2020	Broad tech policy	US DOJ Google antitrust lawsuit	−	No
24-Jun-2021	Digital/corporate	Microsoft Windows 11 announced	?	No
04-Oct-2021	Tech disruption	Facebook/Instagram/WhatsApp global outage	−	Yes
28-Oct-2021	Digital/corporate	Facebook rebrands as Meta / metaverse strategy	?	Yes
24-Mar-2022	Broad tech policy	EU Digital Markets Act political agreement	−	Yes
21-Jun-2022	Tech disruption	Cloudflare global outage	−	Yes
30-Jun-2022	Chip cycle	Micron PC/smartphone demand slowdown	−	Yes
09-Aug-2022	Broad tech policy	US CHIPS and Science Act signed	+	No
07-Sep-2022	Big-tech product	Apple iPhone 14 launch	?	Yes
06-Oct-2022	Chip cycle	AMD PC-demand revenue warning	−	Yes
26-Jan-2023	Chip cycle	Intel PC-cycle earnings shock	−	Yes
05-Jun-2023	Digital/corporate	Apple Vision Pro spatial-computing launch	?	No
06-Sep-2023	Broad tech policy	EU DMA gatekeeper designations	−	No
19-Jul-2024	Tech disruption	CrowdStrike Windows outage	−	No

Notes: The table reports the non-AI technology events used in the placebo exercise. "Exp." denotes the ex ante expected sign of the generic technology shock for events with an unambiguous directional interpretation. Events marked "?" are retained in the placebo chronology but are excluded from the signed dummy and the SAAR calculation. "Market-moving" identifies events for which the maximum absolute standardised movement in broad equity returns or the VIX is at least 0.75 standard deviations. The AI factor is not used in defining this subsample.

Table B.9: Placebo-event SAAR across financial variables

Variable	Placebo SAAR	p-value
AI factor	−0.052	0.874
EQ Returns	−0.138	0.643
ΔVIX	−0.081	0.654
Δ2Y	+0.028	0.937
ΔSpread	−0.237	0.253
ΔILS2Y	+0.432	0.371
ΔILS10Y	+0.116	0.635
ΔNEER	−0.010	0.960

Notes: Coefficients are signed average abnormal responses (SAARs) in standard-deviation units. The signed placebo dummy includes the 12 events with an ex ante directional prior. p-values are based on Newey–West HAC standard errors using the same bandwidth rule as in the baseline event study. The ILS variables are daily changes in 2- and 10-year inflation compensation.

that exercise here by repeating the placebo event study using semiconductor-orthogonalised factors.

$$AI_t = a_k + b_k r_t^k + u_t^k, \qquad k \in \{\texttt{SMH}, \texttt{SOXX}\}, \tag{B.18}$$

and define

$$AI_t^{\perp k} \equiv \widehat{u}_t^k. \tag{B.19}$$

The residual factors therefore remove the linear contemporaneous component of the benchmark AI factor associated with broad semiconductor returns. We standardise each residual factor before re-estimating the event-study regressions. For comparison, we additionally pool the signed AI events used in the baseline analysis into a single signed AI-event dummy and estimate its SAAR using the same specification.

Table B.10: AI versus placebo-event SAAR: semiconductor-orthogonalised factors

	All signed placebos		Market-moving	AI events	
Factor	SAAR	Hit rate	Placebo SAAR	AI SAAR	$p(\text{AI} = \text{P})$
Baseline	−0.052	0.50	+0.180	+1.180	0.002
Orthogonalised to `SMH`	+0.008	0.58	+0.260	+0.977	0.009
Orthogonalised to `SOXX`	−0.008	0.50	+0.232	+1.036	0.006

Notes: SAAR denotes the signed average abnormal response in standard-deviation units. The first two columns refer to the 12 signed placebo events. "Hit rate" is the fraction of signed placebo events for which the factor moves in the ex ante expected direction. "Market-moving" reports the SAAR for the 7 signed placebo events belonging to the market-moving subset. The corresponding p-values are 0.710, 0.564, and 0.608 for the baseline, `SMH`-orthogonalised, and `SOXX`-orthogonalised factors, respectively. "AI SAAR" is obtained from a pooled signed dummy containing the AI events used in the baseline analysis. The final column reports the p-value of a Newey–West HAC Wald test of equality between the AI-event SAAR and the SAAR for the full signed placebo sample. The baseline factor uses 3,709 daily observations, while the semiconductor-orthogonalised specifications use 2,904 observations.

The contrast is sharp. For the benchmark factor, the signed response on placebo dates is −0.05 standard deviations, whereas the pooled response on AI-event dates equals 1.18 standard deviations. A Wald test strongly rejects equality of the two responses ($p = 0.002$). Selecting only placebo dates that generate sizeable movements in broad equities or volatility does not alter this conclusion: the market-moving placebo SAAR is 0.18 and remains statistically insignificant ($p = 0.710$).

The result is also robust to removing broad semiconductor exposure. The placebo SAAR is +0.01 after orthogonalisation with respect to `SMH` and −0.01 after orthogonalisation with respect to `SOXX`. In contrast, the corresponding AI-event SAARs remain sizeable, at 0.98 and 1.04 standard deviations. Equality between the AI and placebo responses is rejected in

both cases, with p-values of 0.009 and 0.006, respectively. Thus, removing the component associated with broad semiconductor returns leaves the directional AI-event signal largely intact.

Placebo categories. Table B.11 decomposes the signed placebo evidence by event category. Generic big-tech product announcements and non-AI digital/corporate innovations are shown for completeness in the underlying regressions but have no signed SAAR, since no directional prior was imposed. None of the three categories with an ex ante sign generates a statistically significant SAAR. The result is particularly informative for semiconductor-cycle events. Although chip-cycle announcements can generate sizeable movements in the long–short portfolio, their *direction* is not systematically captured by the AI factor. The absolute-response AAR for the baseline factor is positive and statistically significant, whereas the signed semiconductor-cycle SAAR is negative and statistically insignificant. The factor moves in the predicted direction on only two of five events, and the corresponding hit rate remains 40% after orthogonalising the factor with respect to either `SMH` or `SOXX`. Semiconductor news can therefore move the portfolio substantially without reproducing the directionally consistent response observed around AI events.

Table B.11: Signed placebo-event responses by category

Category	N_s	Baseline		Orthog. `SMH`		Orthog. `SOXX`	
Broad tech policy	4	−0.064	(0.880)	+0.100	(0.824)	+0.055	(0.900)
Chip cycle	5	−0.226	(0.713)	−0.157	(0.789)	−0.171	(0.769)
Tech disruption	3	+0.254	(0.656)	+0.158	(0.766)	+0.180	(0.734)

Notes: Entries are signed average abnormal responses (SAARs); Newey–West HAC p-values are reported in parentheses. N_s denotes the number of events with an ex ante directional prior. Generic big-tech product and digital/corporate events are excluded from this table because no directional prior is imposed for these categories.

Overall assessment. Taken together, the placebo exercises provide a direct falsification test of the interpretation of the equity factor. First, non-AI technology events do not generate a systematic signed response of the factor, even when attention is restricted to episodes that produce sizeable movements in broad financial markets. The same is true for both 2- and 10-year inflation compensation, indicating that the ILS responses documented around AI events are not a generic feature of salient technology-news dates. Second, the contrast

between AI and placebo events remains statistically significant after removing the component of the factor associated with either `SMH` or `SOXX`. Third, the demanding semiconductor-cycle placebo shows that large chip-sector announcements can move the long–short portfolio, as is natural given its composition, but they do not reproduce the directionally consistent response observed around AI news. The evidence therefore supports the interpretation of the factor as capturing AI-specific information rather than generic mega-cap, broad technology, or semiconductor-cycle news.

C BVAR

C.1 Event-informed posterior weighting

The baseline identification combines contemporaneous sign and magnitude restrictions with narrative restrictions on the sign of the AI news shock around the event dates described in Section 2.1. The latter restrictions discard structural decompositions that are inconsistent with the direction of the AI-related information revealed by these events. Conditional on satisfying these restrictions, however, all admissible structural decompositions would otherwise receive equal posterior weight. This may leave substantial identification uncertainty if the admissible set contains rotations in which the AI news shock plays only a limited role in explaining movements in the AI factor even on dates associated with major AI announcements.

To exploit more fully the information contained in the event chronology, we therefore complement the hard restrictions with a soft event-informed re-weighting of the admissible structural decompositions. The procedure builds on the narrative-identification logic of Antolín-Díaz and Rubio-Ramírez (2018), but differs from a standard narrative restriction in an important respect: rather than imposing an additional zero-one restriction, we use the historical decomposition on the selected event dates to assign different posterior weights to otherwise admissible rotations.

Structural contributions on event dates. Let $m = 1, \ldots, M$ index posterior draws of the reduced-form BVAR together with an admissible structural rotation, and let

$$u_t = B_0^{(m)} \varepsilon_t^{(m)} \tag{C.20}$$

denote the corresponding structural representation, where $\varepsilon_t^{(m)}$ is the vector of mutually orthogonal structural shocks and $B_0^{(m)}$ is the contemporaneous impact matrix. Shock $s = 1$ is the identified AI news shock.

For a variable i, the contribution of structural shock s to the unexpected movement on date t is

$$c_{i,s,t}^{(m)} = B_{0,is}^{(m)} \varepsilon_{s,t}^{(m)}. \tag{C.21}$$

We focus on the contribution to the AI factor, corresponding to $i = AI$. For each event $e \in \mathcal{E}$ occurring at date τ_e, we therefore compute

$$c_{s,e}^{(m)} \equiv c_{AI,s,\tau_e}^{(m)}. \tag{C.22}$$

Importantly, these contributions are computed for all structural shocks, not only for the three economically labelled shocks. Hence,

$$\sum_{s=1}^{N} c_{s,e}^{(m)} = u_{AI,\tau_e}^{(m)}, \tag{C.23}$$

so that the historical decomposition exactly reproduces the reduced-form innovation in the AI factor draw by draw.

AI-event dominance measure. For each event and structural draw, we measure the extent to which the AI shock dominates competing explanations of the AI-factor innovation. Define

$$D_e^{(m)} = \frac{\left|c_{1,e}^{(m)}\right| - \max_{s \neq 1} \left|c_{s,e}^{(m)}\right|}{\sum_{s=1}^{N} \left|c_{s,e}^{(m)}\right|}. \tag{C.24}$$

The statistic is scale invariant and compares the absolute contribution of the AI news shock with that of the largest competing structural disturbance. Positive values indicate that the AI news shock is the largest contributor to the AI-factor innovation on event date e, while negative values indicate that another structural shock accounts for a larger contribution. The normalisation by total absolute structural contributions prevents unusually large event-day

innovations from mechanically receiving disproportionate weight.

For descriptive purposes, we also report the AI share of the absolute innovation,

$$S_e^{(m)} = \frac{\left|c_{1,e}^{(m)}\right|}{\sum_{s=1}^{N} \left|c_{s,e}^{(m)}\right|}. \tag{C.25}$$

This statistic is used only to characterise the historical decomposition; the posterior weighting itself is based on the relative dominance measure in Equation (C.24).

Soft event likelihood. Rather than requiring $D_e^{(m)} > 0$ as an additional hard narrative restriction, we map the dominance statistic into a smooth score,

$$\pi_e^{(m)} = \Lambda\big(\kappa D_e^{(m)}\big) = \frac{1}{1 + \exp\left[-\kappa D_e^{(m)}\right]}, \tag{C.26}$$

where $\kappa > 0$ governs the strength with which the event information discriminates across admissible structural decompositions. When $D_e^{(m)} = 0$, the event receives a score of $1/2$. The score approaches one as the AI news shock becomes increasingly dominant and approaches zero when a competing shock becomes increasingly dominant.

This specification deliberately preserves uncertainty about the structural interpretation of individual announcement days. In particular, the procedure does not require the AI news shock to explain the majority of the observed innovation, nor does it exclude decompositions in which other disturbances also make economically important contributions.

Combining the information across the $E = |\mathcal{E}|$ selected event dates, we construct the event score for draw m as the geometric mean

$$\mathcal{L}_{\mathcal{E}}^{(m)} = \left[\prod_{e \in \mathcal{E}} \pi_e^{(m)}\right]^{1/E}, \tag{C.27}$$

or, equivalently,

$$\log \mathcal{L}_{\mathcal{E}}^{(m)} = \frac{1}{E} \sum_{e \in \mathcal{E}} \log \pi_e^{(m)}. \tag{C.28}$$

The $1/E$ tempering is important because it prevents posterior concentration from increasing mechanically with the number of events included in the chronology. The event score should therefore be interpreted as a probabilistic ranking of admissible decompositions rather than as a fully specified sampling likelihood for the event-study observations.

Posterior weights. Let $\mathcal{A}$ denote the set of BVAR draws satisfying the baseline contemporaneous and narrative restrictions. Conditional on belonging to $\mathcal{A}$, the conventional sign-restricted posterior assigns equal weight to the admissible draws. We instead assign the normalised event-informed weights

$$\omega_m = \frac{\mathcal{L}_{\mathcal{E}}^{(m)}}{\sum_{j \in \mathcal{A}} \mathcal{L}_{\mathcal{E}}^{(j)}}, \qquad m \in \mathcal{A}, \tag{C.29}$$

and set $\omega_m = 0$ for draws outside the admissible set. Thus, the hard restrictions continue to determine the support of the structural posterior, whereas the event information affects the relative probability mass within that support.

For any posterior object $g^{(m)}$, including structural shocks, impulse responses, historical decompositions and forecast-error variance decompositions, posterior summaries are subsequently computed using the weights in Equation (C.29). For example, the event-informed posterior distribution of an impulse response satisfies

$$\Pr_{\omega}\left(IRF_h \leq x\right) = \sum_{m \in \mathcal{A}} \omega_m \mathbf{1}\left\{IRF_h^{(m)} \leq x\right\}. \tag{C.30}$$

Medians and posterior intervals are obtained from the corresponding weighted quantiles.

Choice of the weighting parameter. Our baseline calibration sets $\kappa = 10$. The parameter is selected using only first-stage BVAR diagnostics, prior to examining the downstream macroeconomic responses. With $M = 10{,}000$ admissible structural draws, this calibration yields an effective sample size of approximately 7,281 draws, or 72.8% of the original posterior support. The median width of the 68% posterior interval for the daily AI news shock falls by approximately 10%, while the correlation between the event-weighted and equal-

weight posterior-median daily shock series remains 0.99. Hence, the event information reduces structural identification uncertainty without materially altering the chronology of the identified AI news shock. As sensitivity checks, we consider weaker values $\kappa = 5$ and $\kappa = 2.5$. These imply progressively less posterior reweighting and retain a larger effective share of the original admissible set. The exercise therefore allows us to verify that the results are not driven by an excessively concentrated posterior or by a small number of structural rotations.

C.2 Validation against external AI release dates

This subsection complements the validation of the raw AI factor by examining the identified structural shock around the model-release dates collected by Christensen and Rudebusch (2026) – C&R henceforth. We first inspect daily changes in the AI factor around each release and apply the same 1.5-standard-deviation salience criterion used for narrative selection in Section 3. We then assess the identified shock around qualifying dates that were not imposed as narrative restrictions. The chronology is external, but the validation sample is conditional on the raw-factor salience screen.

Table C.12 reports the full set of 26 C&R dates. For events associated with a qualifying AI-factor move within one calendar day of the release, the table reports that matched daily move and its date. For events that do not satisfy the screen, the table reports the AI-factor change on the event trading day. The final two columns indicate whether the event was already included in the BVAR narrative set and, for events that were not included, whether it independently satisfies the same 1.5-standard-deviation screening rule.

Table C.12: C&R events and raw AI-factor movements

Event	C&R date	Move date	ΔAI (%)	In BVAR	1.5 s.d.
OpenAI ChatGPT (GPT-3.5)	30-Nov-2022	30-Nov-2022	+2.03	Yes	–
Google Bard	06-Feb-2023	07-Feb-2023	+2.14	No	Yes
OpenAI GPT-4 / Anthropic Claude 1	14-Mar-2023	14-Mar-2023	+1.84	Yes	–
Anthropic Claude 2	11-Jul-2023	10-Jul-2023	-2.33	No	Yes
xAI Grok 1	03-Nov-2023	03-Nov-2023	+0.21	No	No
Anthropic Claude 2.1	21-Nov-2023	21-Nov-2023	+0.15	No	No
Google Gemini 1.0 Nano and Pro	06-Dec-2023	06-Dec-2023	-0.47	No	No
Google Gemini 1.5 Pro	15-Feb-2024	15-Feb-2024	-0.74	No	No
Anthropic Claude 3	04-Mar-2024	04-Mar-2024	-2.09	No	Yes
OpenAI GPT-4o	13-May-2024	13-May-2024	-1.13	No	No
xAI Grok 1.5	15-May-2024	15-May-2024	+0.81	No	No
Anthropic Claude 3.5 Sonnet	20-Jun-2024	20-Jun-2024	-2.15	No	Yes
xAI Grok 2	13-Aug-2024	13-Aug-2024	+0.67	No	No

Continued on next page

Event	C&R date	Move date	ΔAI (%)	In BVAR	1.5 s.d.
OpenAI o1-preview	12-Sep-2024	11-Sep-2024	+2.16	No	Yes
Google Gemini 2.0 Flash	11-Dec-2024	11-Dec-2024	+2.69	No	Yes
OpenAI o3-mini	31-Jan-2025	30-Jan-2025	-3.69	No	Yes
xAI Grok 3	17-Feb-2025	18-Feb-2025	-2.08	No	Yes
Anthropic Claude 3.7 Sonnet	24-Feb-2025	24-Feb-2025	-1.24	No	No
OpenAI GPT-4.5	27-Feb-2025	26-Feb-2025	+1.94	No	Yes
Google Gemini 2.5 Pro	25-Mar-2025	26-Mar-2025	-3.37	No	Yes
OpenAI GPT-4.1	14-Apr-2025	14-Apr-2025	-1.90	No	Yes
Anthropic Claude 4	22-May-2025	22-May-2025	+1.36	No	No
xAI Grok 4	09-Jul-2025	09-Jul-2025	+1.65	No	No
Anthropic Claude Opus 4.1	05-Aug-2025	05-Aug-2025	+0.07	No	No
OpenAI GPT-5	07-Aug-2025	07-Aug-2025	+0.79	No	No
Anthropic Claude Sonnet 4.5	29-Sep-2025	29-Sep-2025	+0.12	No	No

Notes: Move date is the date of the reported daily AI-factor change, ΔAI. In BVAR indicates inclusion in the narrative restrictions; 1.5 s.d. indicates whether an external event passes the salience screen. – denotes an event already included in the narrative set.

The raw-data screen selects 13 of the 26 C&R events. Two of these—the ChatGPT launch and the GPT-4 / Claude 1 release date—were already part of the BVAR narrative set. Removing these overlaps leaves 11 salient release dates that were not imposed as narrative restrictions.

Passing the salience screen was necessary but not sufficient for inclusion in the original event study and BVAR narrative selection. We did not retain these releases as separate identifying events when their market impact was difficult to distinguish from other news or when events occurred too close together to permit a clear attribution. For example, the qualifying move associated with Claude 2 occurs on the preceding trading day, as does the move associated with o3-mini. The GPT-4.5 release on 27 February 2025 also follows a qualifying AI-factor move on 26 February. These timing issues do not prevent us from examining the identified shock around the omitted releases as a supplementary validation exercise, but they qualify its interpretation.

C.2.1 Detection of shocks around the retained releases

The next step is to check whether the identified daily AI shock is also unusually large around the 11 salient C&R events that were not used as narrative restrictions. We consider the event day itself and short post-event windows extending one and two trading days after the event. Table C.13 reports the posterior median AI shock over the three windows and the highest percentile reached by its absolute value relative to the corresponding full-sample non-event distribution. Table C.14 then reports the same validation across the full set of

admissible, event-weighted BVAR draws, summarising for each event the posterior probability of exceeding the 75th or 90th percentile in at least one of the three windows.

Table C.13: Posterior median AI shocks around external C&R events

Event	Event day	0 to +1	0 to +2	Max. abs. percentile	Above 75th	Above 90th
Google Bard	1.35	2.74	1.70	95.9	Yes	Yes
Anthropic Claude 2	-0.35	-0.48	0.49	34.3	No	No
Anthropic Claude 3	-1.58	-2.14	-3.16	95.2	Yes	Yes
Anthropic Claude 3.5 Sonnet	-1.07	-1.95	-3.72	97.4	Yes	Yes
OpenAI o1-preview	1.30	0.26	-1.42	85.7	Yes	No
Google Gemini 2.0 Flash	1.98	1.21	2.28	96.3	Yes	Yes
OpenAI o3-mini	0.43	-0.36	-0.70	40.8	No	No
xAI Grok 3	-0.86	-0.78	-2.27	86.1	Yes	No
OpenAI GPT-4.5	-1.62	-0.12	-2.93	93.7	Yes	Yes
Google Gemini 2.5 Pro	0.40	-1.80	-1.94	84.7	Yes	No
OpenAI GPT-4.1	-1.71	-1.76	-4.29	98.6	Yes	Yes

Notes: The first three numerical columns report the posterior median AI shock on the event trading day and cumulated through one and two subsequent trading days. "Max. abs. percentile" is the highest percentile reached by the absolute posterior median shock across the three windows relative to the corresponding full-sample non-event distribution.

Table C.14: Posterior probability of detecting a large AI shock

Event	Above 75th percentile in any window	Above 90th percentile in any window
Google Bard	99.0%	79.9%
Anthropic Claude 2	29.0%	4.9%
Anthropic Claude 3	99.5%	85.2%
Anthropic Claude 3.5 Sonnet	99.9%	94.6%
OpenAI o1-preview	92.2%	18.4%
Google Gemini 2.0 Flash	96.2%	79.2%
OpenAI o3-mini	5.5%	0.0%
xAI Grok 3	75.8%	19.2%
OpenAI GPT-4.5	92.7%	65.6%
Google Gemini 2.5 Pro	71.1%	16.2%
OpenAI GPT-4.1	97.7%	88.2%

Notes: Probabilities are calculated across the full set of admissible, event-weighted BVAR draws. An event is counted as detected when the absolute AI shock exceeds the indicated percentile in at least one of the event-day, one-day post-event, or two-day post-event windows.

The event-level evidence is strong for most of the external dates. Nine of the 11 events produce a posterior median AI shock above the 75th percentile in at least one of the three short windows, and six of the 11 exceed the 90th percentile. The strongest cases at the 90th-percentile threshold include Bard, Claude 3, Claude 3.5 Sonnet, Gemini 2.0 Flash, GPT-4.5 and GPT-4.1. Two events, Claude 2 and o3-mini, are not associated with a particularly large identified AI shock; the timing concerns discussed above are especially relevant in these cases.

Table C.15 summarises the successive screening steps and the main cross-validation statistics. Half of the original C&R chronology passes the raw AI-factor salience screen. After removing the two dates already used in the BVAR narrative restrictions, 11 observations remain for the external validation. The posterior median shock catches more than four-fifths of these events above the 75th percentile and slightly more than half above the 90th percentile.

Table C.15: Summary of the C&R cross-validation exercise

Statistic	Result
C&R chronology	26
Salient events	13
Narrative overlap	2
External events (N_E)	11
Posterior median detection: $\hat{s}_{0.75}$	9 of 11 (81.8%)
Posterior median detection: $\hat{s}_{0.90}$	6 of 11 (54.5%)
Posterior median of $s_{0.75}^{(d)}$	81.8%
68% posterior range of $s_{0.75}^{(d)}$	63.6–83.0%
Posterior median of $s_{0.90}^{(d)}$	54.5%
68% posterior range of $s_{0.90}^{(d)}$	36.4–63.6%

Notes: Salient events pass the 1.5-standard-deviation raw AI-factor screen; narrative overlap counts those already used as BVAR narrative restrictions. N_E is the remaining external sample. $\hat{s}_q$ is the share of these events whose absolute posterior median AI shock exceeds the corresponding non-event q-quantile in at least one window: event day, 0 to +1, or 0 to +2 trading days. $s_q^{(d)}$ is the analogous share within posterior draw d; its median and 68% range use the event-weighted posterior draws.

Across the event-weighted BVAR draws, the median share of retained events exceeding the 75th-percentile threshold is 81.8%, with a 68% range of 63.6–83.0%. At the more demanding 90th-percentile threshold, the median share is 54.5%, with a 68% range of 36.4–63.6%. Thus, the strong detection rate at the lower threshold also appears across BVAR draws, while detection at the higher threshold is more sensitive to identification.

Overall, the exercise provides supporting evidence that the identified shock detects salient releases beyond the imposed narrative dates, particularly at the 75th-percentile threshold. Two features qualify this interpretation. First, the screening window includes preceding-day moves, whereas the shock-validation windows begin on the event trading day; this distinction matters for releases such as Claude 2 and o3-mini. Second, the detection shares condition on large raw-factor movements and allow exceedance in any of three windows. They are descriptive validation statistics, not rejection rates from a test calibrated for this selection

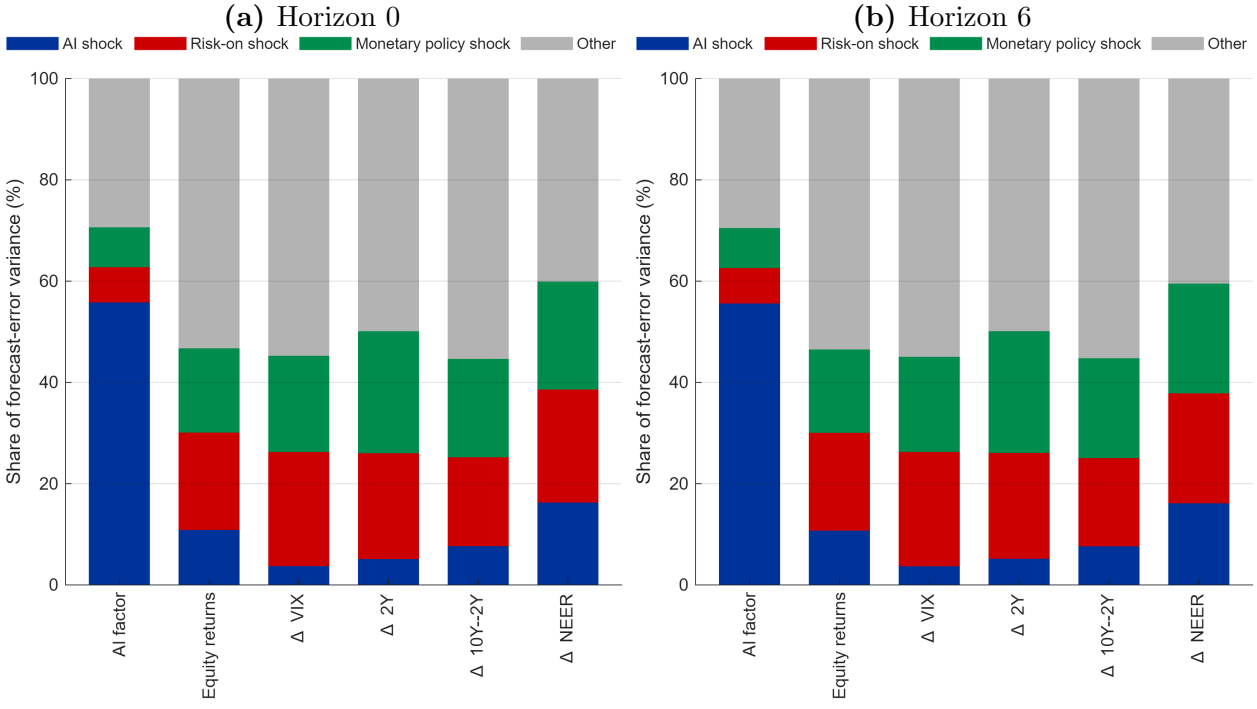
and multiple-window procedure.

C.3 Additional results

C.3.1 Forecast error variance decomposition

The FEVD provides a useful validation of the structural identification. On impact, the identified AI news shock accounts for about 60% of the forecast-error variance of the AI factor, while its contribution to the other financial variables is much smaller: roughly 8% for broad equity returns, 4% for the VIX and the 2-year yield, 6% for the Treasury slope, and 18% for the NEER. Risk-on and monetary policy shocks instead explain considerably larger shares of movements in the broader financial variables, while the residual group of shocks remains quantitatively important. This pattern indicates that the identified AI news shock captures a relatively specific source of variation rather than a generic financial-market disturbance. The decomposition is virtually unchanged at the six-day horizon, consistent with the high-frequency nature of the BVAR: most of the forecast-error variance is determined on impact and subsequent daily propagation is limited.

Figure C.3: Forecast error variance decomposition



Notes: Forecast error variance decomposition of the endogenous variables at horizons 0 (Panel a) and 6 (Panel b).

C.3.2 Historical decomposition

As an additional diagnostic of the identification, we report historical decompositions around the narrative events in Table 4. The event-weighted posterior mean contributions are additive by construction and reproduce the corresponding reduced-form innovation on each event date. The results reinforce the visual evidence in Panel a of Figure C.4: the identified AI news shock contributes with the expected sign on the selected events and is quantitatively important around both favourable and adverse AI news.

For the AI factor, the reported AI share exceeds 50% on eight of the fifteen narrative dates. It is particularly large around the NVIDIA Blackwell announcement (74%), the NVIDIA FY24 Q4 earnings release (62%), the June 2026 US export restrictions on Anthropic frontier models (62%), and the July 2025 US AI Action Plan (61%). The AI share is also slightly above one half around GPT-4 (52%), DeepSeek R1 (54%), NVIDIA Q1 FY25 earnings (53%), and Apple Intelligence (56%). The ChatGPT launch lies just below this threshold, with an AI share of about 49%.

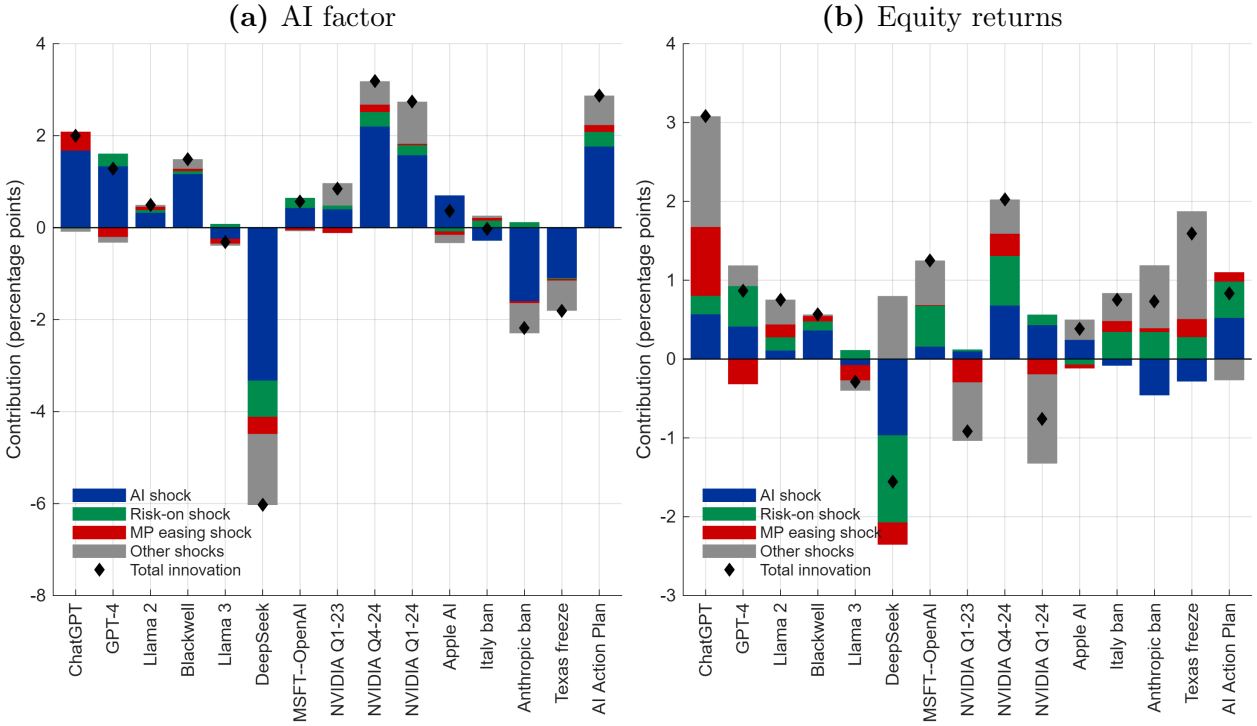
At the same time, the decomposition makes clear that not every event-day movement is predominantly associated with the AI shock. The estimated AI share is about 29% for the Italy ban, 30% for NVIDIA Q1 FY24 earnings, 29% for Microsoft's January 2023 OpenAI investment, and 30% for the Llama 3 release. The Texas data-centre permitting announcement is also more mixed, with an AI share of about 41%. This heterogeneity is useful for the identification strategy because the narrative information is used to *re-weight* admissible structural rotations rather than to impose that the AI shock accounts for the full innovation on every event date.

The largest AI contributions line up with economically salient repricing episodes. The contribution is strongly positive for the ChatGPT launch, GPT-4, Blackwell, NVIDIA FY24 Q4 and Q1 FY25 earnings, and the US AI Action Plan, while it is strongly negative for DeepSeek R1 and the June 2026 US export restrictions. The decomposition also assigns negative AI contributions to the Italy ban, the Llama 3 event, and the September 2026 Texas data-centre permitting announcement, consistent with their adverse classification in the narrative set.

Overall, the historical decomposition supports the interpretation of the long–short factor

as containing economically meaningful AI-specific information while allowing generic risk-on, monetary policy, and residual shocks to contribute materially to event-day market movements. The near-exact additivity of the decomposition further confirms that these contributions exhaust the reduced-form innovation up to numerical precision.

Figure C.4: Historical decompositions around major events



Notes: Historical decomposition of innovations in the AI factor and equity returns on the event days reported in Table 4. Contributions are event-weighted posterior means and are additive by construction.

C.4 Predictability of the monthly AI news shock

As an additional validation exercise, we assess whether lagged US macro-financial conditions predict the identified monthly AI news shock. For each retained structural BVAR draw d, we estimate

$$\varepsilon_t^{AI,(d)} = \alpha^{(d)} + X_{t-1}'\beta^{(d)} + u_t^{(d)}, \tag{C.31}$$

where X_{t-1} contains the standardised values of six US macro-financial predictors: r_{t-1}^{SP}, the previous month's S&P 500 return; $\log VIX_{t-1}$, the log VIX at the end of the previous month; i_{t-1}^{2y} and i_{t-1}^{10y}, the corresponding end-of-month 2-year and 10-year Treasury yields; and g_{t-1}^{IP} and π_{t-1}^{CPI}, industrial production growth and CPI inflation for the previous month. We also

consider an augmented specification that controls for the lagged AI shock,

$$\varepsilon_t^{AI,(d)} = \alpha^{(d)} + \rho^{(d)}\varepsilon_{t-1}^{AI,(d)} + X_{t-1}'\beta^{(d)} + u_t^{(d)}. \tag{C.32}$$

The aligned monthly sample contains 142 observations from January 2015 through October 2026. Because all predictors enter with one lag, the final October 2026 shock observation is conditioned on macro-financial information dated no later than September 2026. We estimate the regressions separately for each of the 10,000 posterior shock histories and summarise the coefficient estimates using the event-informed posterior weights employed in the local projections. These summaries capture variation across shock histories, not the estimation uncertainty within each regression. For the latter, we use Newey–West standard errors with three lags.

Table C.16 reports the weighted 16th, 50th (median), and 84th percentiles of the coefficient estimates across shock histories. The regressions explain little of the variation in the monthly AI shock: the weighted median R^2 is 4.0% in the baseline specification and 5.0% when the lagged shock is included. The lagged AI shock has a median coefficient of 0.089, with a 16–84 percentile range of 0.051 to 0.136.

The results provide little evidence of systematic predictability. As shown in Table C.17, the weighted probability that the joint null of no predictive content from the macro-financial variables is rejected at the 5% level is only 1.4% in the baseline specification and 1.4% when the lagged AI shock is included. Even at the 10% level, the corresponding rejection probabilities are only 5.3% and 5.6%. The lagged AI shock is significant at the 5% level in 9.5% of weighted posterior draws and at the 10% level in 21.4%, indicating limited serial predictability.

Individual predictors also display little systematic predictive power. The largest 5% rejection probabilities in the baseline specification are only about 6.5% for the 2-year Treasury yield, 6.2% for the 10-year yield, and 5.7% for CPI inflation; all other predictors are below 2.4%. The pattern is very similar after including the lagged AI shock. Taken together, the low explanatory power, the small posterior probability of joint rejection, and the limited serial predictability of the shock itself support the interpretation of the identified monthly AI

Table C.16: Predictability of the identified AI shock

Regressor	16th pct.	Median	84th pct.
Lagged macro-financial predictors			
Constant	-0.0041	0.0002	0.0045
r_{t-1}^{SP}	-0.0039	-0.0028	-0.0016
$\log VIX_{t-1}$	-0.0077	-0.0061	-0.0038
i_{t-1}^{2y}	0.0075	0.0160	0.0233
i_{t-1}^{10y}	-0.0251	-0.0181	-0.0098
g_{t-1}^{IP}	0.0012	0.0044	0.0073
π_{t-1}^{CPI}	-0.0106	-0.0061	-0.0009
R^2	0.0170	0.0402	0.0748
Including the lagged AI shock			
Constant	-0.0038	0.0002	0.0041
ε_{t-1}^{AI}	0.0507	0.0886	0.1359
r_{t-1}^{SP}	-0.0047	-0.0032	-0.0019
$\log VIX_{t-1}$	-0.0076	-0.0060	-0.0036
i_{t-1}^{2y}	0.0078	0.0155	0.0222
i_{t-1}^{10y}	-0.0241	-0.0175	-0.0098
g_{t-1}^{IP}	0.0011	0.0042	0.0071
π_{t-1}^{CPI}	-0.0103	-0.0055	-0.0000
R^2	0.0263	0.0502	0.0856

Notes: The table reports event-weighted posterior percentiles across 10,000 retained structural BVAR shock histories. Macro-financial regressors are dated $t-1$. Treasury yields and the VIX are measured at the end of the previous month. Industrial production growth and CPI inflation refer to the previous monthly observation. Predictors are standardised prior to estimation, while the lagged AI shock remains in its original units. Newey–West standard errors with three lags are used within each draw. The R^2 rows report weighted posterior percentiles computed across draw-specific regressions.

Table C.17: Posterior rejection probabilities

	5% level	10% level
Joint test, baseline specification	0.014	0.053
Joint test, including ε_{t-1}^{AI}	0.014	0.056
ε_{t-1}^{AI}	0.095	0.214
Individual 5% rejection probabilities		
r_{t-1}^{SP}	0.000	0.000
$\log VIX_{t-1}$	0.000	0.000
i_{t-1}^{2y}	0.065	0.065
i_{t-1}^{10y}	0.062	0.071
g_{t-1}^{IP}	0.024	0.020
π_{t-1}^{CPI}	0.057	0.043

Notes: The first three rows report the event-weighted fraction of posterior shock histories for which the corresponding null hypothesis is rejected. The lower panel reports predictor-specific 5% rejection probabilities in the baseline specification (first column) and in the specification including the lagged AI shock (second column). Joint tests refer to the null that all six lagged macro-financial predictor coefficients are zero. The lagged-shock row refers to the null $\rho = 0$.

shock as largely unpredictable from lagged US macro-financial conditions.

D Local projections

D.1 Bayesian smooth local projections

This section describes the Bayesian smooth local projection procedure used to estimate the macroeconomic responses to the AI news shock. The estimator has two objectives. First, it regularises the impulse response across adjacent horizons, which is particularly useful at long horizons, where standard local projections are imprecisely estimated. Second, it propagates uncertainty about the structural shock estimated in the daily BVAR through the local projection stage.

D.1.1 Horizon-specific local projections

For a given realisation r of the monthly AI news shock, we estimate the sequence of horizon-specific regressions

$$y_{t+h} - y_{t-1} = \alpha_h + \beta_h^{(r)} \varepsilon_t^{\mathrm{AI},(r)} + \mathbf{x}_t^{(r)\prime} \boldsymbol{\gamma}_h^{(r)} + u_{t+h,h}^{(r)}, \qquad h = 0, \ldots, H, \tag{D.33}$$

where $\mathbf{x}_t^{(r)}$ contains lags of the AI news shock, lags of the dependent variable and the remaining controls (COVID dummy). The coefficient of interest $\beta_h^{(r)}$ is left unrestricted in sign and magnitude at every horizon. The baseline estimates cumulative responses through the dependent variable $y_{t+h} - y_{t-1}$. Let

$$\mathbf{m}^{(r)} = \left(m_0^{(r)}, \ldots, m_H^{(r)} \right)' \tag{D.34}$$

denote the vector of horizon-specific estimates of the impulse-response coefficient and let $\mathbf{V}^{(r)}$ denote the corresponding estimation uncertainty. We employ the Gaussian approximation

$$\mathbf{m}^{(r)} \mid \boldsymbol{\beta}^{(r)} \sim \mathcal{N}\left(\boldsymbol{\beta}^{(r)}, \mathbf{V}^{(r)} \right), \tag{D.35}$$

where

$$\boldsymbol{\beta}^{(r)} = \left(\beta_0^{(r)}, \ldots, \beta_H^{(r)}\right)'. \tag{D.36}$$

The nuisance coefficients entering Equation (D.33) are regularised using Minnesota-type shrinkage, with prior tightness declining with the lag order. The contemporaneous AI-shock coefficient, which defines the impulse response, is not subject to this nuisance-parameter shrinkage.

D.1.2 Second-order random-walk (RW2) prior

Estimating each β_h independently can generate considerable horizon-to-horizon sampling variation, especially because the effective sample size declines with h. We therefore place a second-order random-walk prior on the impulse-response sequence. Let $\mathbf{D}_2$ denote the second-difference matrix, such that

$$\mathbf{D}_2\boldsymbol{\beta}^{(r)} = \begin{pmatrix} \beta_2^{(r)} - 2\beta_1^{(r)} + \beta_0^{(r)} \\ \vdots \\ \beta_H^{(r)} - 2\beta_{H-1}^{(r)} + \beta_{H-2}^{(r)} \end{pmatrix}. \tag{D.37}$$

The smoothing prior is

$$\mathbf{D}_2\boldsymbol{\beta}^{(r)} \mid q_r^2 \sim \mathcal{N}\left(\mathbf{0}, q_r^2\mathbf{I}\right). \tag{D.38}$$

Equivalently, the smoothing precision is $\lambda_r = 1/q_r^2$. Small values of q_r^2 imply a smoother response, whereas larger values allow greater curvature across horizons. Importantly, Equation (D.38) does not impose the sign, level, monotonicity or location of the peak of the impulse response; it penalises only abrupt changes in its slope.

Rather than fixing the amount of smoothing, we estimate it hierarchically. We specify

$$q_r^2 \sim \mathcal{IG}(a_q, b_q), \tag{D.39}$$

where the baseline prior uses $a_q = 2.5$ and is parameterised to imply $\mathbb{E}[q_r^2] = 1$. Conditional on q_r^2, the posterior for $\boldsymbol{\beta}^{(r)}$ is Gaussian. Combining Equation (D.35) and Equation (D.38),

its precision matrix is

$$\mathbf{P}_r = \mathbf{V}^{(r)-1} + \frac{1}{q_r^2}\mathbf{D}_2'\mathbf{D}_2, \tag{D.40}$$

and therefore

$$\boldsymbol{\beta}^{(r)} \mid q_r^2, \mathbf{m}^{(r)} \sim \mathcal{N}\left(\boldsymbol{\mu}_r, \mathbf{P}_r^{-1}\right), \tag{D.41}$$

$$\boldsymbol{\mu}_r = \mathbf{P}_r^{-1}\mathbf{V}^{(r)-1}\mathbf{m}^{(r)}. \tag{D.42}$$

The conditional posterior for q_r^2 is inverse-Gamma,

$$q_r^2 \mid \boldsymbol{\beta}^{(r)} \sim \mathcal{IG}\left(a_q + \frac{H-1}{2},\ b_q + \frac{1}{2}\left\|\mathbf{D}_2\boldsymbol{\beta}^{(r)}\right\|^2\right), \tag{D.43}$$

which gives a simple Gibbs sampler alternating between Equation (D.42) and Equation (D.43).

Because the first-stage BVAR itself produces a distribution of admissible shock histories, learning the smoothing parameter from a single arbitrarily selected structural draw would be undesirable. We therefore calibrate the hierarchical distribution using a representative subset of the posterior shock histories and pool the resulting draws of q^2 with equal weight. In the baseline implementation we use 500 shock histories for this calibration, run 300 Gibbs iterations for each calibration draw, and discard the first 150 iterations as burn-in. The resulting hyperposterior is then propagated through the full local projection exercise.[24]

D.1.3 Propagation of first- and second-stage uncertainty

The AI news shock entering Equation (D.33) is not observed. Let

$$\mathcal{E} = \left\{\boldsymbol{\varepsilon}^{(r)}, \omega_r\right\}_{r=1}^{M} \tag{D.44}$$

denote the event-informed posterior distribution of daily structural AI-shock histories obtained from the BVAR, where ω_r is the posterior weight attached to draw r. Each daily shock history is first aggregated to the monthly frequency,

$$\varepsilon_m^{(r)} = \sum_{t \in m} \varepsilon_t^{(r)}, \tag{D.45}$$

[24]The posterior estimates of the hyperparameters are reported in Table D.18 below.

before any posterior summary is taken. This is important because aggregation and posterior quantiles do not commute. The event weights are preserved after monthly aggregation. Since the downstream Bayesian local projection routine operates on equally weighted shock histories, we transform the weighted posterior into an equally weighted sample using systematic importance resampling. If R histories are required, the resulting indices

$$r_1, \ldots, r_R \sim_{\text{SIR}} \{\omega_1, \ldots, \omega_M\} \tag{D.46}$$

are distributed according to the event-informed posterior probabilities. Higher-weight structural rotations consequently enter the LP stage more often, while low-weight rotations remain represented according to their posterior mass. Repeated indices are therefore an intentional feature of the importance resampling rather than duplicate observations to be removed.

For every resampled shock history r, we estimate Equation (D.33) and generate B draws

$$\boldsymbol{\beta}^{(r,b)} \sim p\left(\boldsymbol{\beta} \mid \boldsymbol{\varepsilon}^{(r)}, \mathbf{Y}\right), \qquad b = 1, \ldots, B. \tag{D.47}$$

The unconditional posterior of the impulse response can therefore be written using the law of total probability as

$$\begin{aligned} p(\boldsymbol{\beta} \mid \mathbf{Y}) &= \int p\left(\boldsymbol{\beta} \mid \boldsymbol{\varepsilon}, \mathbf{Y}\right) p\left(\boldsymbol{\varepsilon} \mid \mathcal{I}_{\text{BVAR}}\right) d\boldsymbol{\varepsilon} \\ &\simeq \frac{1}{R} \sum_{r=1}^{R} p\left(\boldsymbol{\beta} \mid \boldsymbol{\varepsilon}^{(r)}, \mathbf{Y}\right), \end{aligned} \tag{D.48}$$

where $\mathcal{I}_{\text{BVAR}}$ denotes the information and identifying restrictions used in the first-stage BVAR. The Monte Carlo approximation to Equation (D.48) is consequently

$$\left\{\boldsymbol{\beta}^{(r,b)} : r = 1, \ldots, R,\ b = 1, \ldots, B\right\}. \tag{D.49}$$

Variation across r captures first-stage uncertainty concerning the structural decomposition and the estimated AI-shock history. Variation across b captures second-stage uncertainty concerning the local projection coefficients conditional on that history. Posterior medians and

pointwise highest-density posterior sets are computed directly from this pooled distribution.

This treatment is important for inference because the structural shock is a generated regressor (Pagan, 1984). A conventional two-step procedure that first replaces the latent shock by a single posterior estimate and subsequently treats that estimate as observed would condition on the first-stage estimate and discard uncertainty about the structural decomposition. Our procedure instead marginalises over this first-stage uncertainty. Thus, while the estimation remains modular rather than a fully joint BVAR–LP model, the reported uncertainty bands incorporate both uncertainty about the generated shock and conditional LP estimation uncertainty. In the baseline specification we use $R = 10{,}000$ event-weighted monthly shock histories and retain $B = 50$ posterior local projection draws for each shock history, yielding up to 500,000 draws from the mixture posterior at each horizon. The baseline horizon is 60 months and the model contains twelve lags.

D.1.4 Posterior estimates of the hierarchical RW2 hyperparameters

As discussed above, the Bayesian local projections impose a hierarchical second-order random-walk (RW2) prior on the sequence of shock-response coefficients across horizons. The degree of smoothing is not fixed ex ante, but is estimated from the data through the hyperparameter q^2, with the corresponding smoothing precision given by $\lambda = 1/q^2$. Higher values of λ correspond to stronger smoothing across horizons. Table D.18 reports the posterior estimates of these hyperparameters for the baseline specification ($p = 12$), separately for US aggregates, euro area aggregates, and the US sectoral-network responses.

The estimates are relatively homogeneous across variables and applications. Posterior median values of λ are generally concentrated in a fairly narrow range, implying similar but not identical degrees of smoothing across outcomes. Production and real-activity variables tend to imply somewhat lower smoothing precision than price variables, while there is no evidence that the hierarchical procedure selects extreme smoothing parameters for any particular series. This provides reassurance that the shape of the estimated impulse responses is not driven by markedly different or highly restrictive smoothing assumptions across variables.

Table D.18: Posterior estimates of hierarchical RW2 hyperparameters

Variable	λ	q^2
US aggregates		
PCE core	6.18 [4.52, 8.25]	0.162 [0.121, 0.221]
CPI headline	5.96 [4.39, 7.99]	0.168 [0.125, 0.228]
CPI core	6.01 [4.40, 8.04]	0.166 [0.124, 0.227]
PPI headline	5.89 [4.29, 7.87]	0.170 [0.127, 0.233]
Industrial production	4.87 [3.46, 6.68]	0.205 [0.150, 0.289]
Real GDP	4.84 [3.43, 6.61]	0.207 [0.151, 0.291]
Employment	5.19 [3.70, 7.08]	0.193 [0.141, 0.270]
Hours	5.29 [3.78, 7.18]	0.189 [0.139, 0.264]
Hourly earnings	5.58 [4.04, 7.53]	0.179 [0.133, 0.247]
CAPEX orders	5.18 [3.70, 7.06]	0.193 [0.142, 0.270]
Capacity utilisation	4.98 [3.55, 6.81]	0.201 [0.147, 0.282]
Euro area aggregates		
Equity prices	4.98 [3.54, 6.82]	0.201 [0.147, 0.282]
Gas prices	5.28 [3.80, 7.18]	0.190 [0.139, 0.263]
Inflation expectations (10Y)	5.51 [3.97, 7.45]	0.182 [0.134, 0.252]
Inflation expectations (2Y)	5.13 [3.67, 6.98]	0.195 [0.143, 0.273]
OIS rate (10Y)	5.52 [4.00, 7.43]	0.181 [0.135, 0.250]
OIS rate (2Y)	6.09 [4.47, 8.10]	0.164 [0.123, 0.224]
Employment	5.63 [4.06, 7.59]	0.178 [0.132, 0.246]
Energy prices	5.27 [3.78, 7.12]	0.190 [0.140, 0.264]
HICP core	6.01 [4.38, 8.02]	0.166 [0.125, 0.228]
HICP headline	5.91 [4.30, 7.91]	0.169 [0.126, 0.232]
Hours worked	5.18 [3.70, 7.06]	0.193 [0.142, 0.270]
Import prices	5.65 [4.10, 7.58]	0.177 [0.132, 0.244]
Industrial production	4.80 [3.41, 6.60]	0.208 [0.152, 0.293]
PPI	5.83 [4.26, 7.84]	0.171 [0.128, 0.235]
Exchange rate vs USD	4.90 [3.47, 6.72]	0.204 [0.149, 0.288]
EA–US spread (2Y)	5.09 [3.62, 6.94]	0.197 [0.144, 0.276]
EA–US spread (10Y)	5.23 [3.74, 7.11]	0.191 [0.141, 0.267]
US sectoral aggregates		
Prices – upstream, high exposure	5.72 [4.14, 7.69]	0.175 [0.130, 0.242]
Prices – upstream, low exposure	5.67 [4.14, 7.65]	0.176 [0.131, 0.242]
Prices – upstream differential	5.13 [3.66, 7.00]	0.195 [0.143, 0.273]
Prices – downstream, high exposure	5.80 [4.22, 7.77]	0.173 [0.129, 0.237]
Prices – downstream, low exposure	5.67 [4.13, 7.61]	0.176 [0.131, 0.242]
Prices – downstream differential	5.32 [3.84, 7.19]	0.188 [0.139, 0.261]
Production – upstream, high exposure	5.15 [3.68, 7.03]	0.194 [0.142, 0.272]
Production – upstream, low exposure	5.04 [3.60, 6.86]	0.198 [0.146, 0.278]
Production – upstream differential	4.93 [3.51, 6.73]	0.203 [0.149, 0.285]
Production – downstream, high exposure	5.11 [3.64, 6.98]	0.196 [0.143, 0.275]
Production – downstream, low exposure	5.27 [3.79, 7.16]	0.190 [0.140, 0.264]
Production – downstream differential	5.31 [3.82, 7.18]	0.188 [0.139, 0.262]
Employment – upstream, high exposure	5.12 [3.66, 7.00]	0.195 [0.143, 0.273]
Employment – upstream, low exposure	5.16 [3.65, 7.07]	0.194 [0.141, 0.274]
Employment – upstream differential	5.35 [3.84, 7.25]	0.187 [0.138, 0.260]
Employment – downstream, high exposure	5.43 [3.92, 7.35]	0.184 [0.136, 0.255]
Employment – downstream, low exposure	5.13 [3.66, 7.00]	0.195 [0.143, 0.273]
Employment – downstream differential	5.07 [3.64, 6.91]	0.197 [0.145, 0.275]

Notes: Posterior estimates of the hyperparameters governing the hierarchical second-order random-walk (RW2) prior used to smooth the local-projection coefficients across horizons. $\lambda = 1/q^2$ is the implied smoothing precision; q^2 is the innovation variance of the RW2 process. The 16th and 84th percentiles are in square brackets. Estimates refer to the baseline specification with 12 monthly lags.

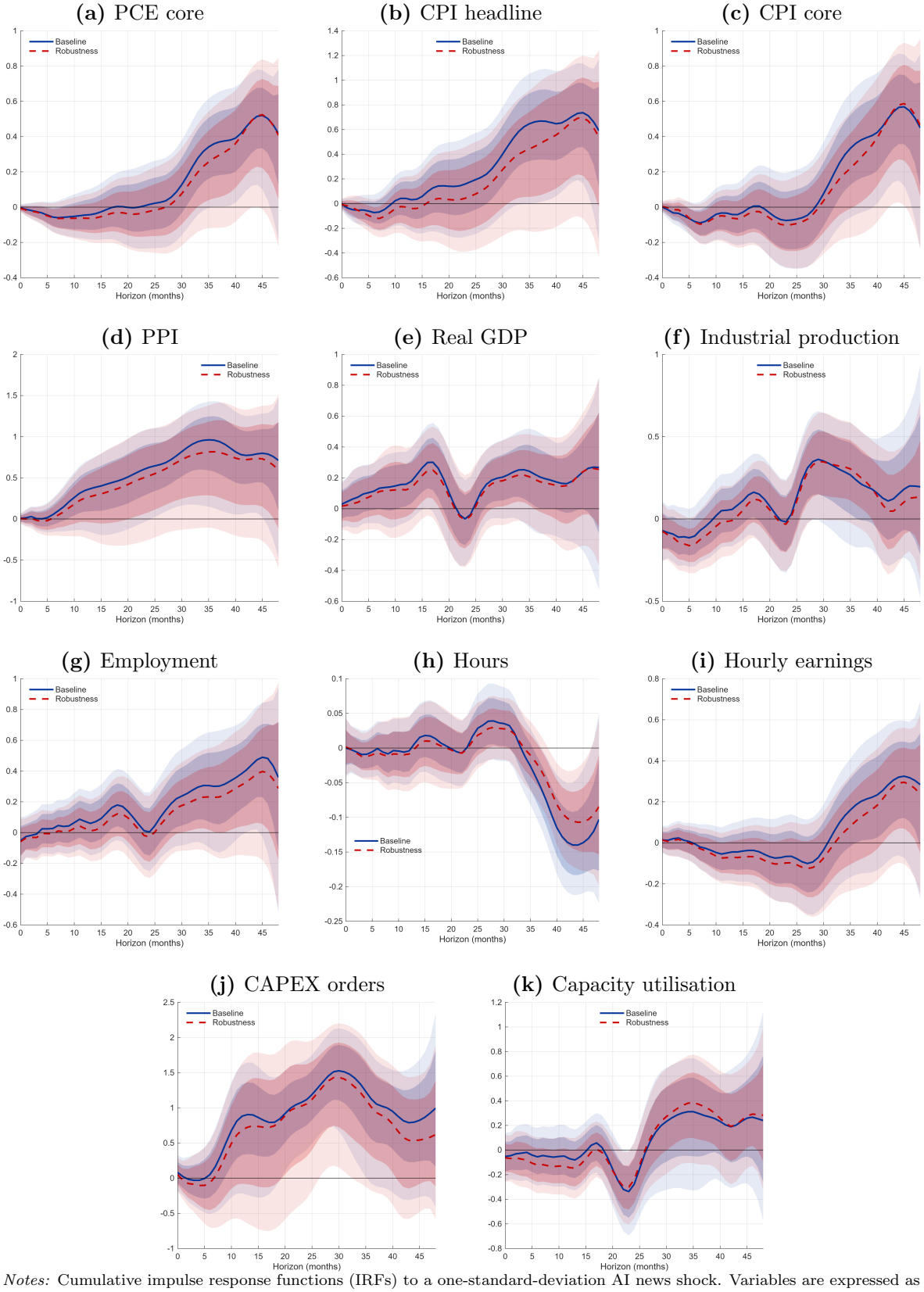
D.2 Robustness checks

D.2.1 Shorter lag length

As a first robustness exercise, we assess the sensitivity of the local projection estimates to the choice of lag length. While the baseline specification includes $p = 12$ monthly lags, we re-estimate Equation (5) using $p = 6$. The shorter lag structure provides a useful check that the estimated propagation patterns are not mechanically driven by the relatively rich dynamic specification adopted in the baseline. It also reduces the number of parameters estimated at each horizon and therefore mitigates the loss of effective observations that becomes increasingly relevant at longer horizons in our relatively short monthly sample. All other elements of the estimation procedure, including the identified AI-shock draws, the shock normalisation, the cumulative specification of the dependent variable, and the two-layer resampling procedure, are kept unchanged. Figure D.5, Figure D.6, and Figure D.7 compare the resulting impulse responses with those obtained under the specification with $p = 12$.

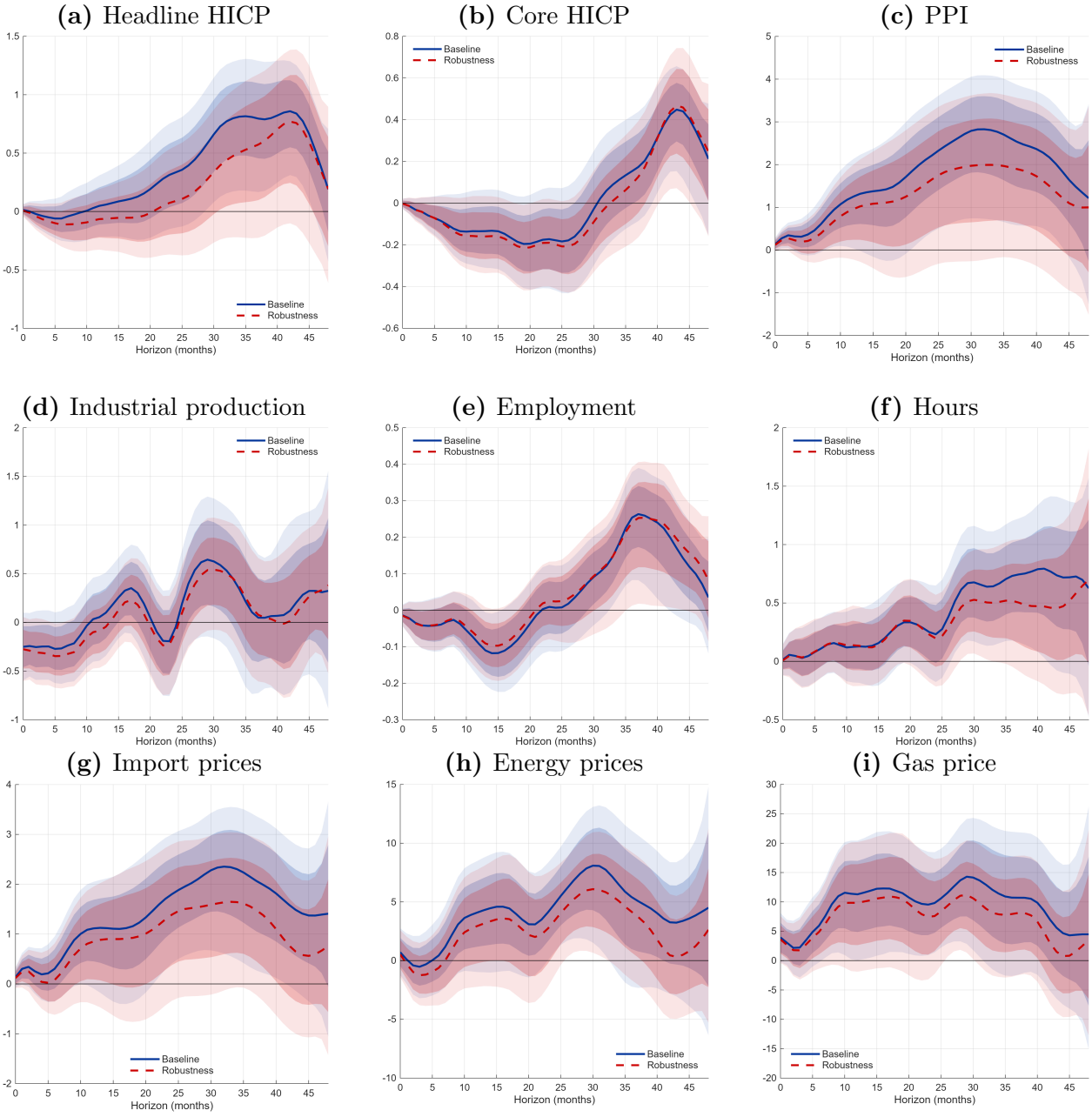
Overall, the results are broadly stable across the two lag specifications. For the US, the shorter-lag estimates preserve the main features of the baseline response: prices rise over the medium run, capital expenditure expands relatively early, production and capacity utilisation initially display adjustment effects and subsequently recover, while the later increase in employment and earnings is accompanied by a decline in hours. The responses obtained with $p = 6$ are generally somewhat smaller and become more volatile at the longest horizons, but the timing and signs of the main adjustment phases remain broadly unchanged. A similar conclusion emerges for the euro area. The positive responses of headline prices, producer prices, import prices, and energy costs are preserved, as is the delayed improvement in real activity. The financial block is also broadly preserved: the euro depreciation, the medium-run increase in inflation expectations and nominal rates, and the phased response of the EA–US interest rate differentials are all reproduced under the shorter lag structure. Taken together, these results indicate that the domestic and cross-border propagation patterns documented in the baseline are not sensitive to the lag choice.

Figure D.5: Impact of AI news shocks on US aggregates



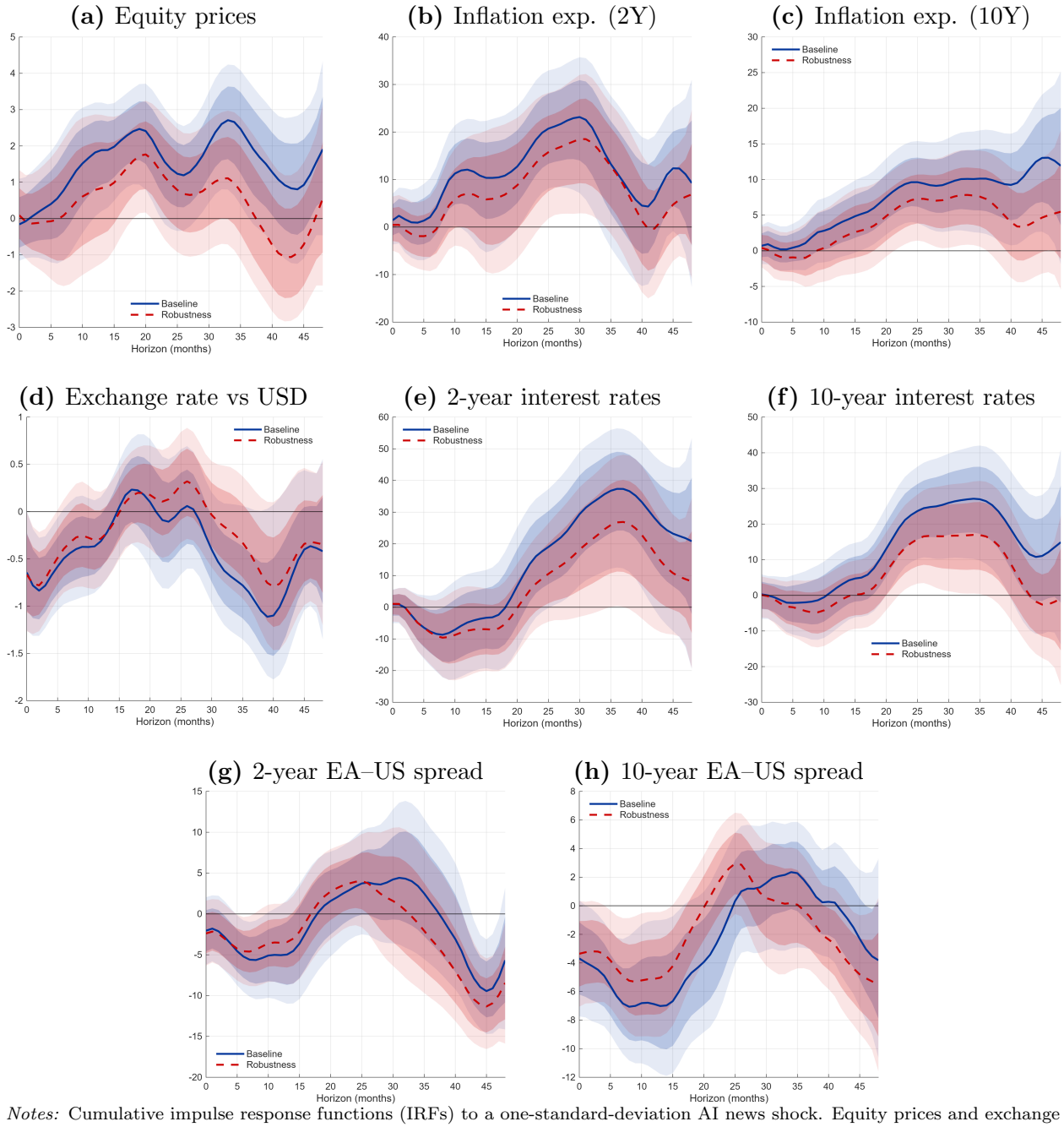
Notes: Cumulative impulse response functions (IRFs) to a one-standard-deviation AI news shock. Variables are expressed as percent deviations from the baseline ($t-1$). Darker (lighter) shaded areas represent 68% (90%) highest-density posterior sets. Blue (red) lines indicate the response with $p = 12$ (6).

Figure D.6: Impact of AI news shocks on euro area *real and price* aggregates



Notes: Cumulative impulse response functions (IRFs) to a one-standard-deviation AI news shock. Variables are expressed as percent deviation from the baseline $(t-1)$. Darker (lighter) shaded areas represent 68% (90%) highest-density posterior sets. Blue (red) lines indicate the response with $p = 12$ (6).

Figure D.7: Impact of AI news shocks on euro area *financial* aggregates



Notes: Cumulative impulse response functions (IRFs) to a one-standard-deviation AI news shock. Equity prices and exchange rate are expressed as percent deviation from the baseline $(t-1)$, rates, spreads and inflation expectations in basis points. Darker (lighter) shaded areas represent 68% (90%) highest-density posterior sets. Blue (red) lines indicate the response with $p = 12$ (6).

D.2.2 No cross-horizon smoothing

As an additional robustness exercise, we assess the extent to which the estimated impulse responses depend on the hierarchical smoothing imposed across local-projection horizons. To do so, we re-estimate the baseline specification without the second-order random-walk (RW2) prior on the sequence of horizon-specific response coefficients, while leaving the remaining elements of the Bayesian specification unchanged. The comparison therefore isolates the contribution of cross-horizon regularisation to the estimated aggregate response profiles.

For the US (Figure D.8), the unsmoothed estimates closely track the baseline responses across most variables. Removing the RW2 prior introduces visibly more horizon-to-horizon variation and generally widens the posterior sets, especially at longer horizons, but it leaves the main dynamic patterns largely unchanged. This is particularly clear for core capital-goods orders, where the unsmoothed median follows almost the same delayed and persistent expansion as the smoothed baseline. Real GDP and industrial production likewise retain the characteristic medium-run strengthening, with only modest additional short-run irregularity.

The labour-market responses display the same degree of continuity. Employment continues to rise gradually at medium and longer horizons, while hourly earnings show a similarly delayed positive response. Average hours preserve the baseline pattern as well, including the subsequent decline at longer horizons. Capacity utilisation is somewhat more irregular without smoothing, but its median response remains closely aligned with the baseline throughout most of the horizon.

The price block provides an equally informative comparison. The unsmoothed responses of core PCE, headline and core CPI, and producer prices all remain close to their smoothed counterparts. In particular, the gradual build-up in consumer-price responses and the earlier, stronger increase in producer prices are already present in the horizon-specific estimates. The main effect of the RW2 prior is therefore to attenuate local fluctuations and tighten posterior uncertainty rather than to generate the positive medium-run price response.

Taken together, the US results provide strong evidence that the principal features of the baseline transmission mechanism are not an artefact of cross-horizon smoothing. The delayed expansion in investment and real activity, the subsequent strengthening of employment and earnings, and the medium-run increase in prices are all visible when each horizon is

estimated independently. The hierarchical RW2 prior mainly improves precision and produces a more regular representation of response paths that are already evident in the underlying horizon-by-horizon estimates.

For the euro area (Figures D.9 and D.10), the matched comparison leads to the same conclusion. The unsmoothed medians are less regular and their posterior sets are generally wider, but the response paths remain strikingly close to the smoothed baseline for most real, price, and financial variables.

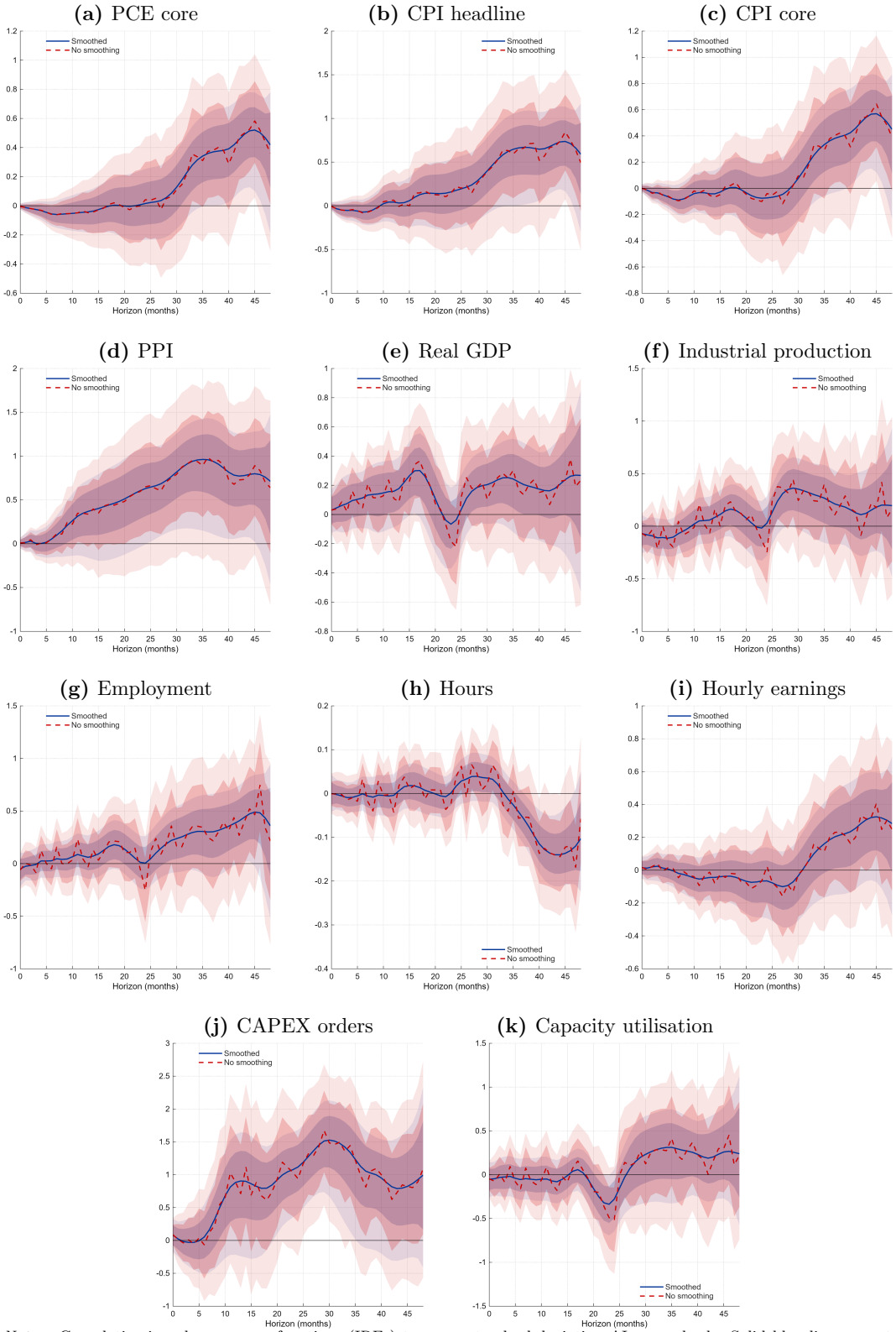
On the real side, employment reproduces the baseline sequence closely: an initial weakening is followed by a sustained medium-run increase before the response fades towards the end of the horizon. Industrial production also retains its short-run weakness, subsequent recovery, and later positive response, although the unsmoothed path is naturally more jagged around the turning points. Hours worked display somewhat greater local variation, but the broader medium-run increase remains visible. The similarity of the two sets of medians indicates that the delayed strengthening of euro area activity does not depend on cross-horizon pooling.

The price responses are likewise robust. Headline HICP, core HICP, producer prices, import prices, gas prices, and the broader energy-price measure all preserve the main timing and shape of the baseline responses. The unsmoothed specification reproduces the delayed rise in headline and core inflation, the stronger response of producer and import prices, and the medium-run increase in energy-related prices. The largest differences are confined to individual horizons, where the independently estimated coefficients fluctuate more sharply and their posterior sets widen. These fluctuations do not alter the overall sequencing of the price response.

The financial block shows similarly close correspondence. Euro area equity prices, two- and ten-year inflation-linked rates, and two- and ten-year OIS rates all display unsmoothed median paths that closely follow the smoothed baseline, including the medium-run rise in nominal and inflation-linked rates. The euro area–US yield differentials also preserve their characteristic initial decline and subsequent reversal, despite greater horizon-specific volatility. The exchange-rate response follows the same broad sequence as in the baseline, with only modest additional irregularity in the unsmoothed estimates.

Overall, the euro area comparison reinforces the US evidence. Across a broad set of

Figure D.8: Impact of AI news shocks on US aggregates—smoothed versus unsmoothed local projections

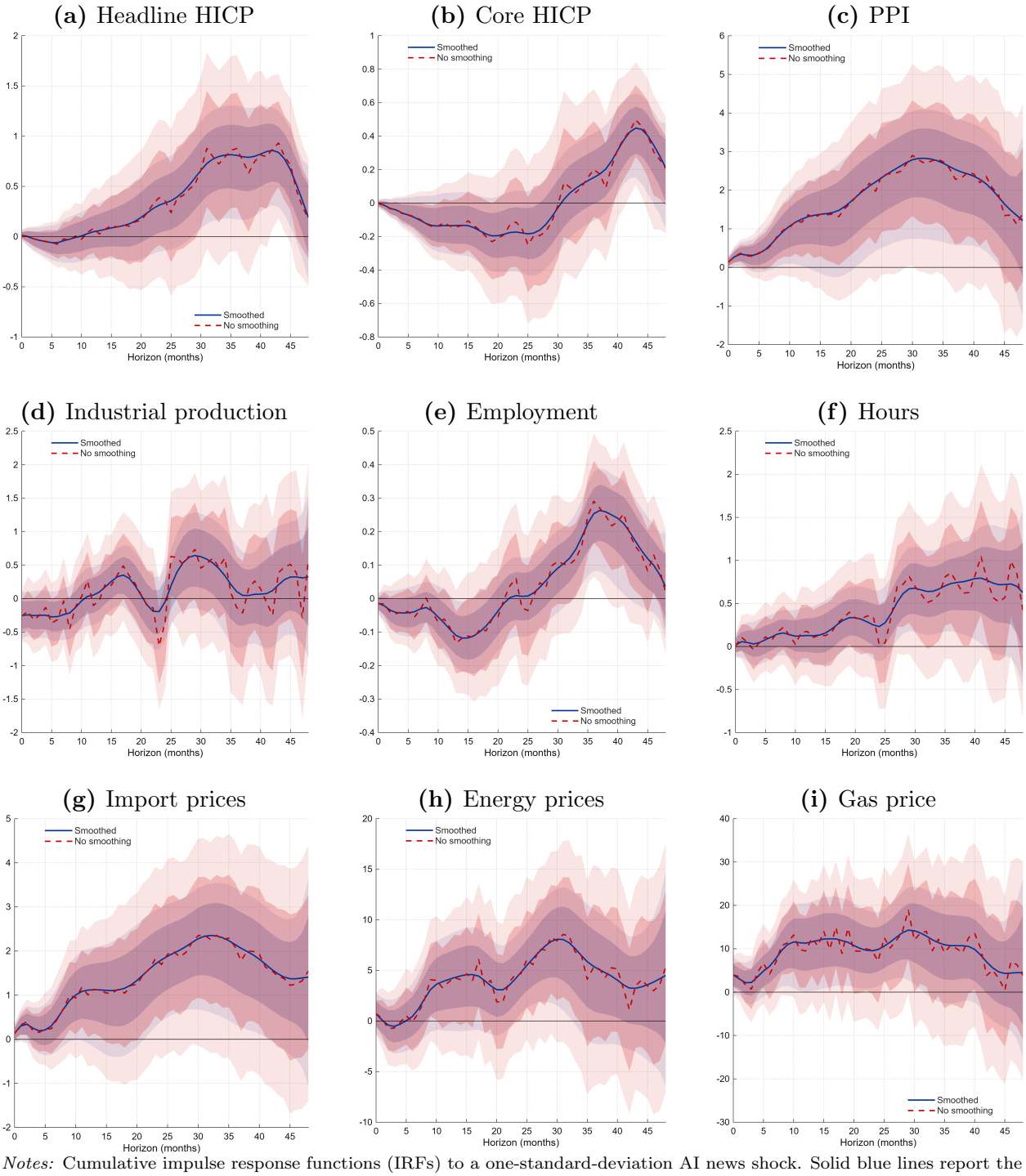


Notes: Cumulative impulse response functions (IRFs) to a one-standard-deviation AI news shock. Solid blue lines report the baseline estimates with hierarchical RW2 smoothing across horizons; dashed red lines report the corresponding estimates without cross-horizon smoothing. Darker (lighter) shaded areas represent 68% (90%) highest-density posterior sets. Variables are expressed as percent deviations from the baseline ($t-1$).

macroeconomic and financial variables, removing the RW2 prior increases local volatility and posterior uncertainty but leaves the substantive response patterns largely intact. The delayed strengthening of activity and inflation, the rise in energy and external prices, the adjustment in euro area interest rates, and the dynamics of the cross-country yield differentials are already present in the independently estimated horizon-specific coefficients.

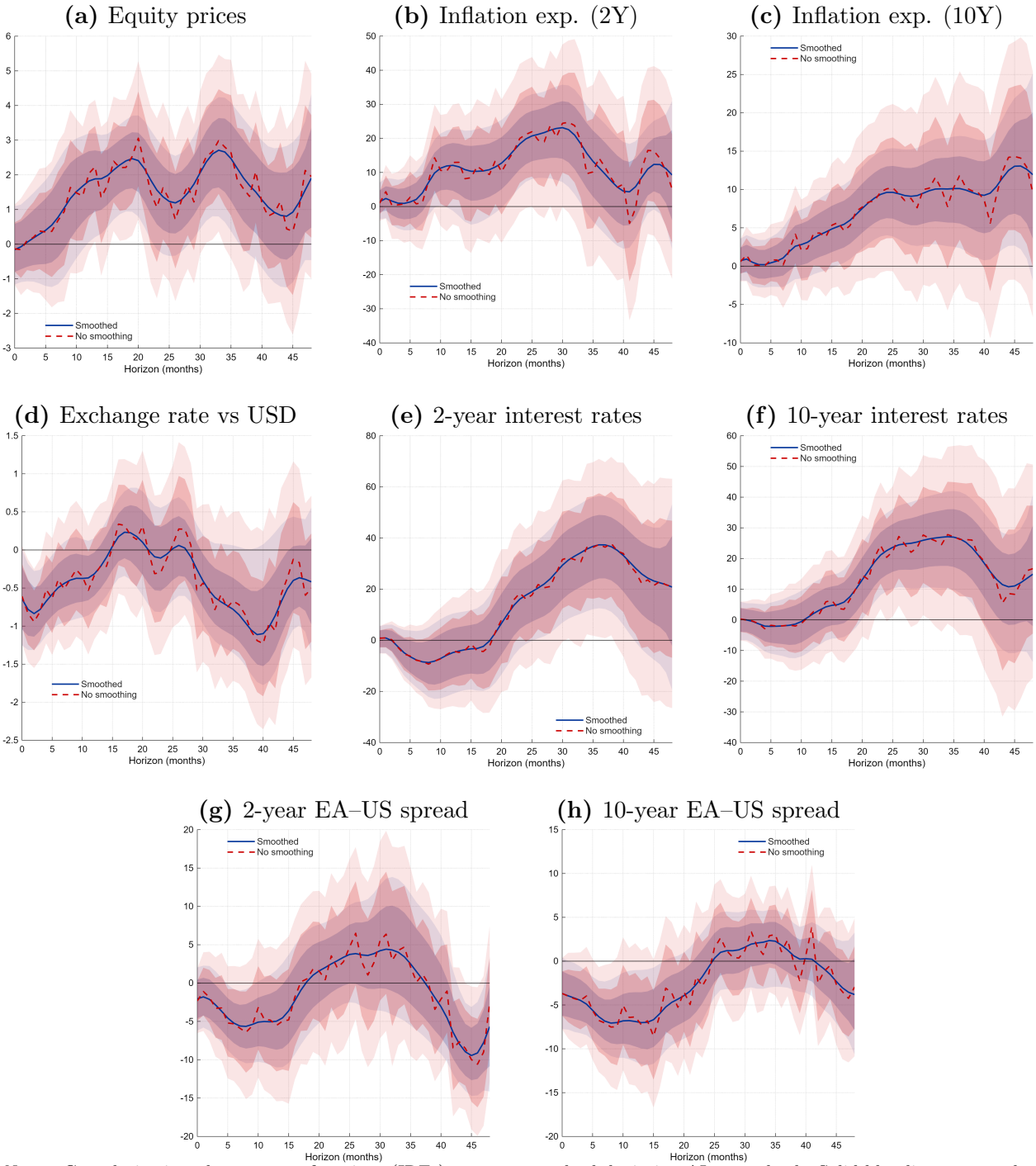
Taken together, the US and euro area results indicate that the hierarchical RW2 prior acts primarily as a regularisation device. Across both economies, the unsmoothed median responses remain closely aligned with the smoothed baseline, while the main differences concern horizon-to-horizon variability and the width of posterior sets. The central economic patterns documented in the baseline therefore do not appear to be generated by the smoothing prior. Instead, cross-horizon regularisation improves precision and interpretability by extracting smoother response profiles from an underlying signal that is already visible in the unrestricted horizon-specific estimates.

Figure D.9: Impact of AI news shocks on euro area *real and price* aggregates—smoothed versus unsmoothed local projections



Notes: Cumulative impulse response functions (IRFs) to a one-standard-deviation AI news shock. Solid blue lines report the baseline estimates with hierarchical RW2 smoothing across horizons; dashed red lines report the corresponding estimates without cross-horizon smoothing. Variables are expressed as percent deviations from the baseline $(t-1)$. Darker (lighter) shaded areas represent 68% (90%) highest-density posterior sets.

Figure D.10: Impact of AI news shocks on euro area *financial* aggregates—smoothed versus unsmoothed local projections



Notes: Cumulative impulse response functions (IRFs) to a one-standard-deviation AI news shock. Solid blue lines report the baseline estimates with hierarchical RW2 smoothing across horizons; dashed red lines report the corresponding estimates without cross-horizon smoothing. Equity prices and exchange rate are expressed as percent deviations from the baseline ($t-1$), while rates, spreads and inflation expectations are expressed in basis points. Darker (lighter) shaded areas represent 68% (90%) highest-density posterior sets.

D.2.3 Excluding 2020–2022

We re-estimate Equation (5) after removing observations from 2020–2022, a period shaped by the pandemic, reopening, geopolitical tensions, and the energy shock. This is a demanding sample restriction: it removes three years of observations and, together with the lag structure, substantially reduces the effective estimation sample. We therefore report responses only through month 24, beyond which posterior uncertainty becomes especially large.

For the US (Figure D.11), the restricted sample continues to display a short-run increase in consumer prices. Headline and core CPI and core PCE prices rise over approximately the first year, and their responses are generally stronger than in the baseline over these short horizons. The responses subsequently weaken and move back towards zero, turning negative at the end of the reported horizon. Producer prices display a similar but more volatile pattern: the response is positive initially, falls below zero around the middle of the horizon, briefly recovers, and becomes negative again by month 24. Hence, the evidence for a short-run inflationary effect is not driven exclusively by 2020–2022, whereas the persistent increase in prices observed in the full sample is more sensitive to the inclusion of those years.

The real and labour-market responses are more sample-sensitive. Real GDP remains mildly positive during the first year but subsequently weakens and turns negative, while industrial production is initially weaker than in the baseline and remains close to zero thereafter. Employment follows the baseline relatively closely at short horizons but declines after roughly one year in the restricted sample. Hourly earnings show a short-run increase before weakening and eventually turning negative, whereas average hours remain comparatively close to zero and display substantial uncertainty. Capacity utilisation declines throughout much of the horizon and is somewhat weaker than in the baseline at intermediate horizons.

The clearest US departure continues to concern investment. Core capital-goods orders decline through much of the first 18 months when 2020–2022 is excluded, in sharp contrast with the persistent increase in the full-sample estimates. The response subsequently rebounds and becomes positive towards the end of the reported horizon, but remains well below the baseline path. The strong and persistent investment response that characterises the full sample should therefore be interpreted with some caution: its magnitude and timing are sensitive to the inclusion of the pandemic and reopening period, even though the late-horizon

recovery indicates that the response is not uniformly negative outside those years.

For the euro area (Figures D.12 and D.13), the restricted-sample estimates also preserve some of the short-run price responses while substantially altering their persistence. Headline and core HICP rise over the first year, with core inflation in particular becoming positive rather than displaying the initial weakness observed in the baseline. Producer and import prices increase at short horizons but subsequently fall back and become negative around the middle of the horizon before partially recovering. Energy and gas prices remain highly variable, although positive responses are still visible over parts of the horizon. Thus, the broad conclusion that US AI news can generate external and upstream price pressures in the euro area survives the sample restriction, but the persistence of these effects is considerably less robust.

Euro area real activity becomes noticeably weaker when 2020–2022 is removed. Industrial production remains subdued and turns significantly negative around the middle of the horizon before recovering. Employment and hours worked also decline at medium horizons, in contrast with the delayed strengthening observed in the full sample. The longer-run domestic expansion in the baseline is therefore particularly sensitive to the inclusion of the pandemic, reopening, and energy-crisis years.

The financial responses are more heterogeneous. Euro area equity prices still rise at medium horizons, although by less than in the baseline, while the euro continues to depreciate against the dollar. Inflation compensation remains generally positive, particularly at longer maturities, but displays greater short-run variation. The largest change concerns nominal rates and relative interest-rate differentials. In the restricted sample, two- and ten-year euro area OIS rates rise already at relatively short horizons, whereas the baseline two-year rate initially declines. Correspondingly, the EA–US differentials no longer display the same persistent short-run movement in favour of the United States. The ten-year differential becomes positive comparatively early, and the two-year differential reverses strongly at later horizons. The precise relative-rate dynamics documented in the baseline are therefore sensitive to the inclusion of 2020–2022.

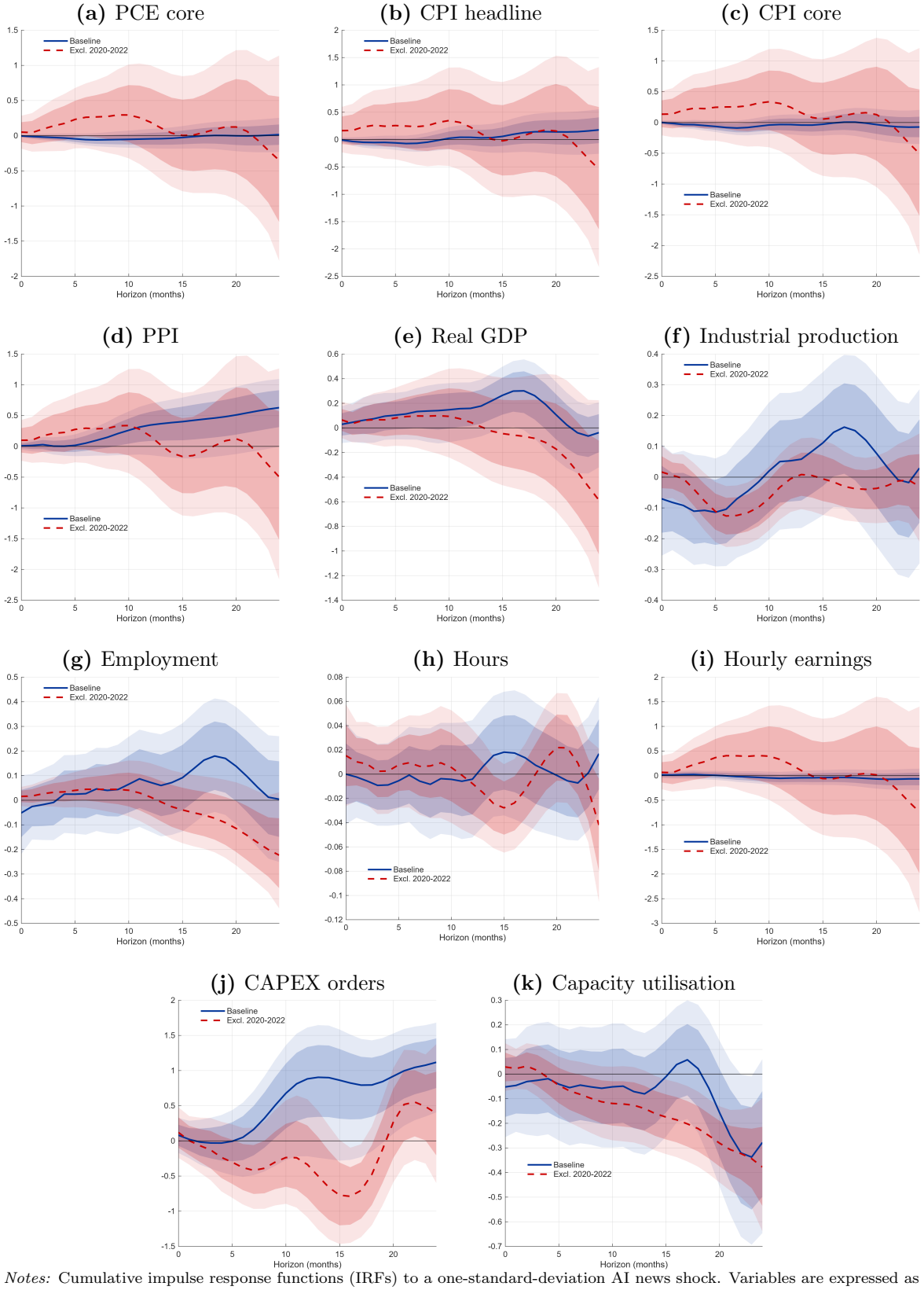
These results qualify, rather than overturn, the baseline interpretation. The short-run transmission of AI news through prices, the exchange rate, and financial markets remains

visible outside the pandemic and reopening period. By contrast, the persistence of the US investment response, the delayed euro area expansion in activity and employment, and especially the detailed path of EA–US interest-rate differentials are materially more sample-dependent. This distinction is also reflected in the substantially wider posterior sets in the restricted sample: for many variables the two specifications remain statistically difficult to distinguish despite differences in their posterior median paths. The robustness exercise therefore provides stronger support for the paper's short-run financial and price-transmission mechanisms than for the exact magnitude and persistence of the longer-run real and interest-rate responses.

For the euro area (Figures D.12 and D.13), several early price and cost responses remain visible. Headline and core HICP, producer prices, import prices, and energy prices generally retain an inflationary short-run profile, although the restricted-sample medians are less persistent and the posterior intervals widen considerably. Industrial production and the labour-market variables remain imprecisely estimated, and their medium-run paths differ more clearly from the baseline. The evidence therefore supports the presence of cross-border price transmission outside 2020–2022, while providing weaker support for the precise timing and magnitude of the real response.

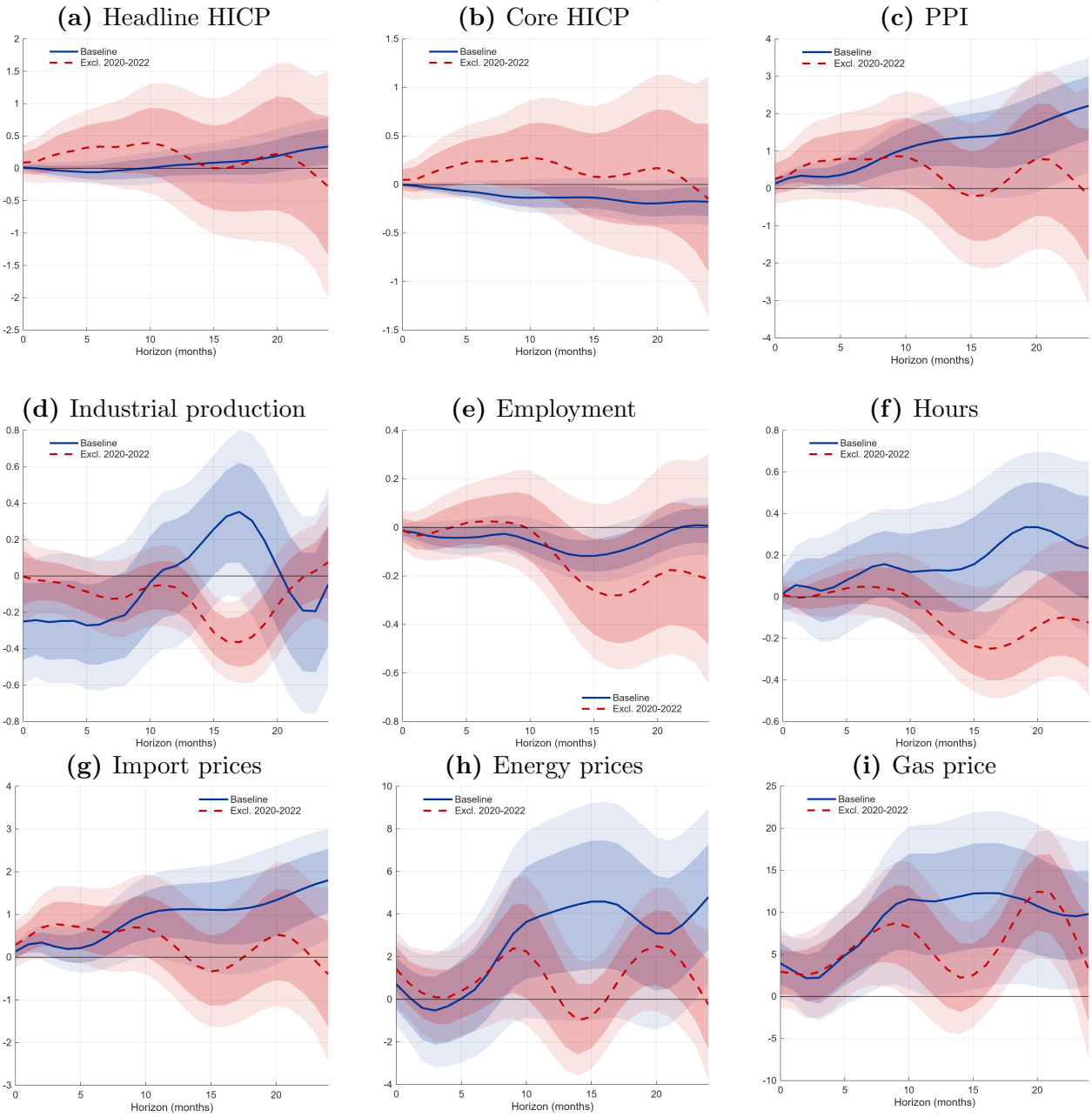
The financial results are more sample-sensitive. The responses of equity prices, the exchange rate, OIS rates, inflation expectations, and the EA–US spreads change materially in magnitude and, at some horizons, in sign. In particular, the restricted-sample rate responses do not reproduce the baseline phasing in a stable manner. The broad evidence of international financial transmission remains, but its dynamic profile depends importantly on the inclusion of the exceptional monetary and financial conditions prevailing in 2020–2022. This qualification is consistent with the wider posterior sets and is the reason the main text treats the euro area financial robustness results more cautiously than the price responses.

Figure D.11: Impact of AI news shocks on US aggregates excluding 2020–2022



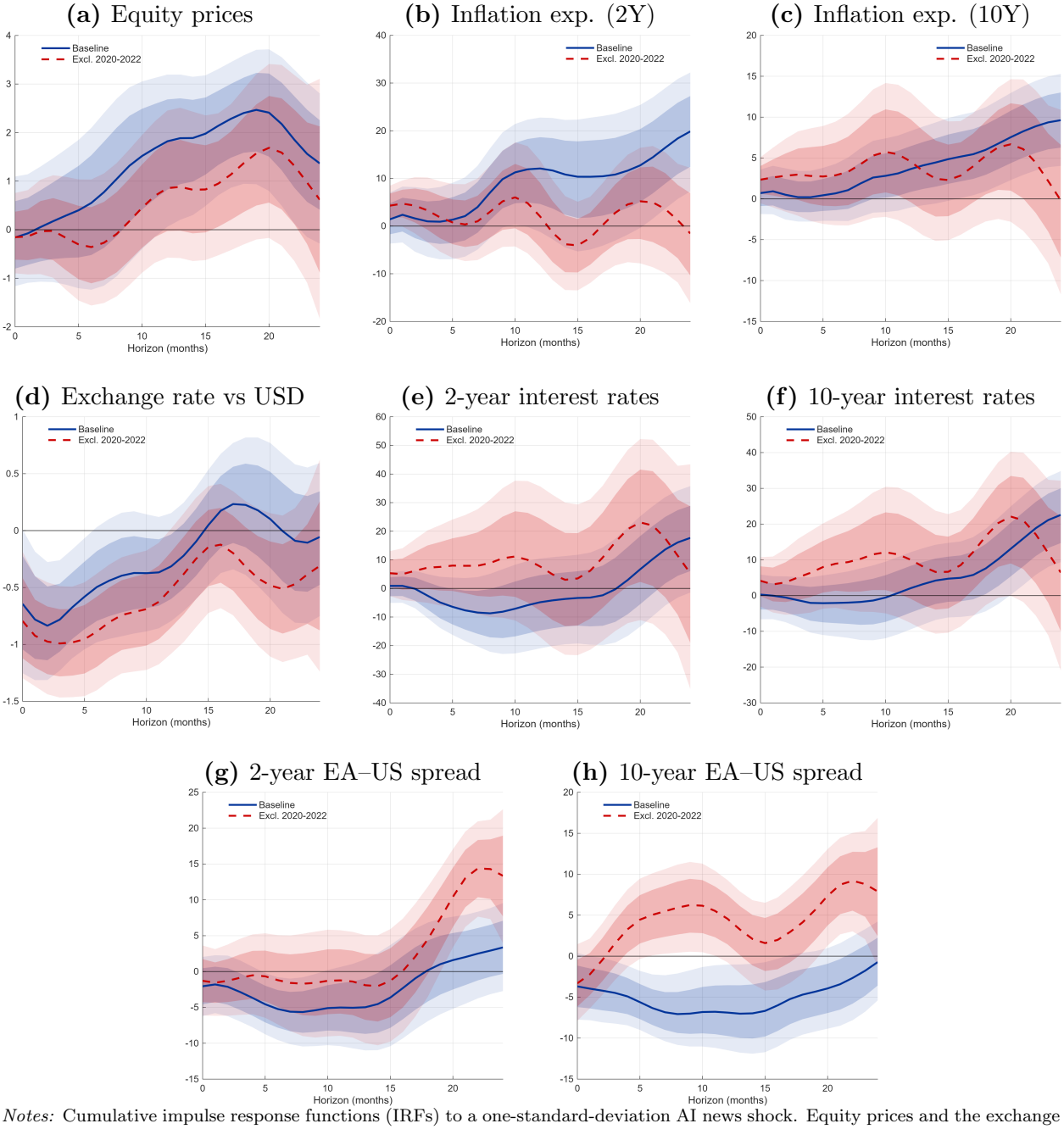
Notes: Cumulative impulse response functions (IRFs) to a one-standard-deviation AI news shock. Variables are expressed as percent deviations from the baseline $(t-1)$. Darker (lighter) shaded areas represent 68% (90%) highest-density posterior sets.

Figure D.12: Impact of AI news shocks on euro area *real/price* aggregates excluding 2020–2022



Notes: Cumulative impulse response functions (IRFs) to a one-standard-deviation AI news shock. Variables are expressed as percent deviations from the baseline ($t-1$). Darker (lighter) shaded areas represent 68% (90%) highest-density posterior sets.

Figure D.13: Impact of AI news shocks on euro area *financial* aggregates excluding 2020–2022



Notes: Cumulative impulse response functions (IRFs) to a one-standard-deviation AI news shock. Equity prices and the exchange rate are expressed as percent deviations from the baseline ($t-1$); rates, spreads, and inflation expectations are expressed in basis points. Darker (lighter) shaded areas represent 68% (90%) highest-density posterior sets.

D.2.4 Sectoral results with different exposure thresholds

We assess whether the sectoral results depend on the thresholds used to define high- and low-exposure industries. We reconstruct the upstream and downstream aggregates using sectors above the 90th percentile and below the 10th percentile of the relevant exposure distribution, rather than the 75th and 25th percentile cut-offs used in the baseline, and re-estimate the local projections without changing the remaining specification (Figures D.14 and D.15).

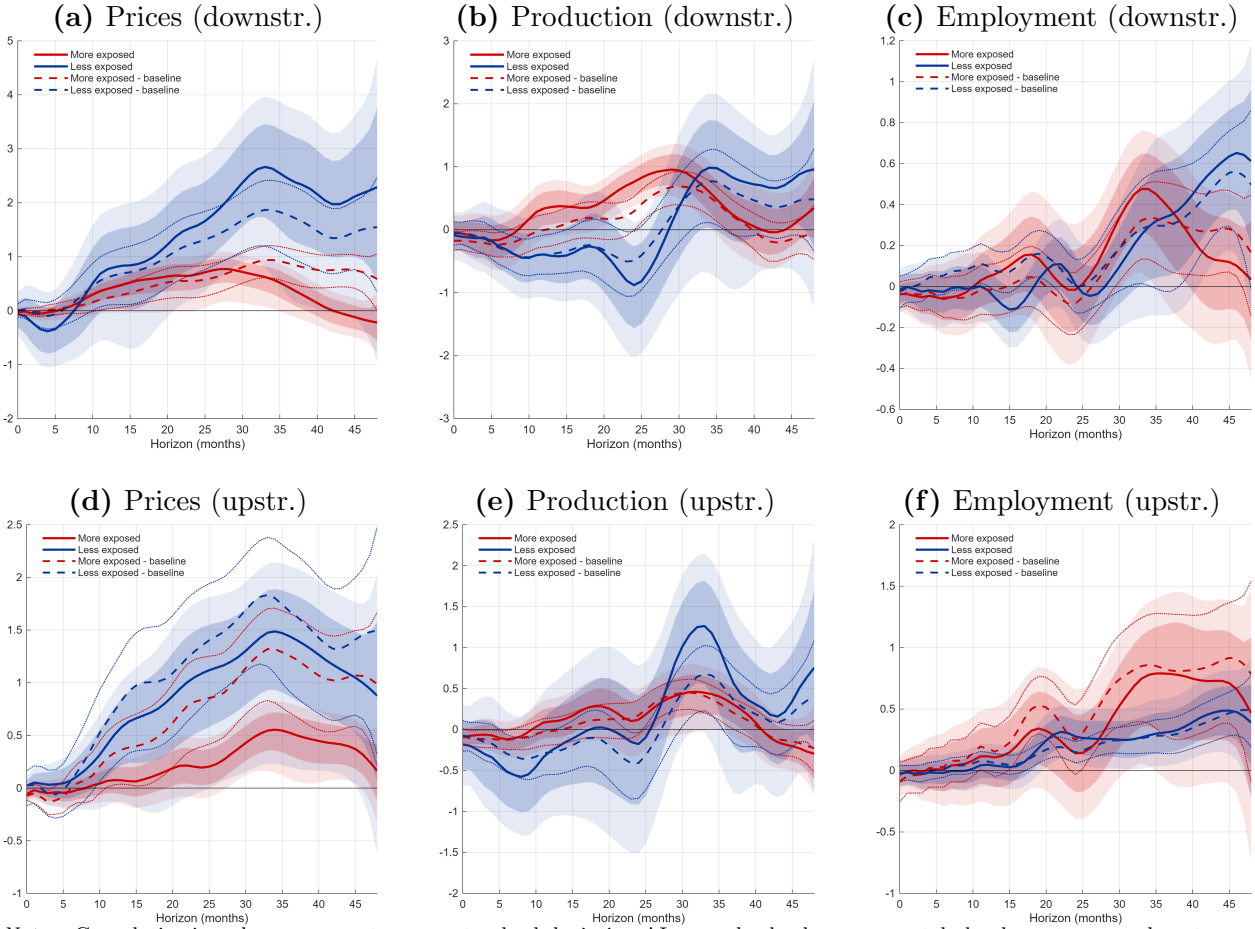
As expected, the narrower groups produce wider posterior sets, since fewer industries enter each aggregate. At the same time, the main qualitative patterns remain visible and, in some cases, become more pronounced. The producer-price results are especially robust. Less-exposed sectors continue to experience substantially larger price increases than more-exposed sectors, both upstream and downstream. Moreover, the high-minus-low differentials are more negative than in the baseline over large parts of the medium and long horizon, with the effect particularly strong downstream. Thus, isolating the tails of the exposure distribution sharpens the evidence that inflationary pressures are concentrated outside the sectors most directly linked to the AI production network.

The production results also preserve the baseline concentration-to-diffusion pattern, although the dynamics are somewhat more uneven. Upstream, more-exposed sectors continue to perform relatively better at shorter horizons, but this advantage weakens, briefly reverses around the middle of the horizon, reappears, and then turns negative again at longer horizons. Downstream, the pattern is clearer: more-exposed sectors outperform early on, but the differential reverses decisively at longer horizons, with less-exposed sectors accounting for a larger share of the response once the shock diffuses beyond the AI ecosystem. Relative to the baseline, the downstream reversal appears sharper.

The employment results continue to display a marked upstream–downstream asymmetry. Among upstream suppliers, highly exposed sectors still show the stronger relative response over much of the medium run, and this advantage is at least as pronounced as in the baseline, although it narrows and may even reverse slightly at the very end of the horizon. Downstream, by contrast, the employment differential shifts more clearly towards less-exposed sectors: after a limited short-run advantage for more-exposed industries, the high-minus-low differential turns negative and remains below the baseline over most of the medium and long horizon.

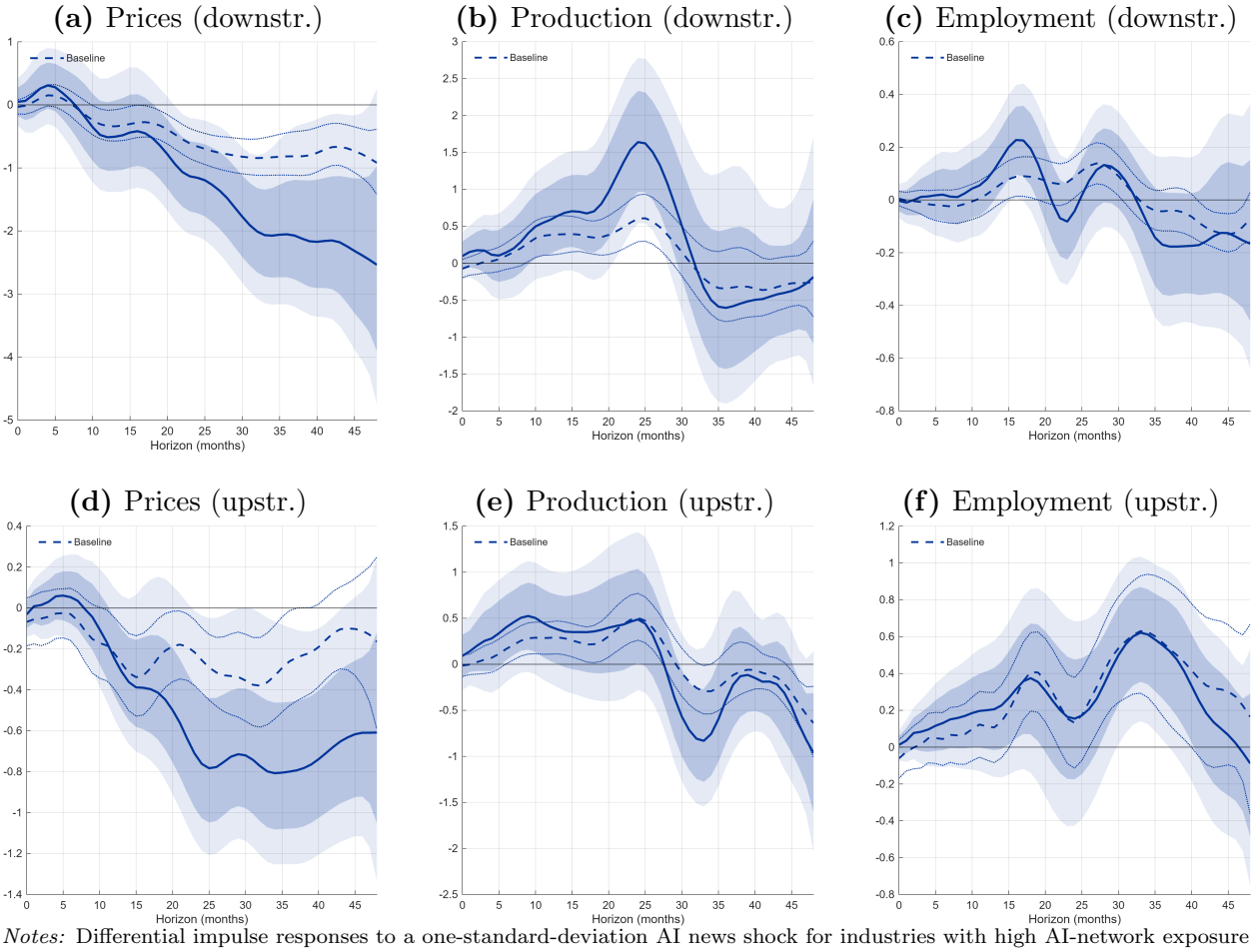
Overall, the sectoral transmission mechanism is not driven by the baseline threshold choice. If anything, focusing on the most and least exposed sectors reinforces the interpretation of the AI news shock as initially concentrated around the AI production network and subsequently diffusing to the broader economy. The narrower-threshold exercise therefore strengthens, rather than weakens, the baseline evidence on network heterogeneity.

Figure D.14: Impact of AI news shock on sectors



Notes: Cumulative impulse responses to a one-standard-deviation AI news shock, shown separately by downstream and upstream input–output exposure to AI-related sectors. Darker (lighter) shaded areas represent 68% (90%) highest-density posterior sets for the robustness exercise; dashed lines show the baseline median and 68% posterior set. Variables are expressed as percentage deviations from the baseline ($t-1$).

Figure D.15: *Differential* impact of AI news shock on sectors



Notes: Differential impulse responses to a one-standard-deviation AI news shock for industries with high AI-network exposure relative to industries with low exposure. Darker (lighter) shaded areas represent 68% (90%) highest-density posterior sets for the robustness exercise; dashed lines show the baseline median and 68% posterior set. Responses are expressed in percentage points.